



UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

Course of Robot Control

Prof. Antonella Ferrara



Fourth Part

Outline of the Course

Basic Terminology and Concepts

Reference Frames and Rotation Matrices

Robot Direct Kinematics: the Geometric Approach

Robot Direct Kinematics: the Denavit-Hartenberg Convention Based Approach

Robot Inverse Kinematics

Robot Differential Kinematics

Robot Dynamical Model

Robot Control



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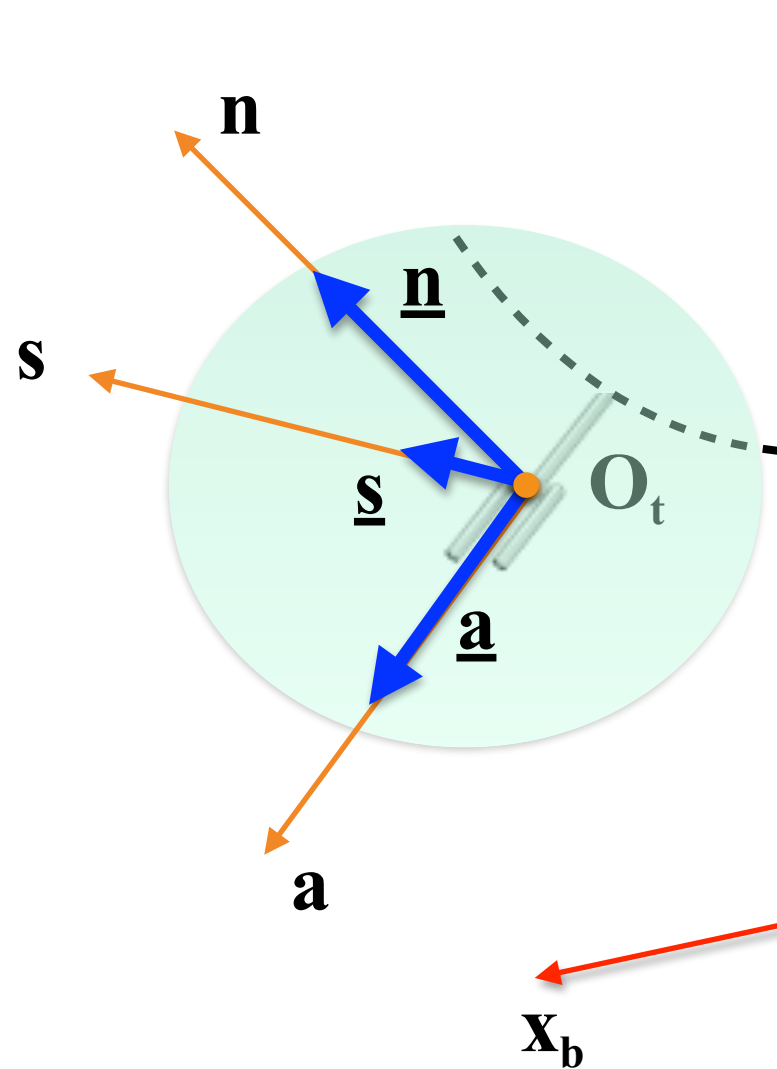


Robot Inverse Kinematics: Some preliminary concepts

In a previous lecture

Orientation of a Rigid Object in 3D Space

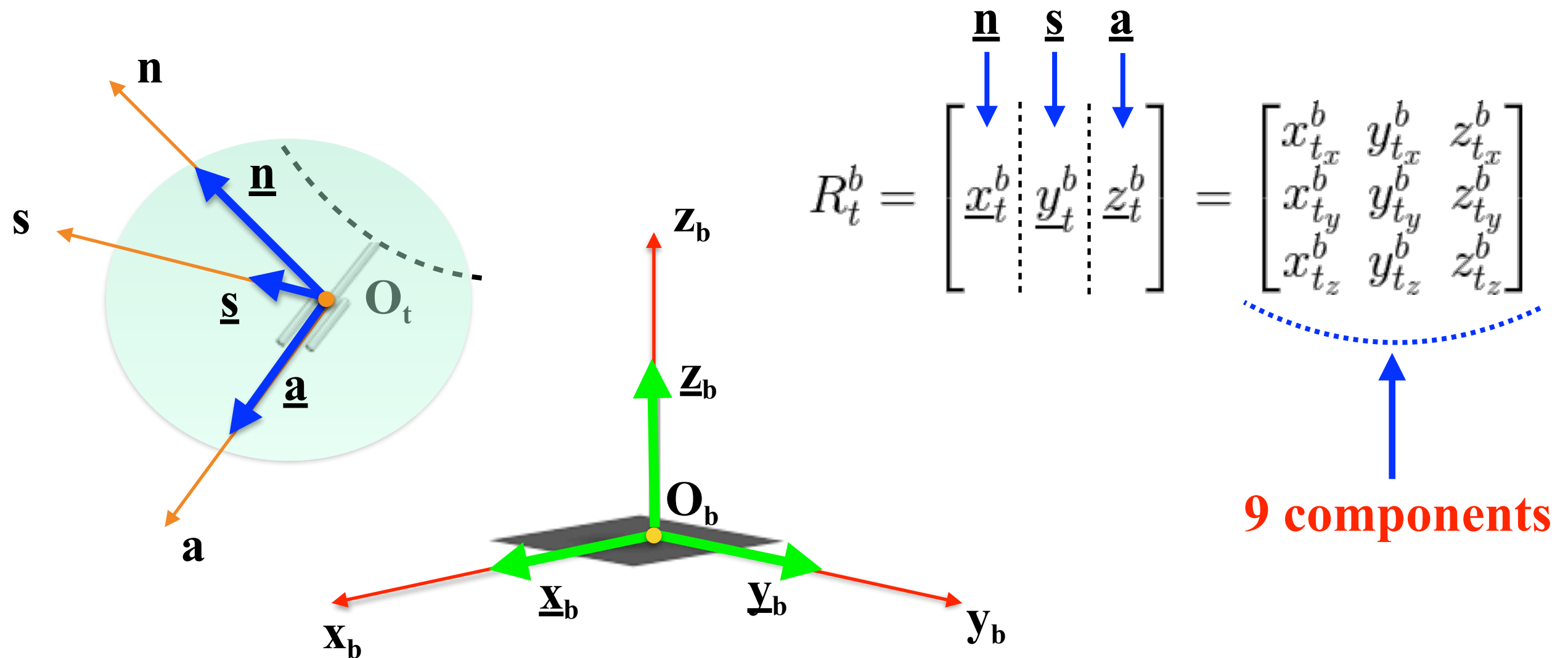
Rotation Matrix



$$R_t^b = \begin{matrix} \begin{matrix} \underline{n} \\ \underline{s} \\ \underline{a} \end{matrix} \\ \begin{matrix} \downarrow \\ \downarrow \\ \downarrow \end{matrix} \\ \begin{bmatrix} \underline{x}_t^b & \underline{y}_t^b & \underline{z}_t^b \end{bmatrix} \end{matrix} = \begin{bmatrix} x_{tx}^b & y_{tx}^b & z_{tx}^b \\ x_{ty}^b & y_{ty}^b & z_{ty}^b \\ x_{tz}^b & y_{tz}^b & z_{tz}^b \end{bmatrix}$$

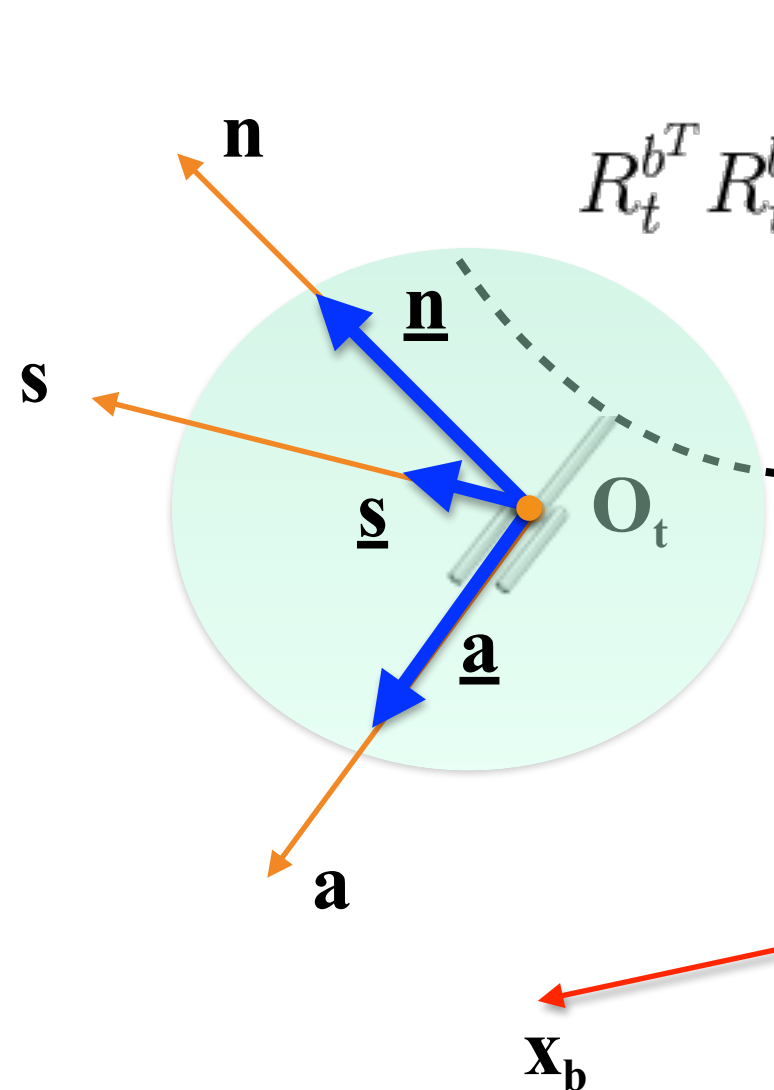
Orientation of a Rigid Object in 3D Space

Rotation Matrix



Orientation of a Rigid Object in 3D Space

Rotation Matrix



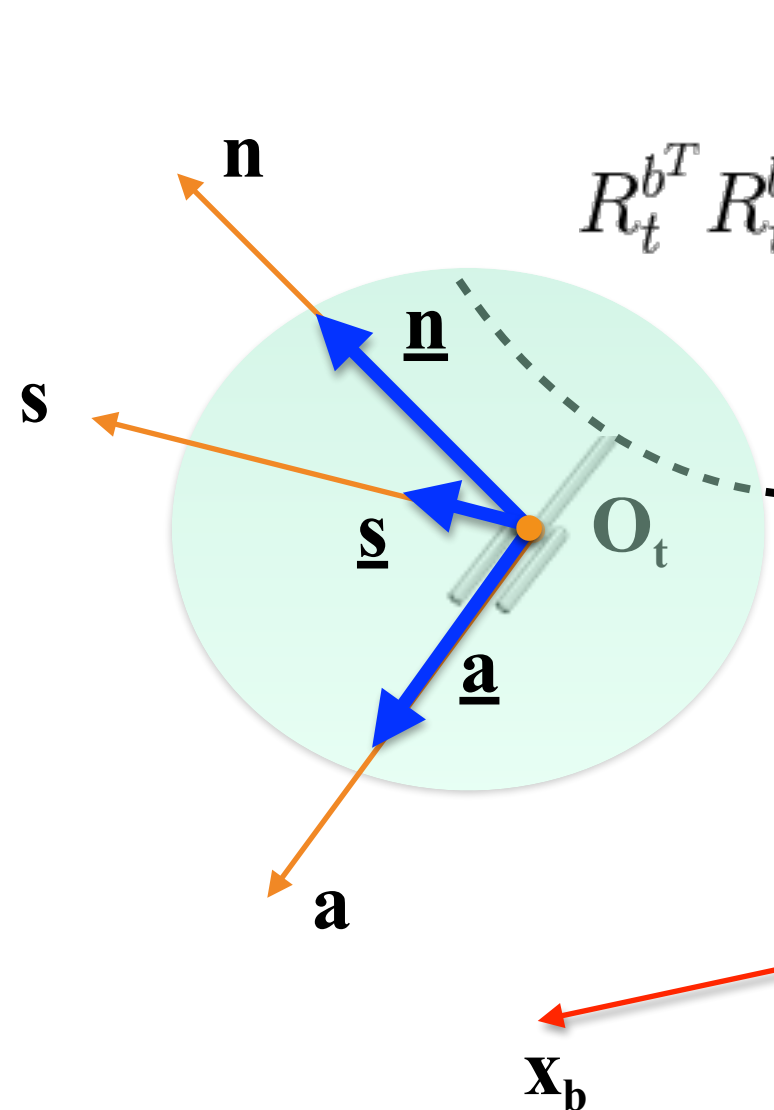
$$R_t^{b^T} R_t^b = \begin{bmatrix} \underline{x}_t^{b^T} \\ \underline{y}_t^{b^T} \\ \underline{z}_t^{b^T} \end{bmatrix} \cdot \begin{bmatrix} \underline{x}_t^b & \underline{y}_t^b & \underline{z}_t^b \end{bmatrix} = \begin{bmatrix} \underline{x}_t^{b^T} \underline{x}_t^b & \underline{x}_t^{b^T} \underline{y}_t^b & \underline{x}_t^{b^T} \underline{z}_t^b \\ \underline{y}_t^{b^T} \underline{x}_t^b & \underline{y}_t^{b^T} \underline{y}_t^b & \underline{y}_t^{b^T} \underline{z}_t^b \\ \underline{z}_t^{b^T} \underline{x}_t^b & \underline{z}_t^{b^T} \underline{y}_t^b & \underline{z}_t^{b^T} \underline{z}_t^b \end{bmatrix}$$

$$R_t^{b^T} R_t^b = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$R_t^{b^T} = R_t^{b^{-1}}$$

Rotation matrices are orthogonal square matrices

Orientation of a Rigid Object in 3D Space



$R_t^{bT} R_t^b = \begin{bmatrix} \underline{x}_t^{bT} \\ \underline{y}_t^{bT} \\ \underline{z}_t^{bT} \end{bmatrix} \cdot \begin{bmatrix} \underline{x}_t^b & \underline{y}_t^b & \underline{z}_t^b \end{bmatrix} = \begin{bmatrix} \underline{x}_t^{bT} \underline{x}_t^b & \underline{x}_t^{bT} \underline{y}_t^b & \underline{x}_t^{bT} \underline{z}_t^b \\ \underline{y}_t^{bT} \underline{x}_t^b & \underline{y}_t^{bT} \underline{y}_t^b & \underline{y}_t^{bT} \underline{z}_t^b \\ \underline{z}_t^{bT} \underline{x}_t^b & \underline{z}_t^{bT} \underline{y}_t^b & \underline{z}_t^{bT} \underline{z}_t^b \end{bmatrix}$

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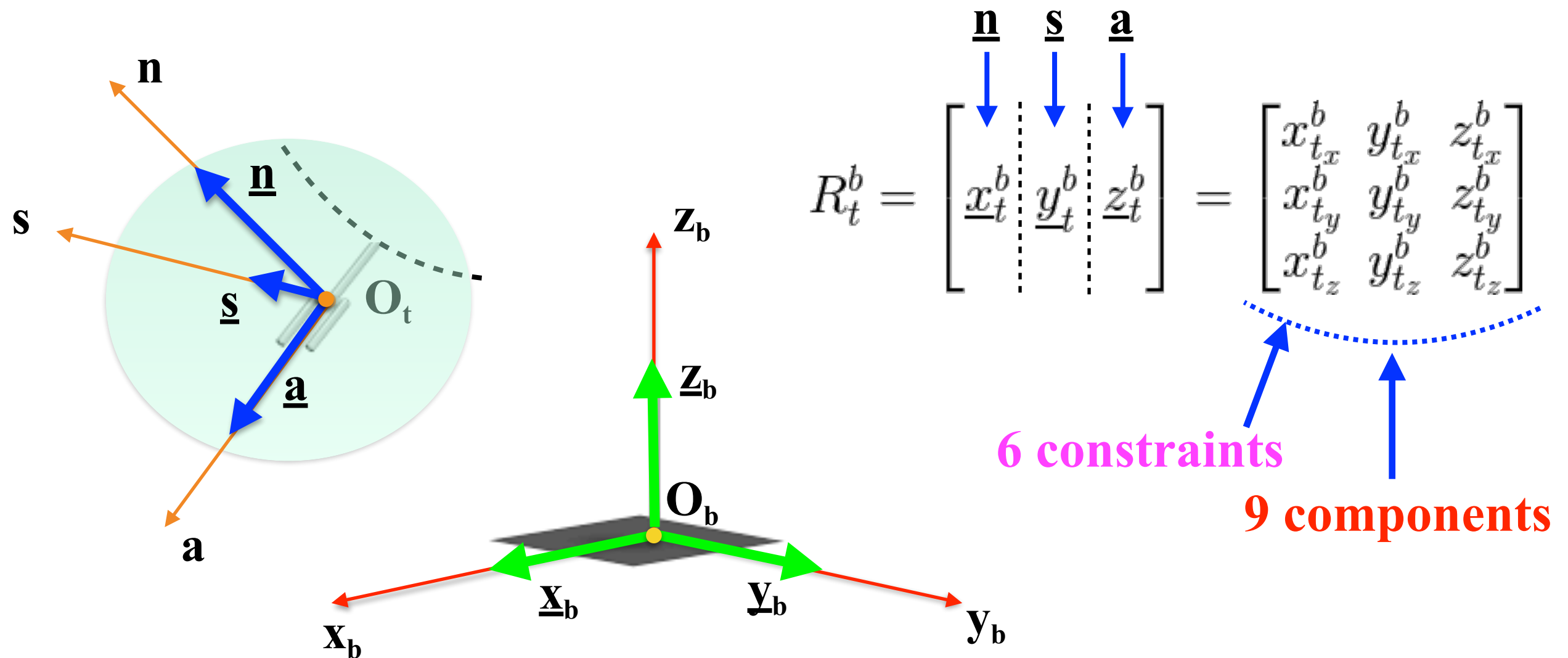
$R_t^{bT} = R_t^{b^{-1}}$

6 constraints

Rotation matrices are orthogonal square matrices

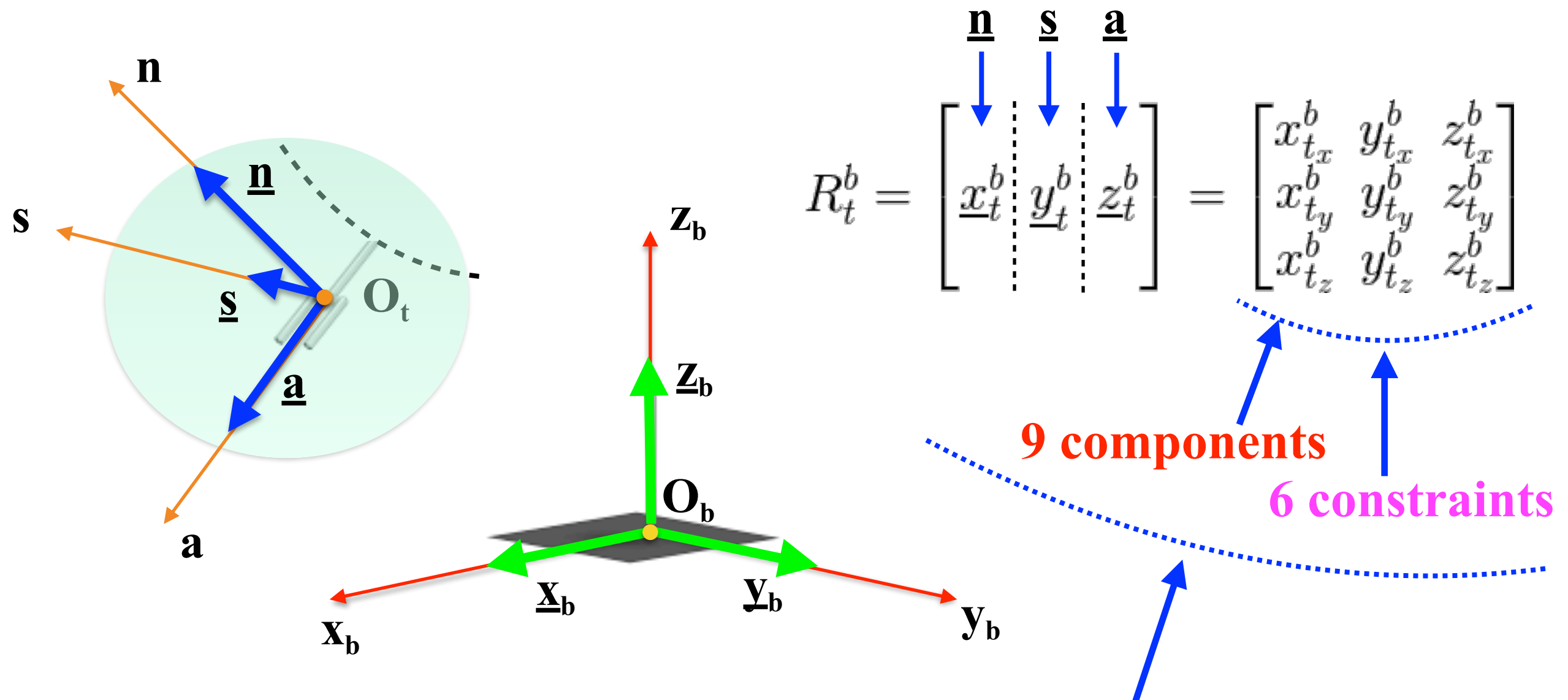
Orientation of a Rigid Object in 3D Space

Rotation Matrix



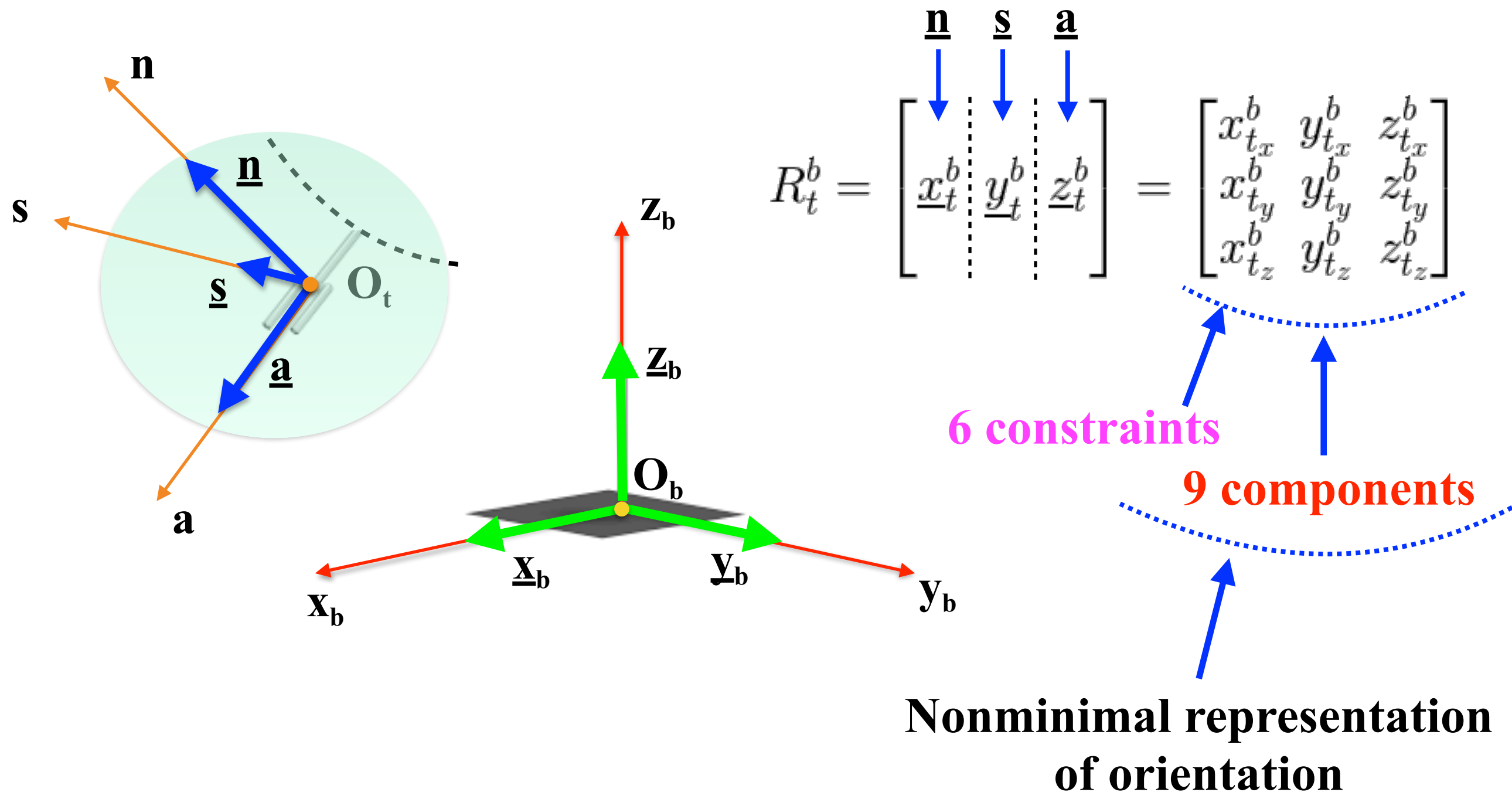
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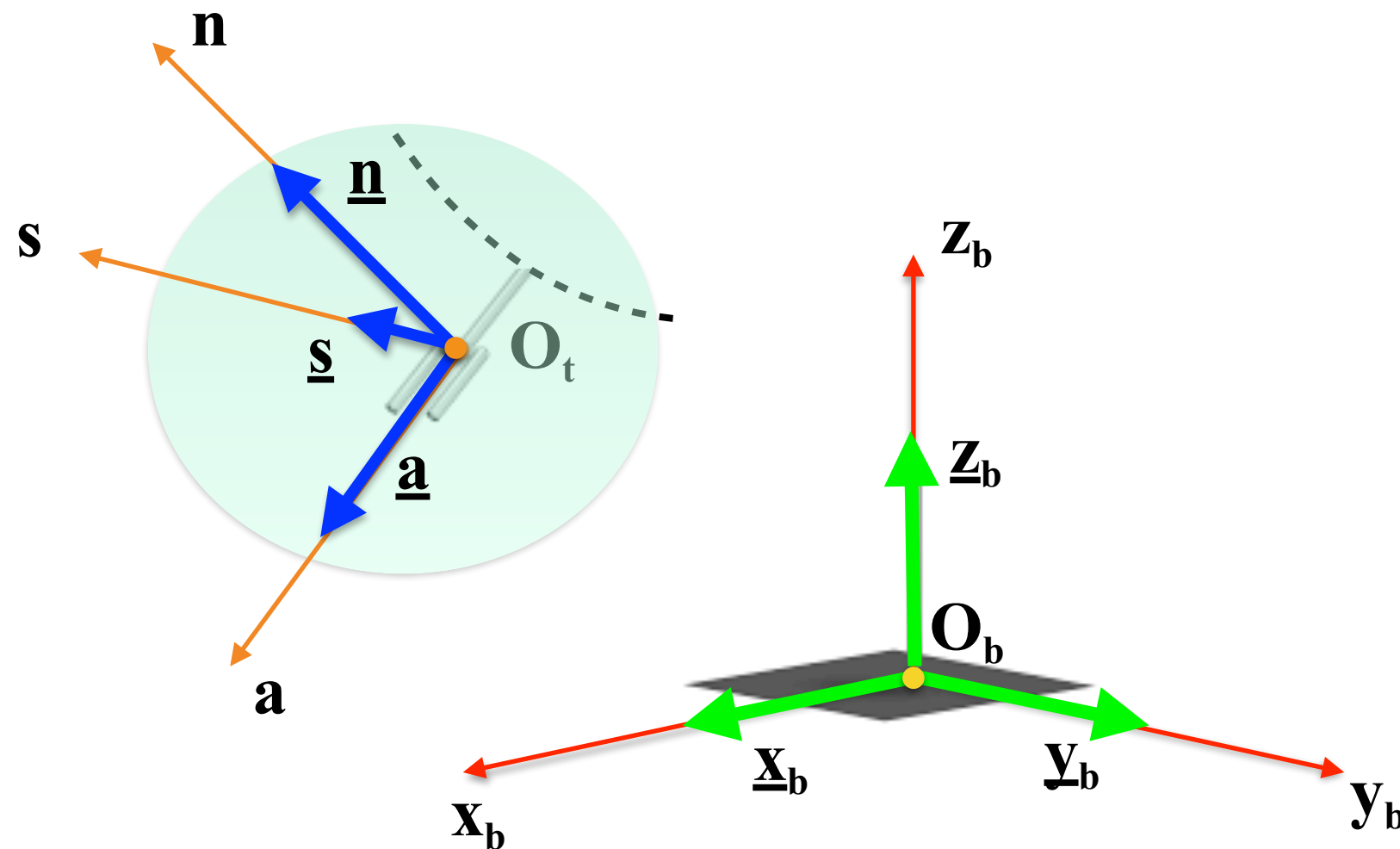
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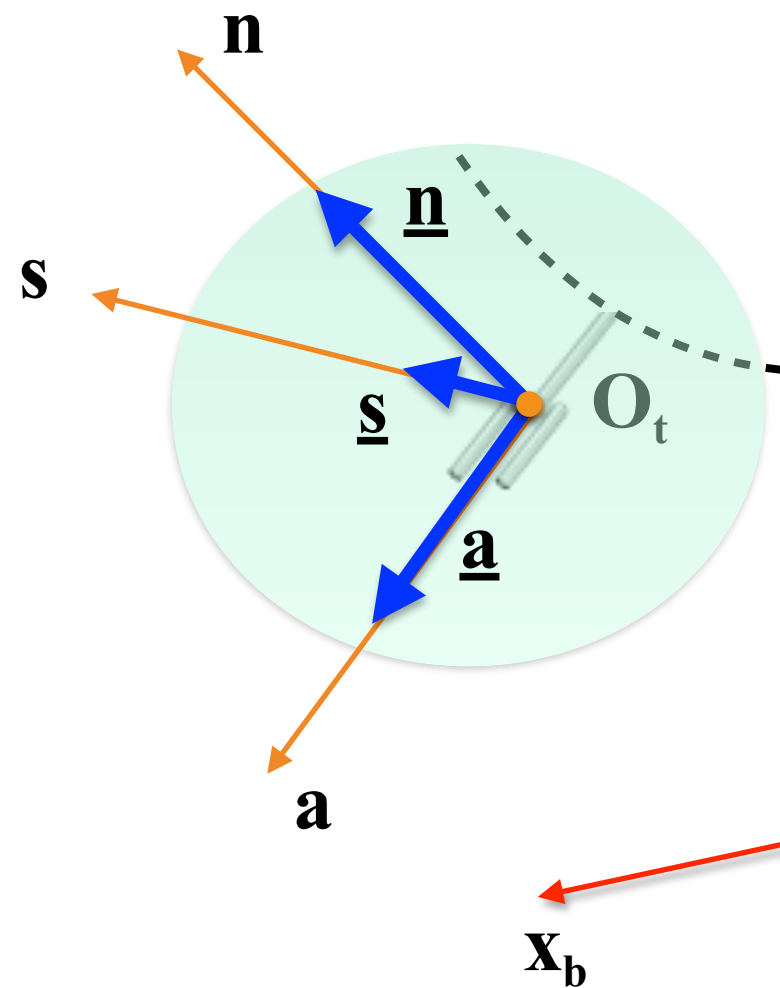
Minimal Description of the Orientation of a Rigid Object in 3D Space

Only 3 independent quantities are needed

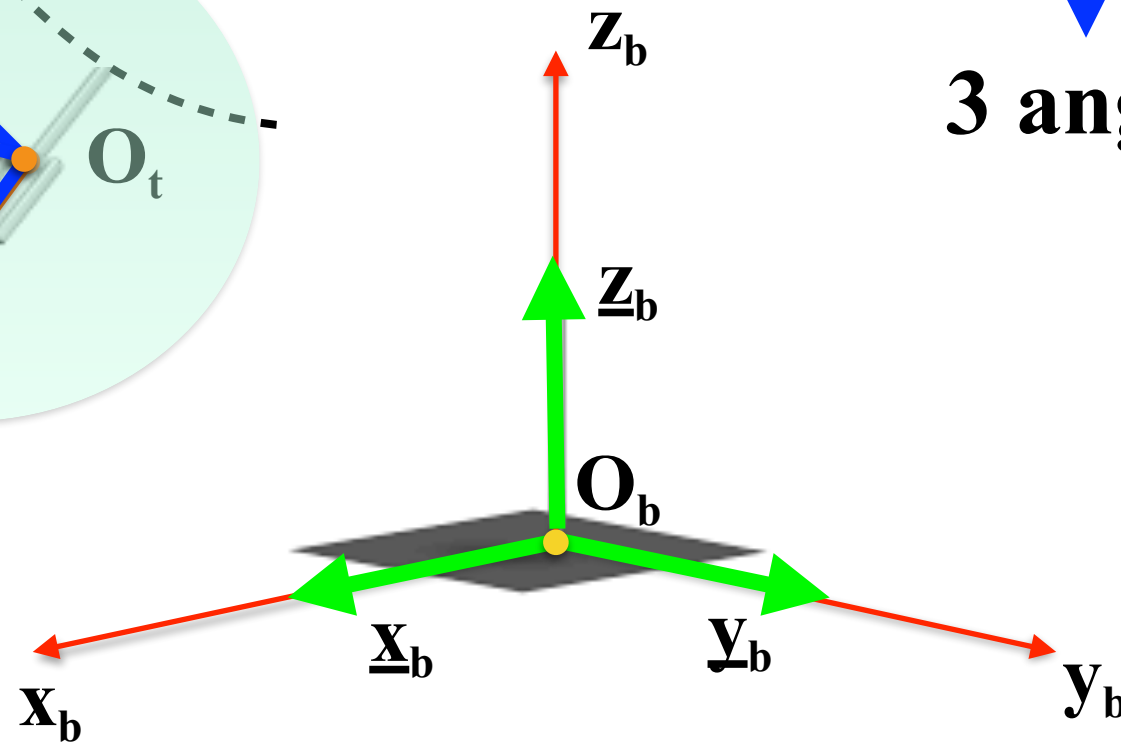


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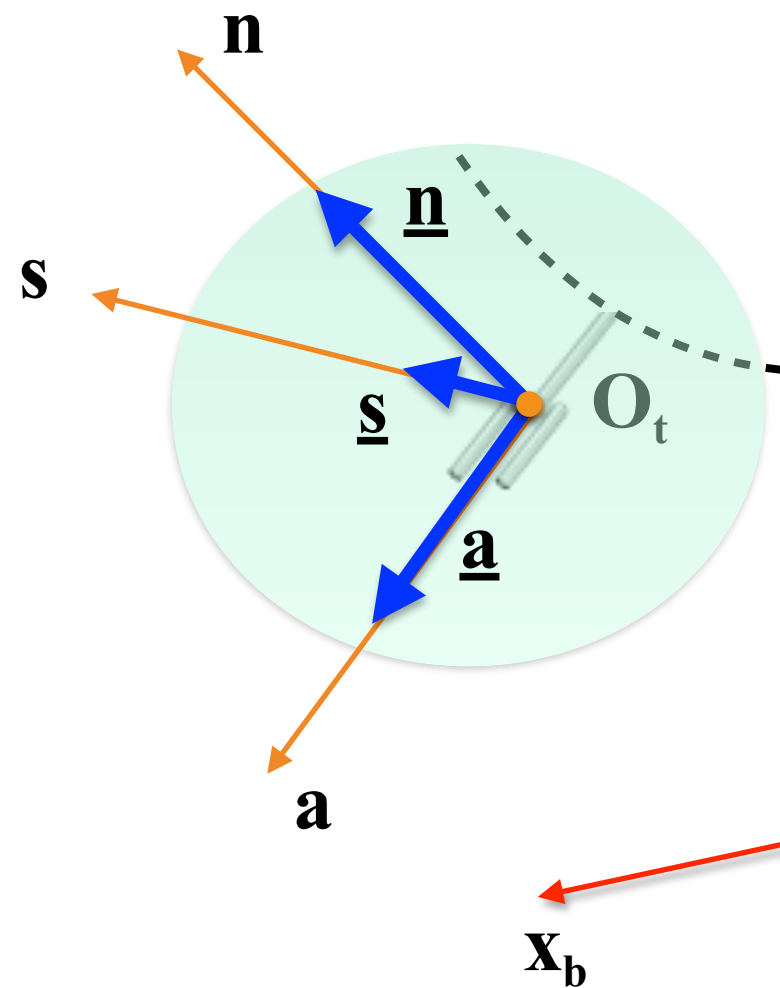


3 angles $\phi = [\varphi \ \vartheta \ \psi]^T$



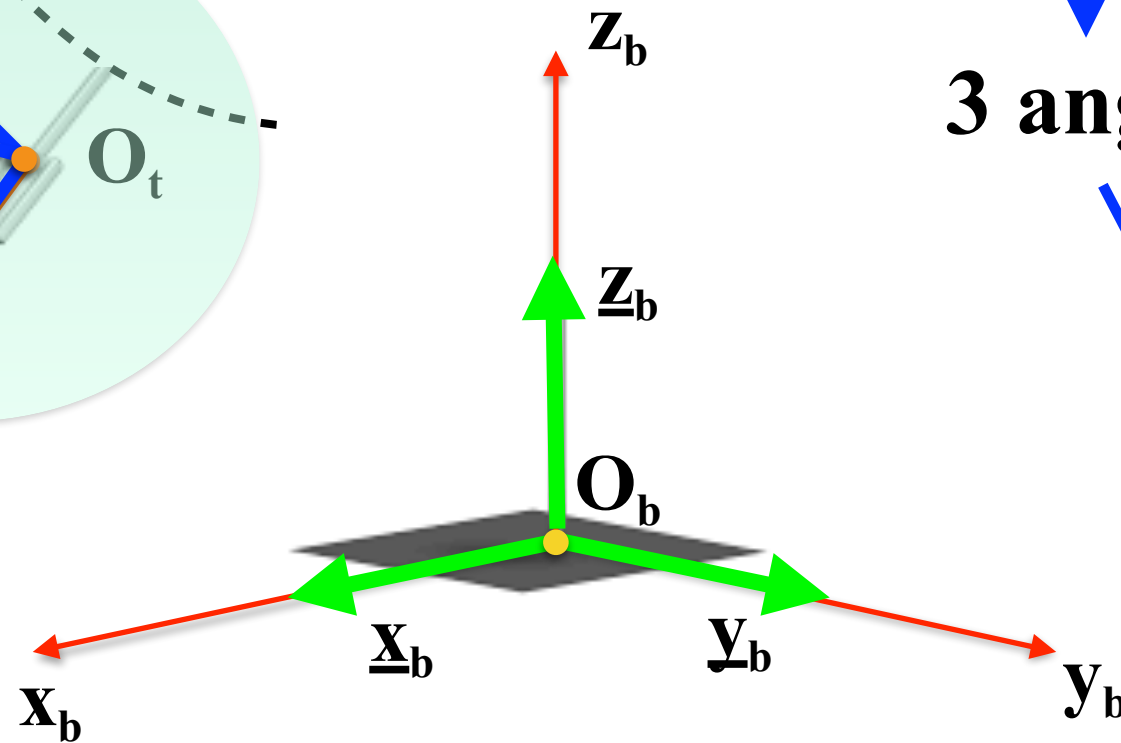
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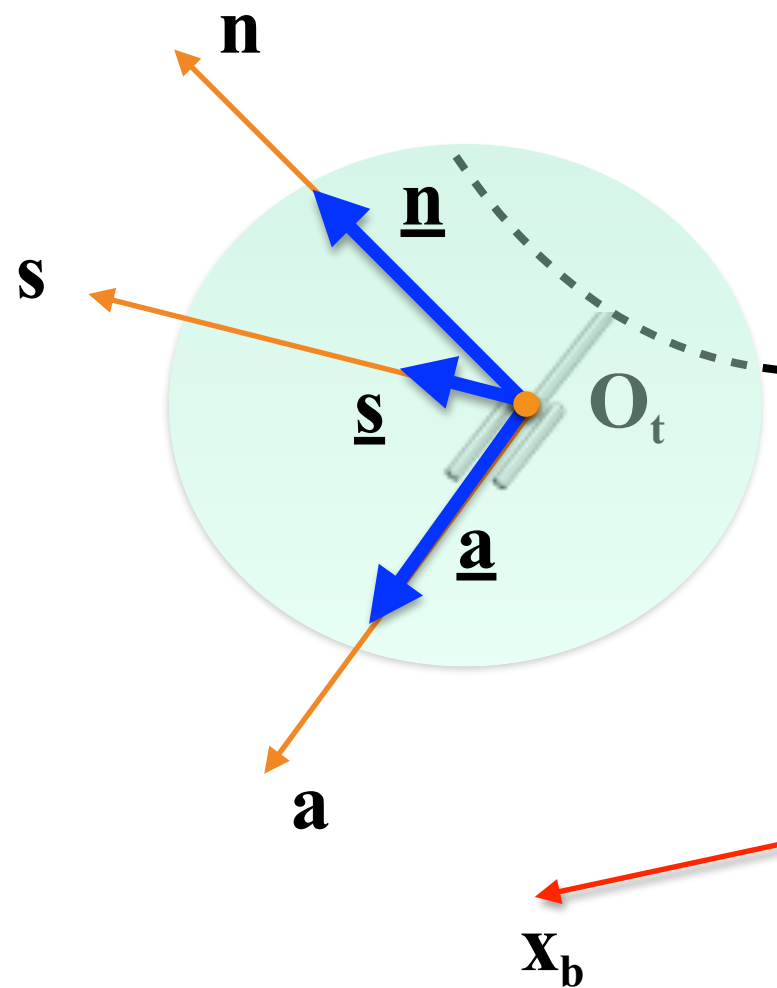
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Euler's Angles



Minimal Description of the Orientation of a Rigid Object in 3D Space

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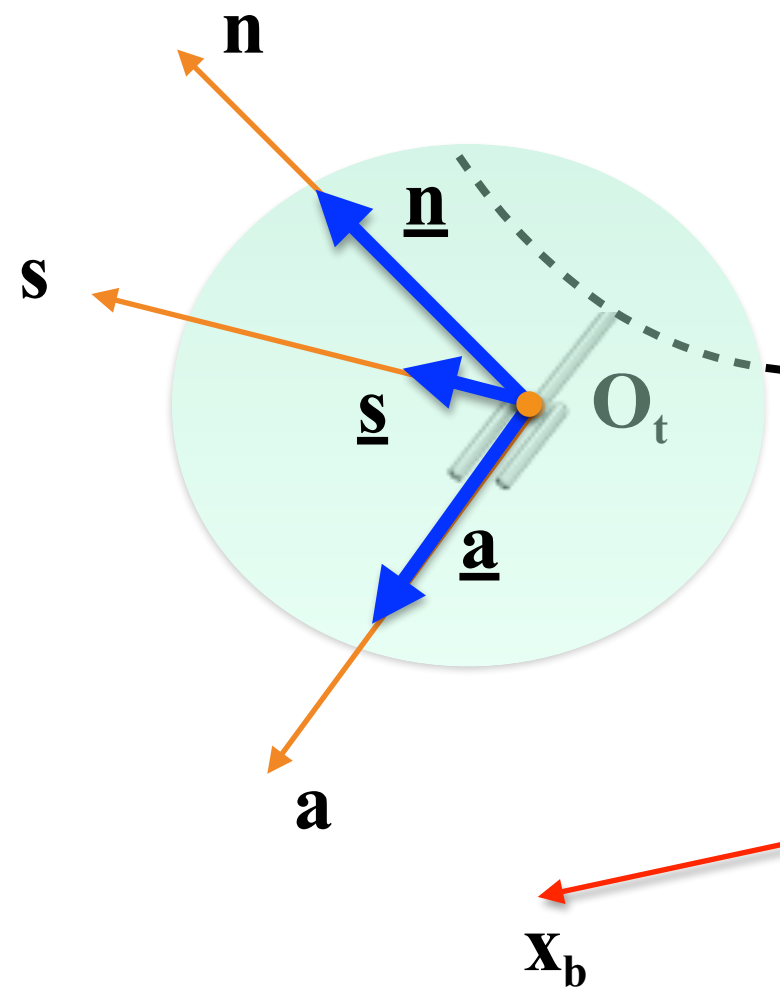
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Euler's Angles

they are not unique

Minimal Description of the Orientation of a Rigid Object in 3D Space

Only 3 independent quantities are needed



3 angles $\phi = [\varphi \ \vartheta \ \psi]^T$

Euler's Angles

12 possible sets of Euler's angles to describe a rigid object orientation

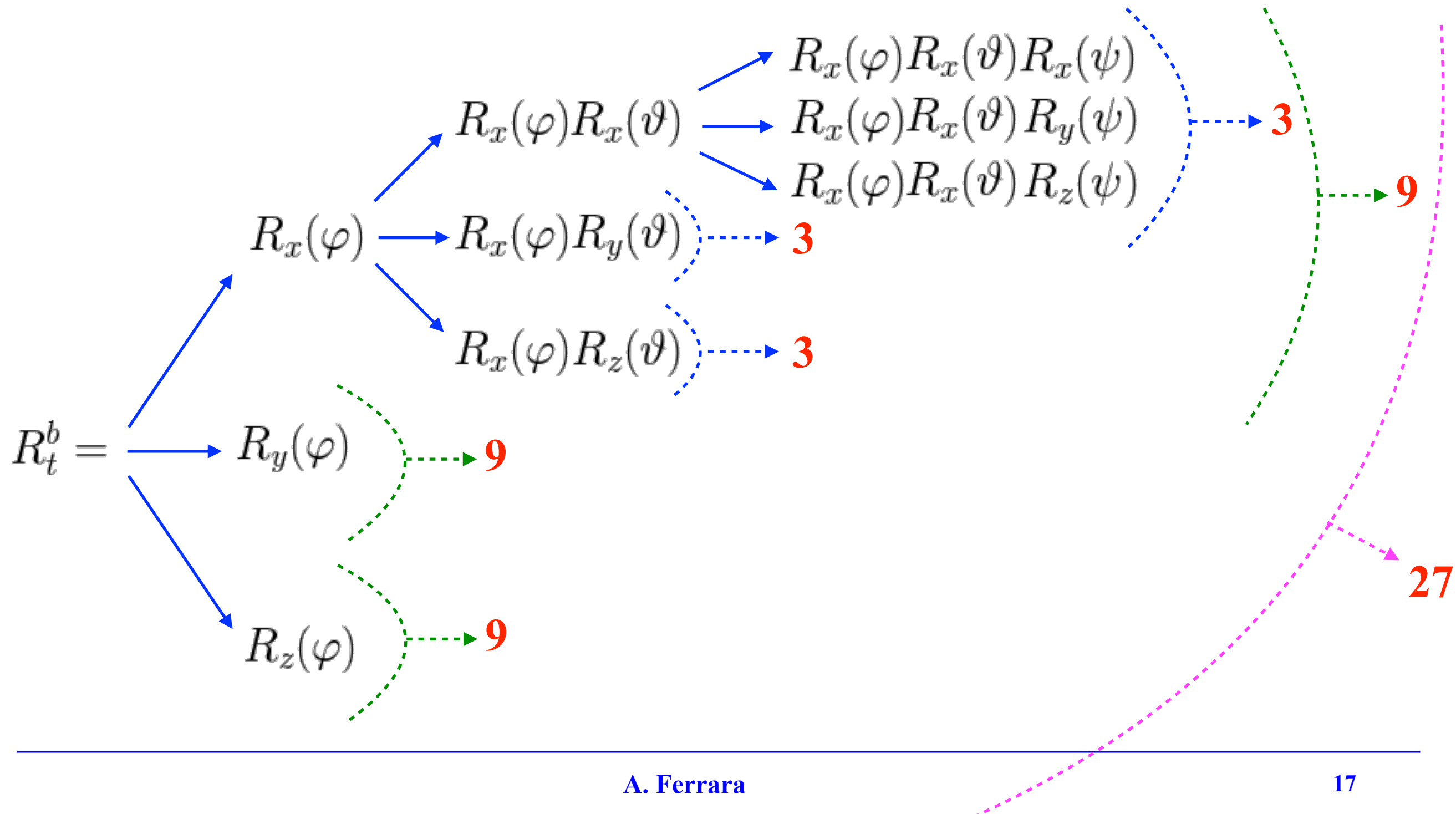
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Minimal Description of Orientation of a Rigid Object in 3D Space: Euler's Angles

The idea: the description of a rigid object orientation via a rotation matrix can be obtained by composing 3 elementary rotation matrices

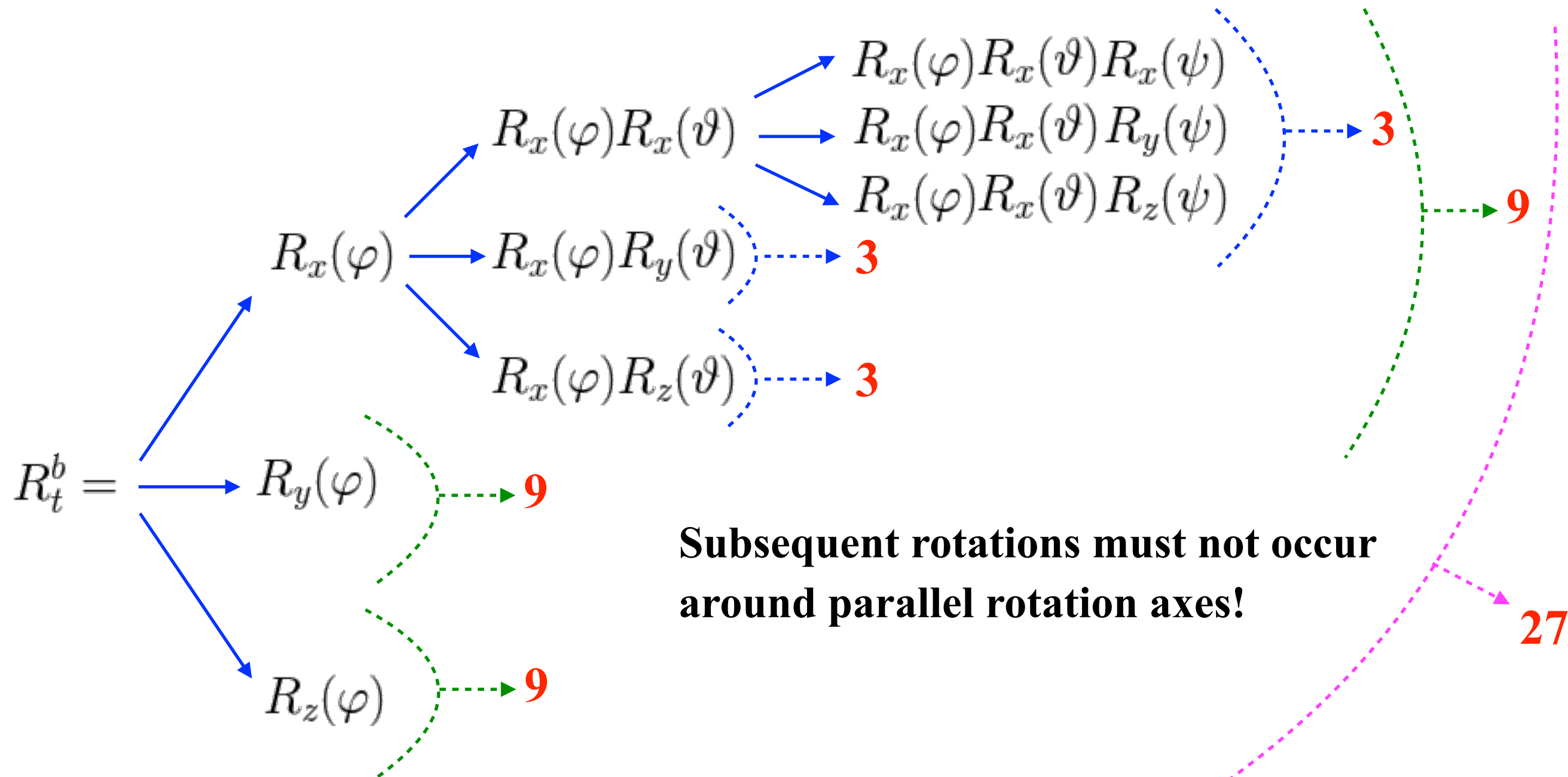
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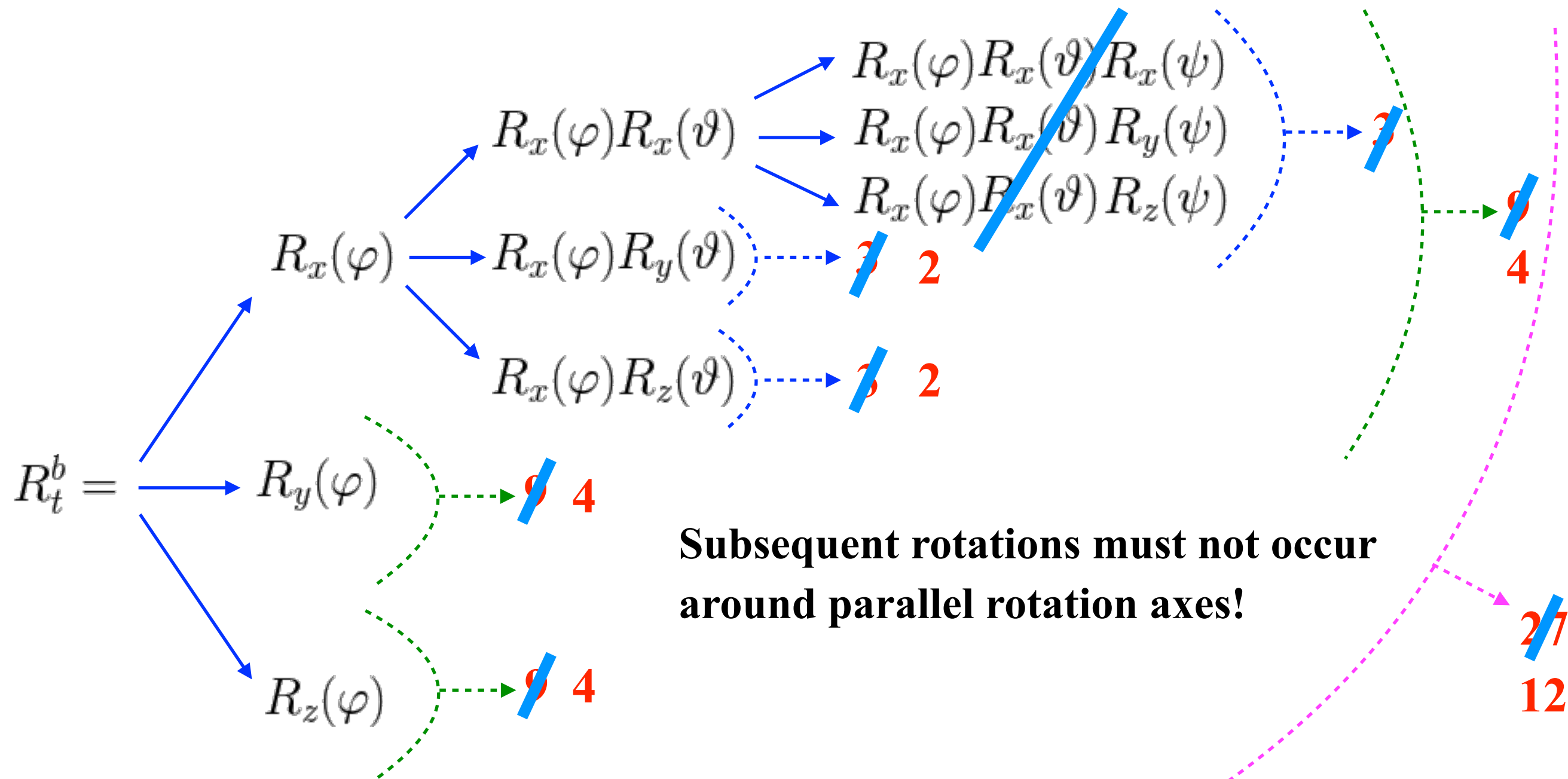
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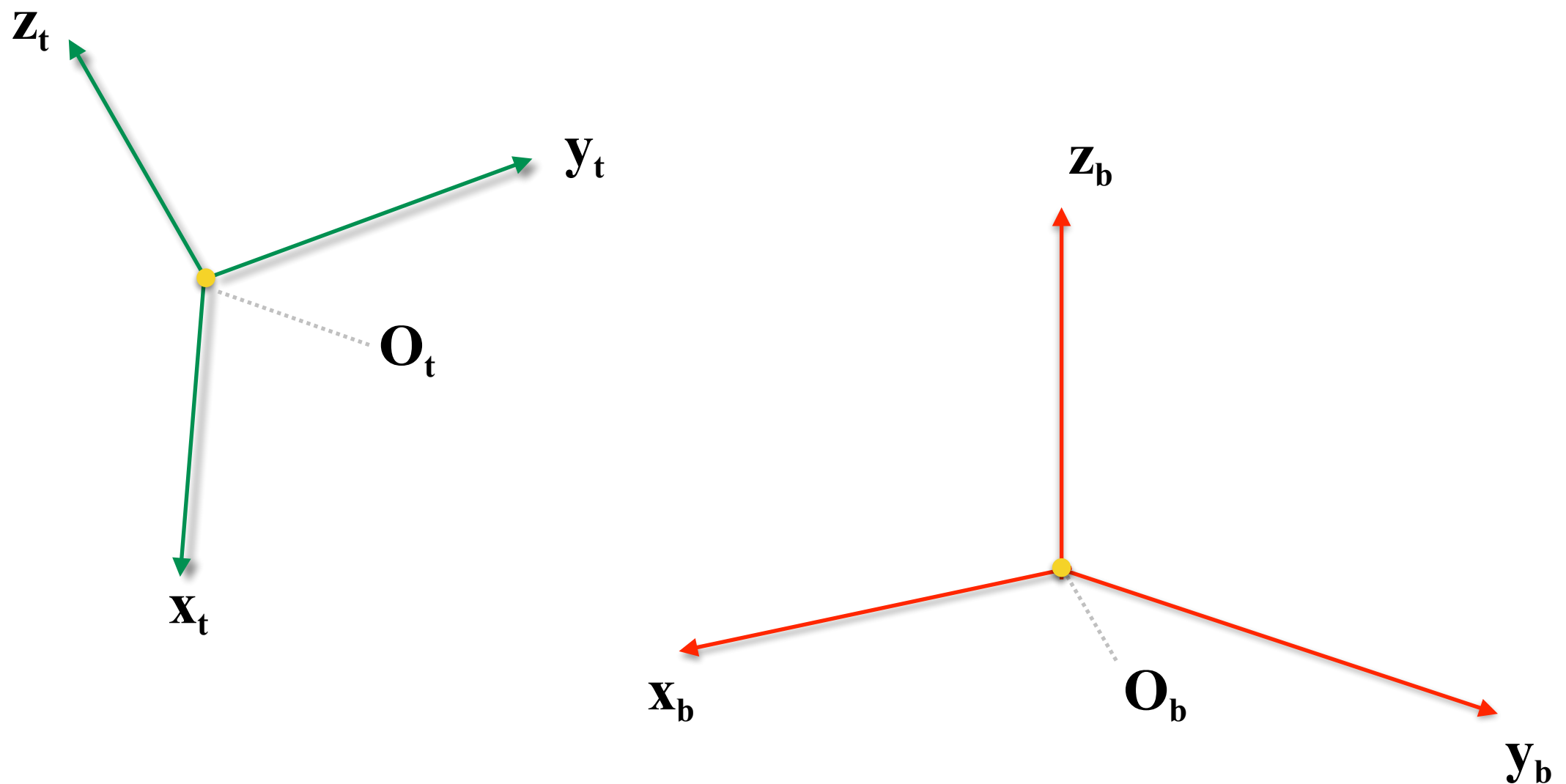
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Minimal Description of Orientation: Euler Angles Definition

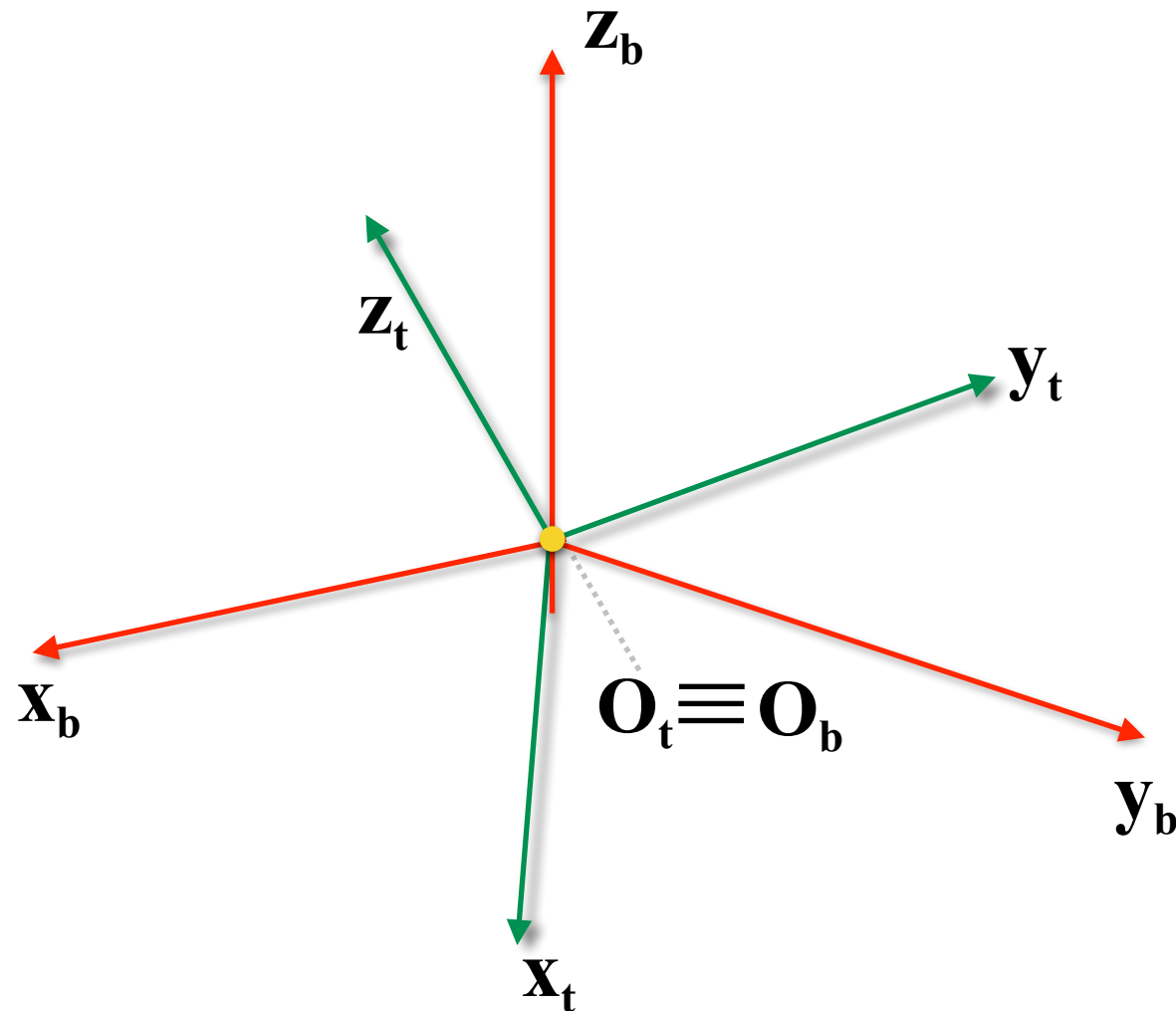
Minimal Description of Orientation of a Rigid Object in 3D Space: Euler's Angles

Example: Describe the orientation of the Tool Reference Frame w.r.t. the Base Reference Frame using the ZYZ Euler's Angles Set $\phi_{ZYZ} = [\varphi \ \vartheta \ \psi]^T$



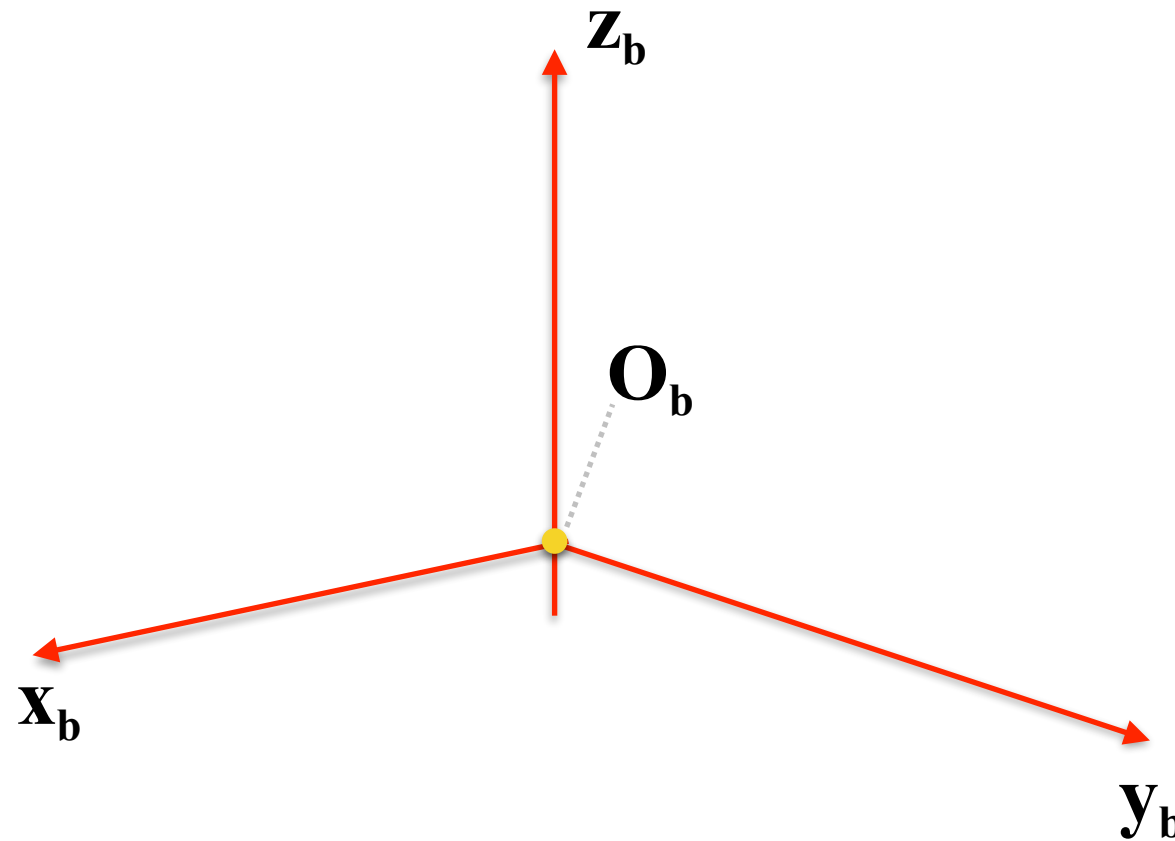
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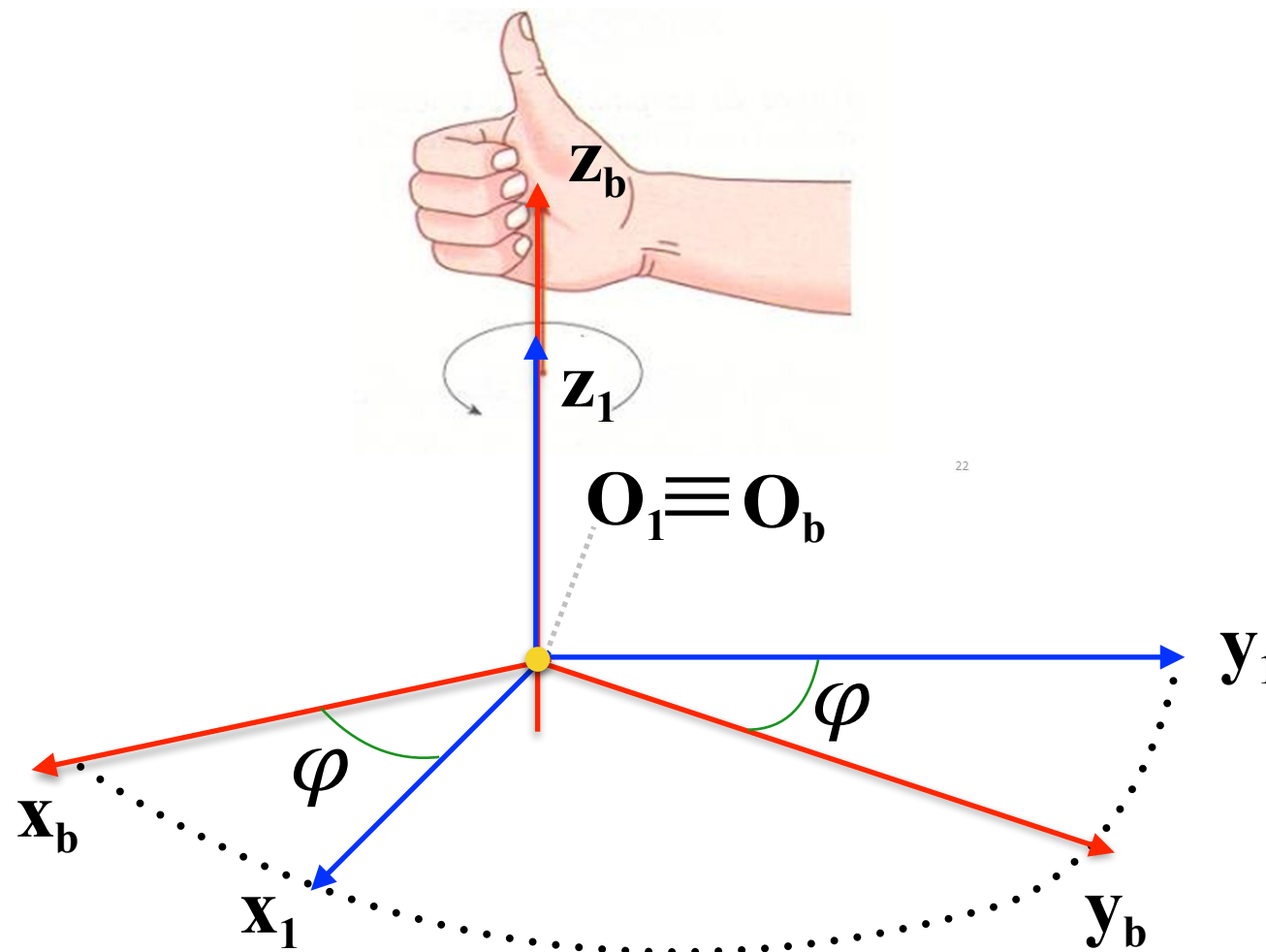
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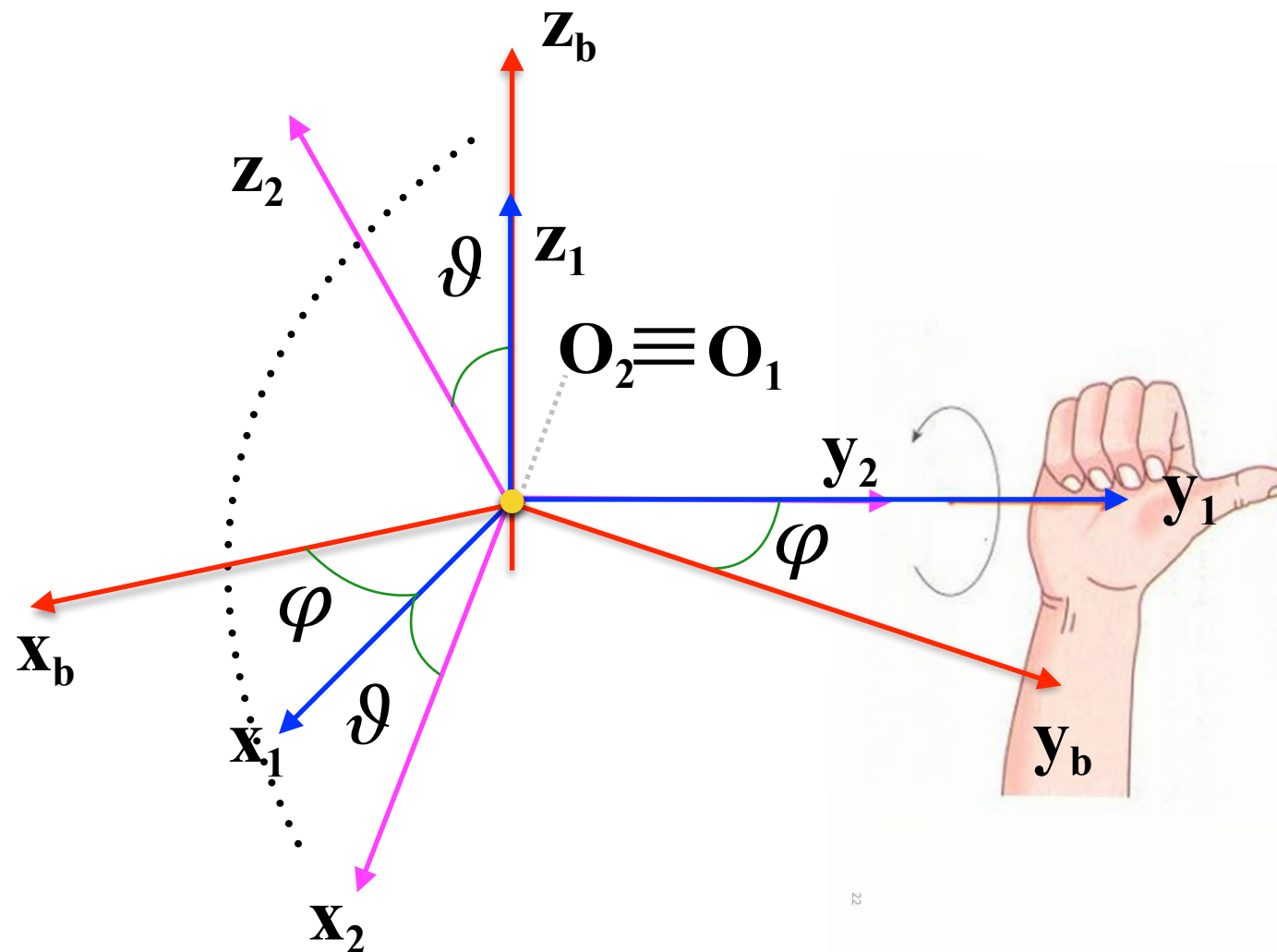
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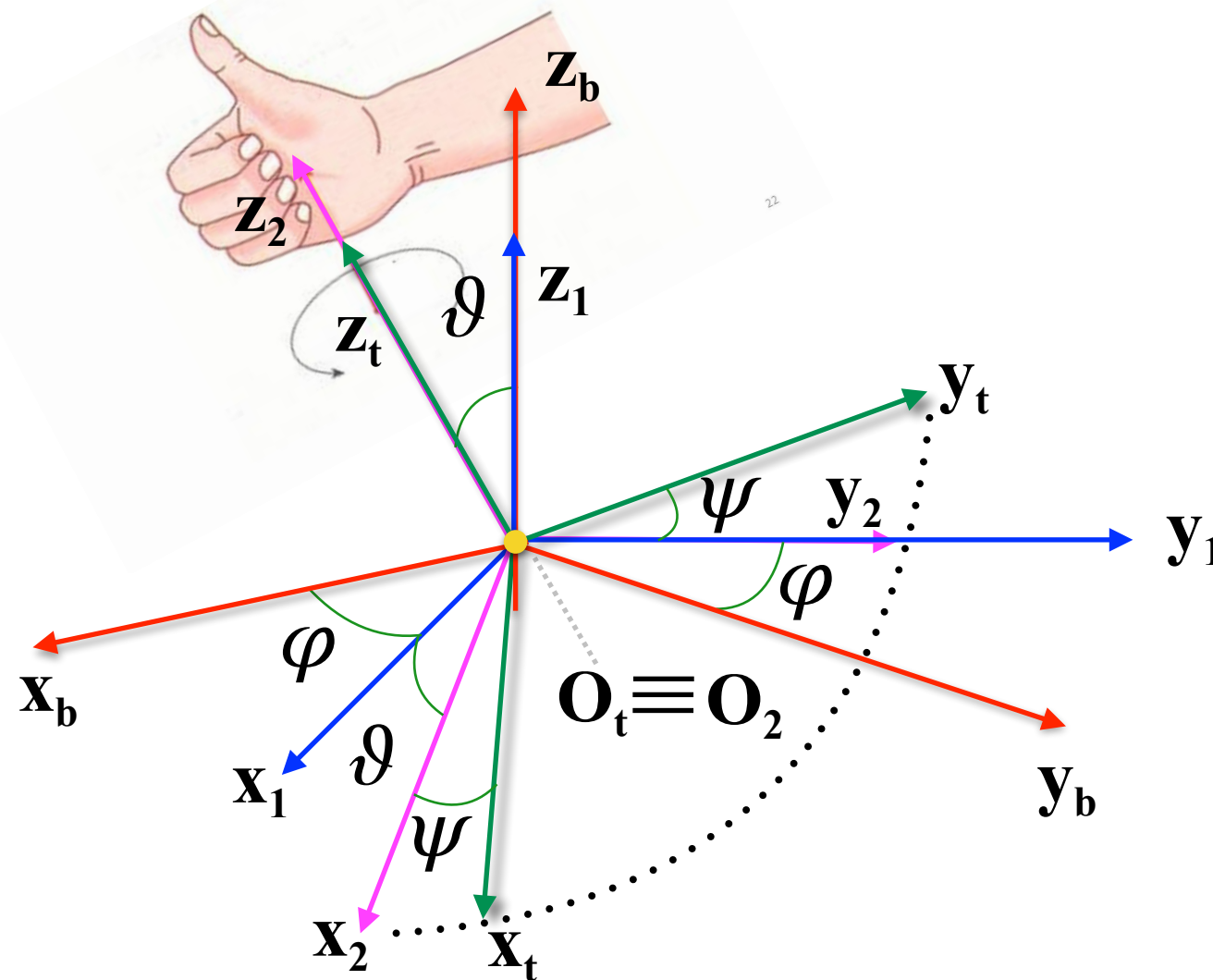
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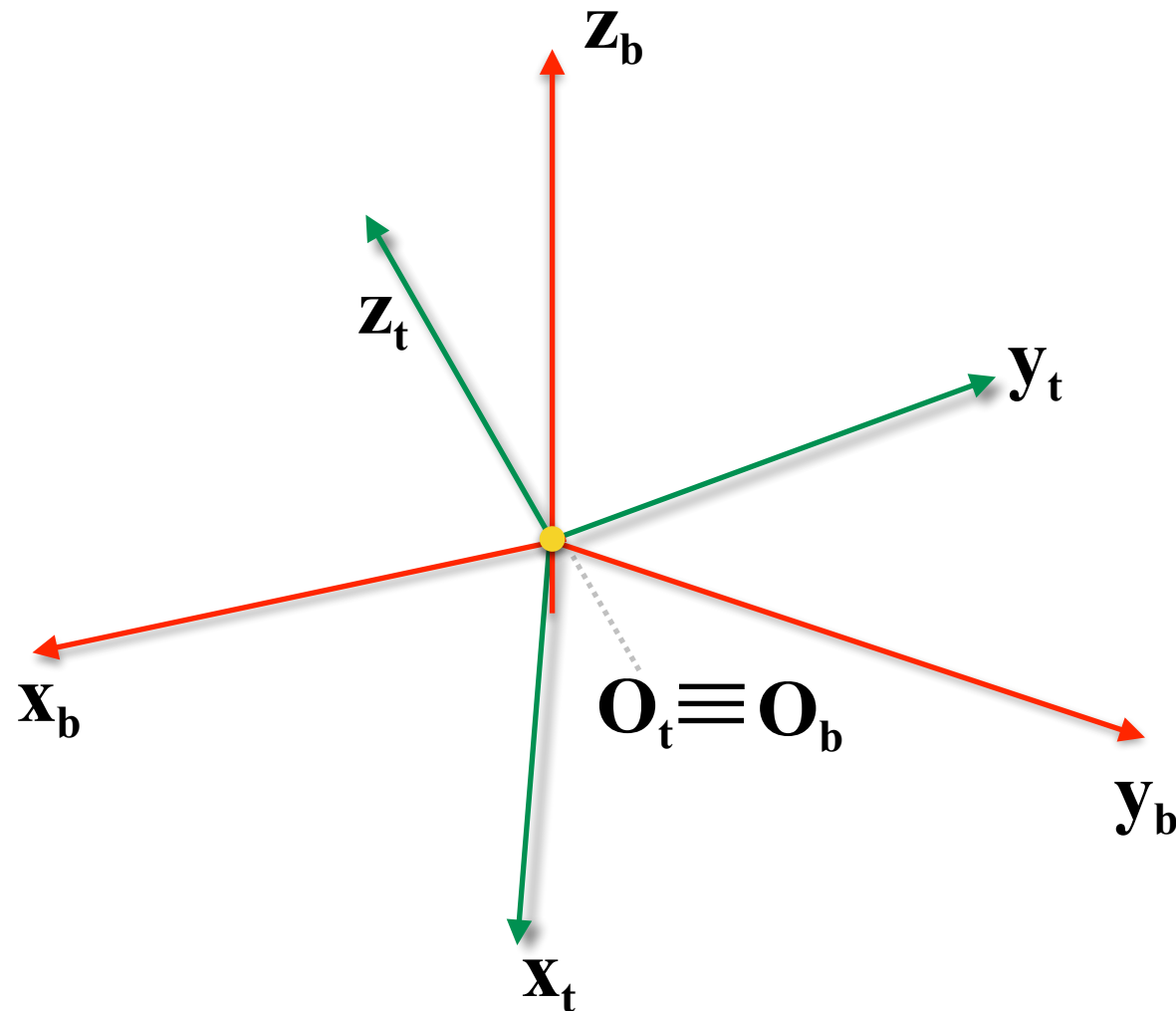
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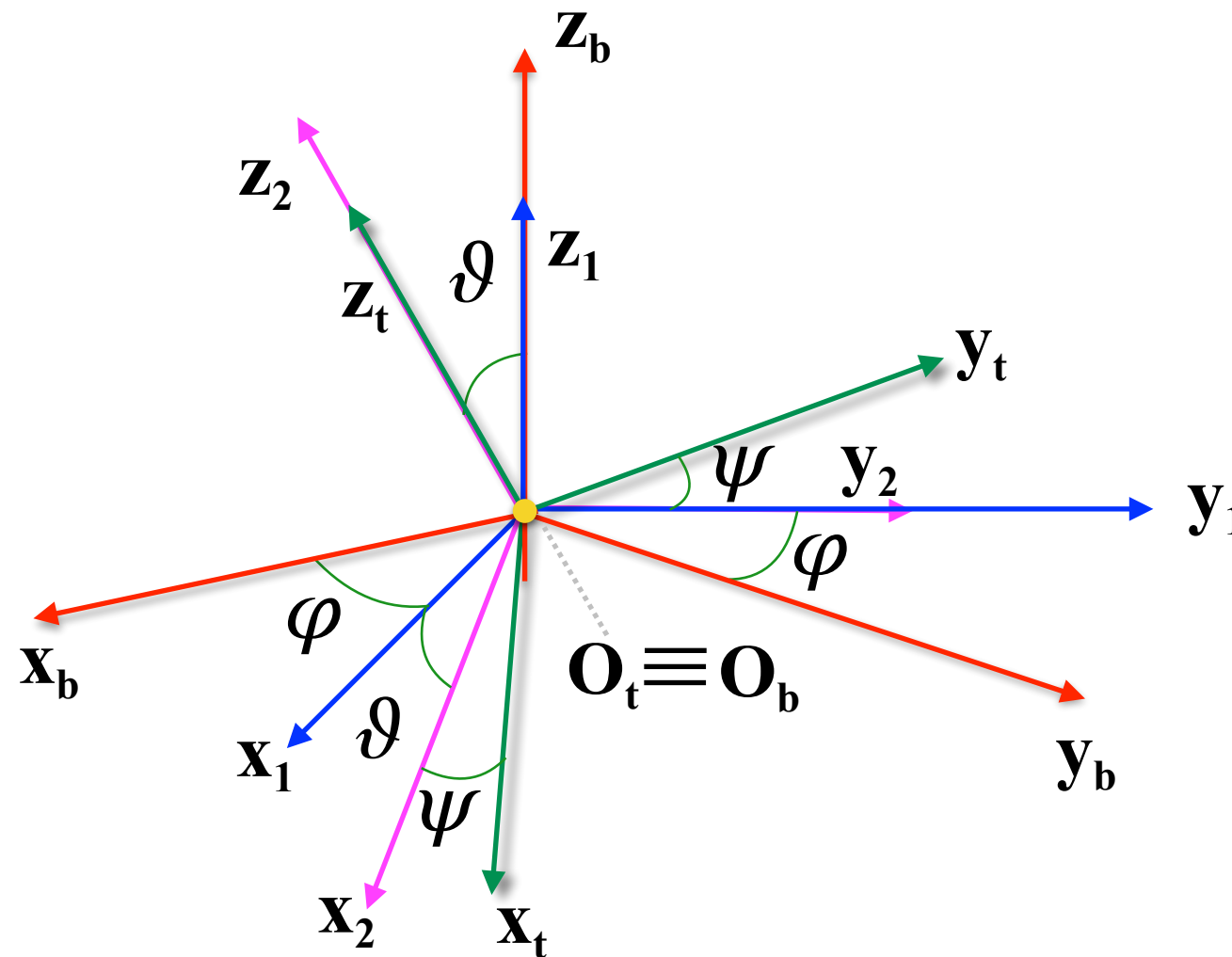
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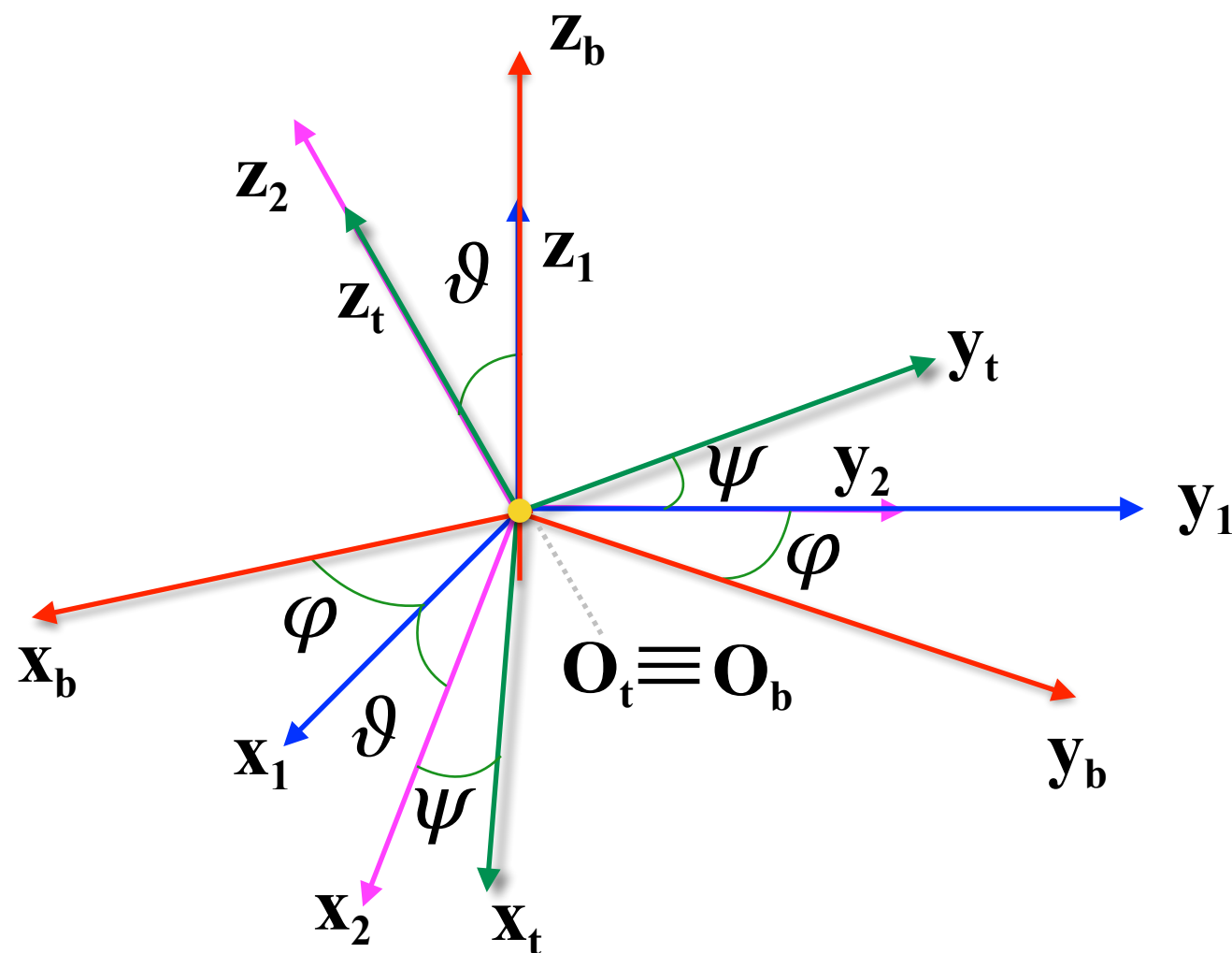
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Minimal Description of Orientation of a Rigid Object in 3D Space: Euler's Angles

ZYZ Euler's Angles

(all rotations w.r.t. the “current reference frame”)



$$\phi_{ZYZ} = [\varphi \ \vartheta \ \psi]^T$$

ZYX Euler's Angles (all rotations w.r.t. a “fixed reference frame” attached to the object center of mass)



Roll, Pitch and Yaw

$$\phi_{ZYX} = [\varphi \ \vartheta \ \psi]^T$$



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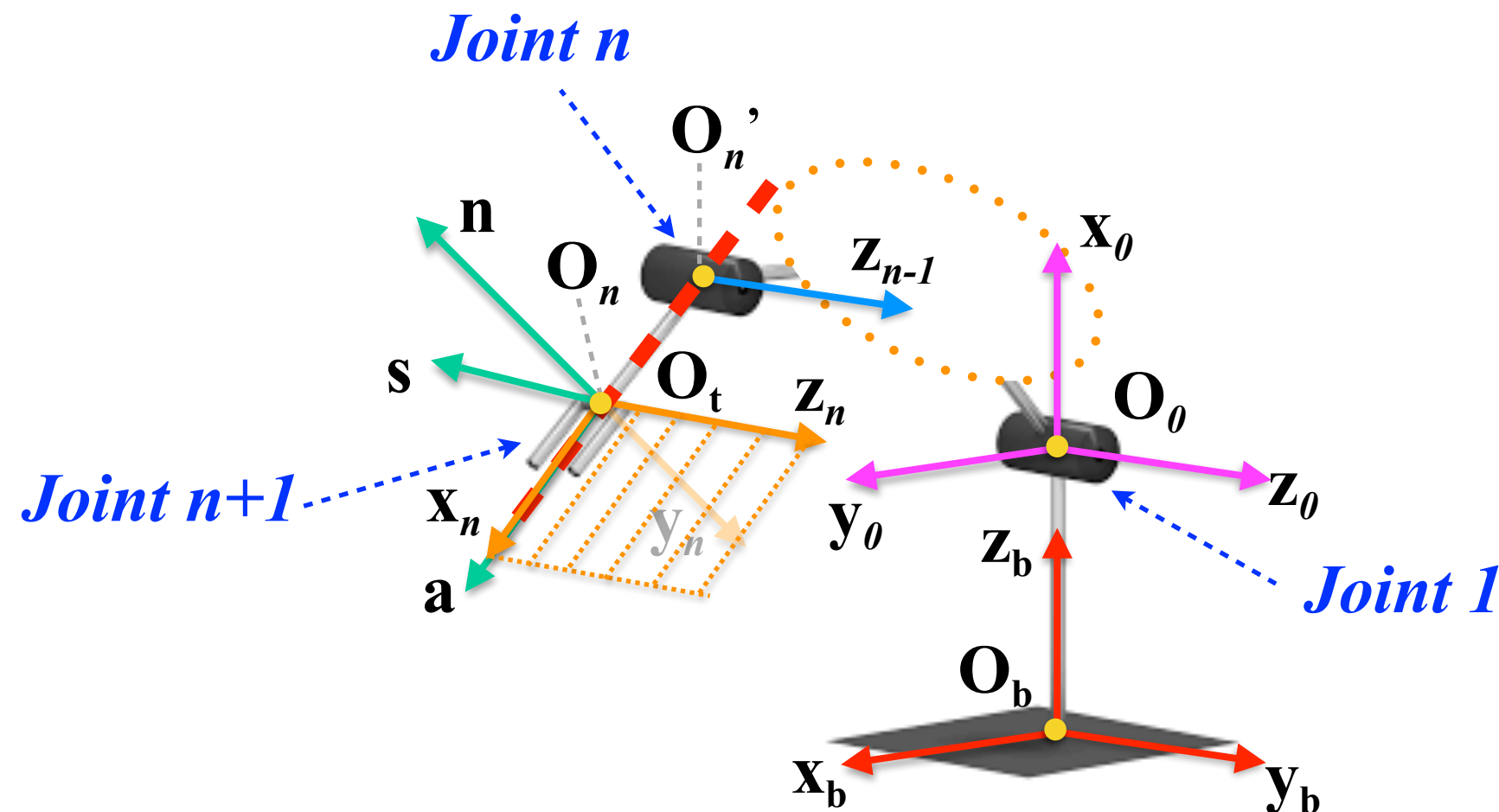
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A new pose description

In a previous lecture....

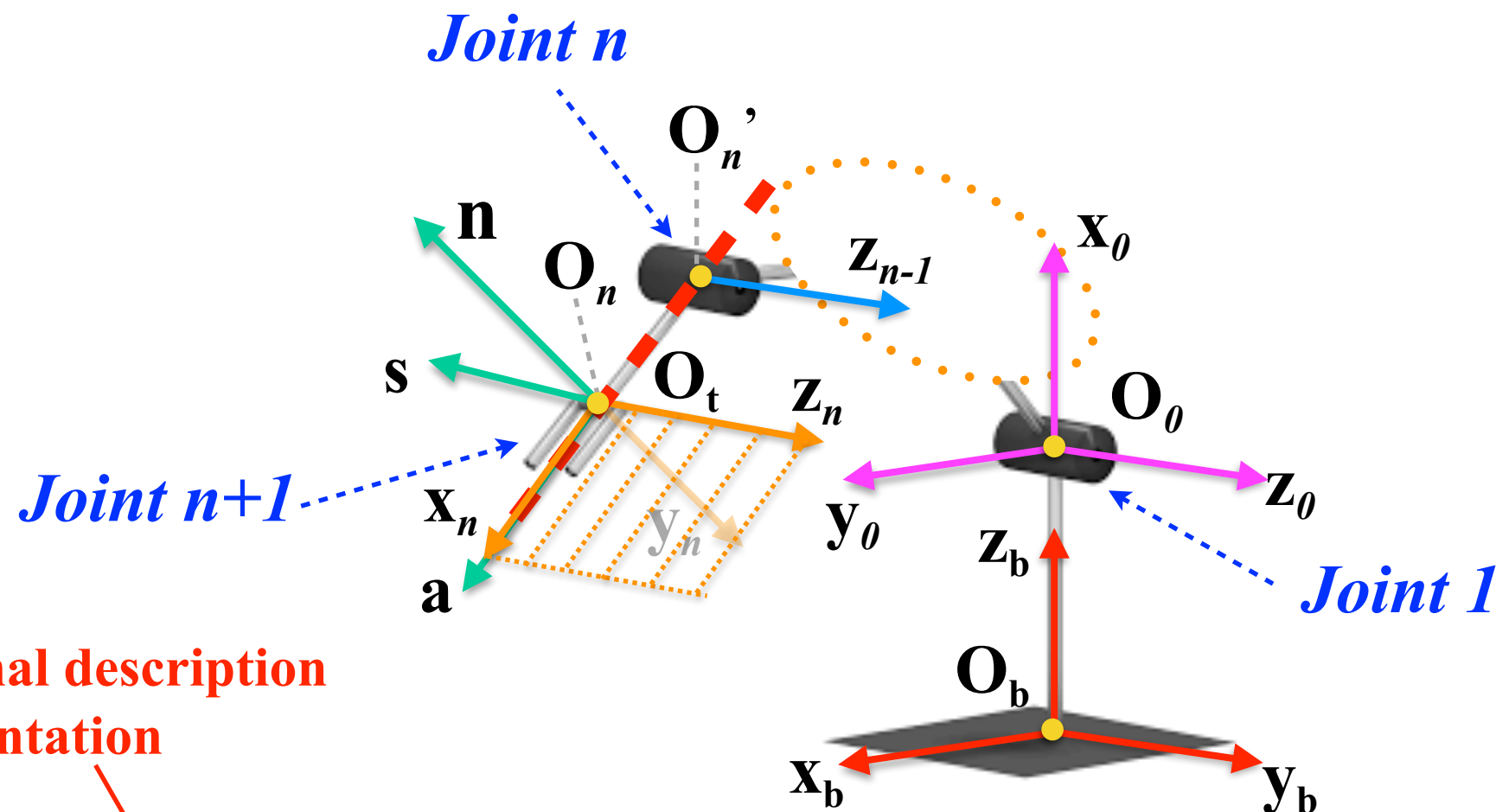
Description of the Direct Kinematics of Robot Manipulators via the D-H Approach



$$T_t^b(\underline{q}) = A_0^b \cdot \begin{bmatrix} c_{\theta_1} & -s_{\theta_1} c_{\alpha_1} & s_{\theta_1} s_{\alpha_1} & a_1 c_{\theta_1} \\ s_{\theta_1} & c_{\theta_1} c_{\alpha_1} & -c_{\theta_1} s_{\alpha_1} & a_1 s_{\theta_1} \\ 0 & s_{\alpha_1} & c_{\alpha_1} & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} c_{\alpha_i} & s_{\theta_i} s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i} c_{\alpha_i} & -c_{\theta_i} s_{\alpha_i} & a_i s_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} c_{\theta_n} & -s_{\theta_n} c_{\alpha_n} & s_{\theta_n} s_{\alpha_n} & a_n c_{\theta_n} \\ s_{\theta_n} & c_{\theta_n} c_{\alpha_n} & -c_{\theta_n} s_{\alpha_n} & a_n s_{\theta_n} \\ 0 & s_{\alpha_n} & c_{\alpha_n} & d_n \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot A_t^n$$

In a previous lecture....

Description of the Direct Kinematics of Robot Manipulators via the D-H Approach

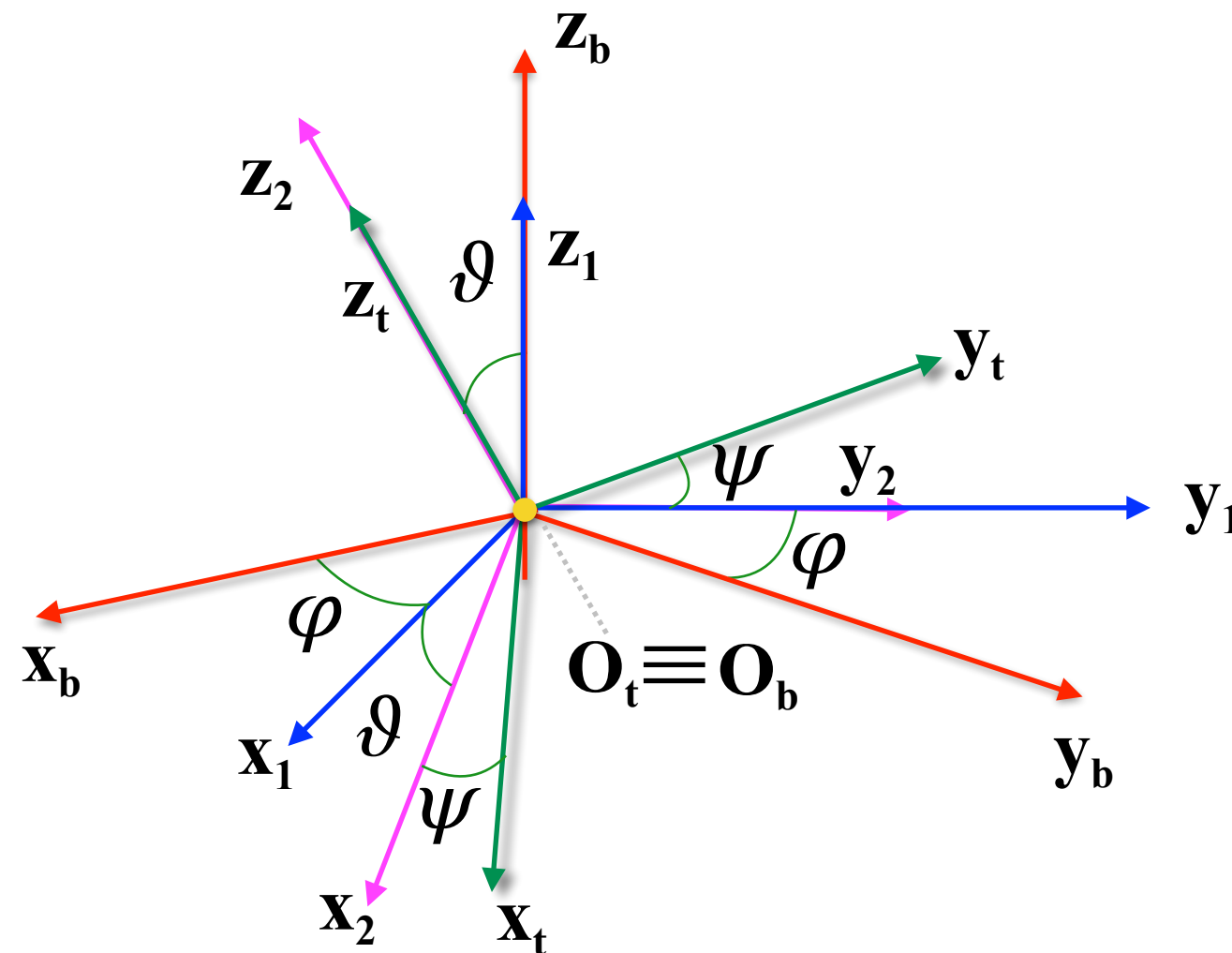


Nonminimal description of the orientation

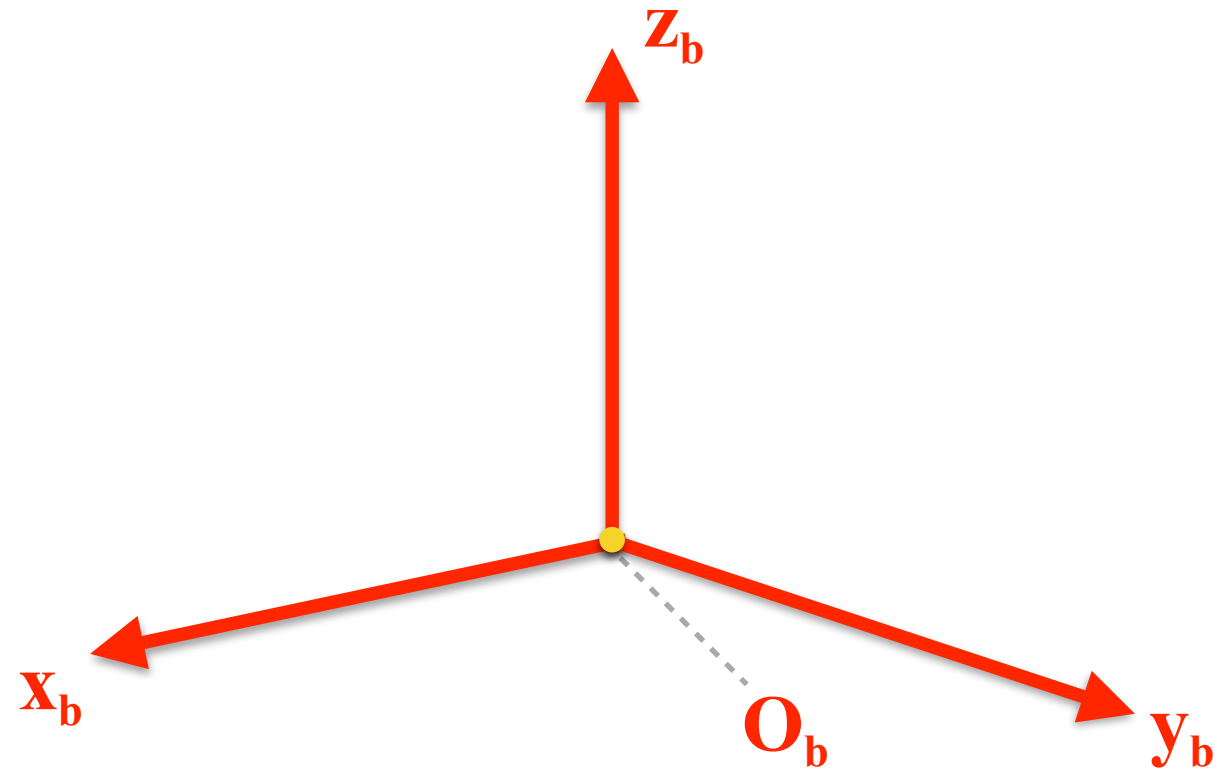
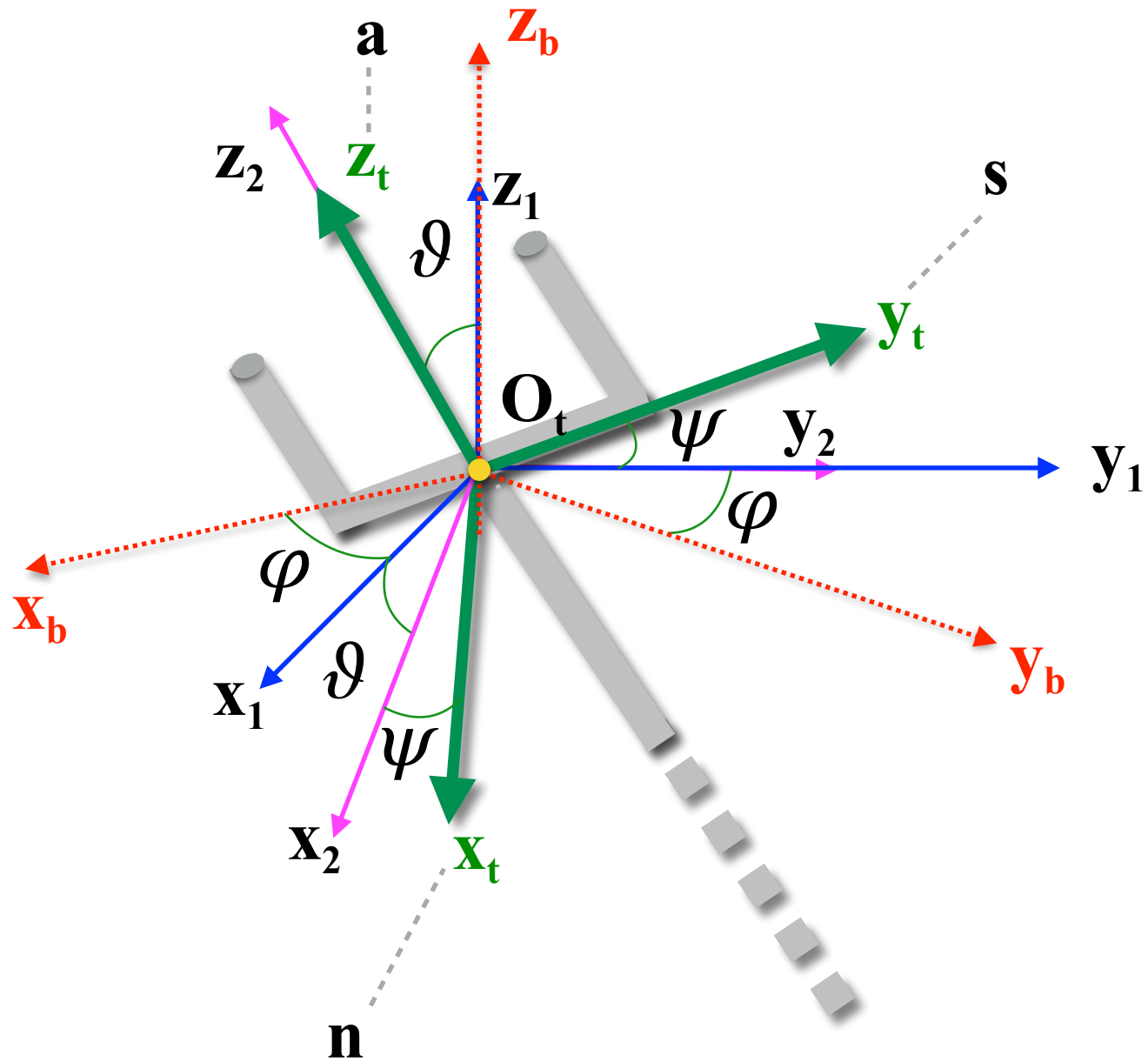
$$T_t^b(\underline{q}) = A_0^b \cdot \begin{bmatrix} c\theta_1 & -s\theta_1 c\alpha_1 & s\theta_1 s\alpha_1 & a_1 c\theta_1 \\ s\theta_1 & c\theta_1 c\alpha_1 & -c\theta_1 s\alpha_1 & a_1 s\theta_1 \\ 0 & s\alpha_1 & c\alpha_1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} c\theta_n & -s\theta_n c\alpha_n & s\theta_n s\alpha_n & a_n c\theta_n \\ s\theta_n & c\theta_n c\alpha_n & -c\theta_n s\alpha_n & a_n s\theta_n \\ 0 & s\alpha_n & c\alpha_n & d_n \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot A_t^n$$

Minimal Description of Orientation of a Rigid Object in 3D Space: Euler's Angles

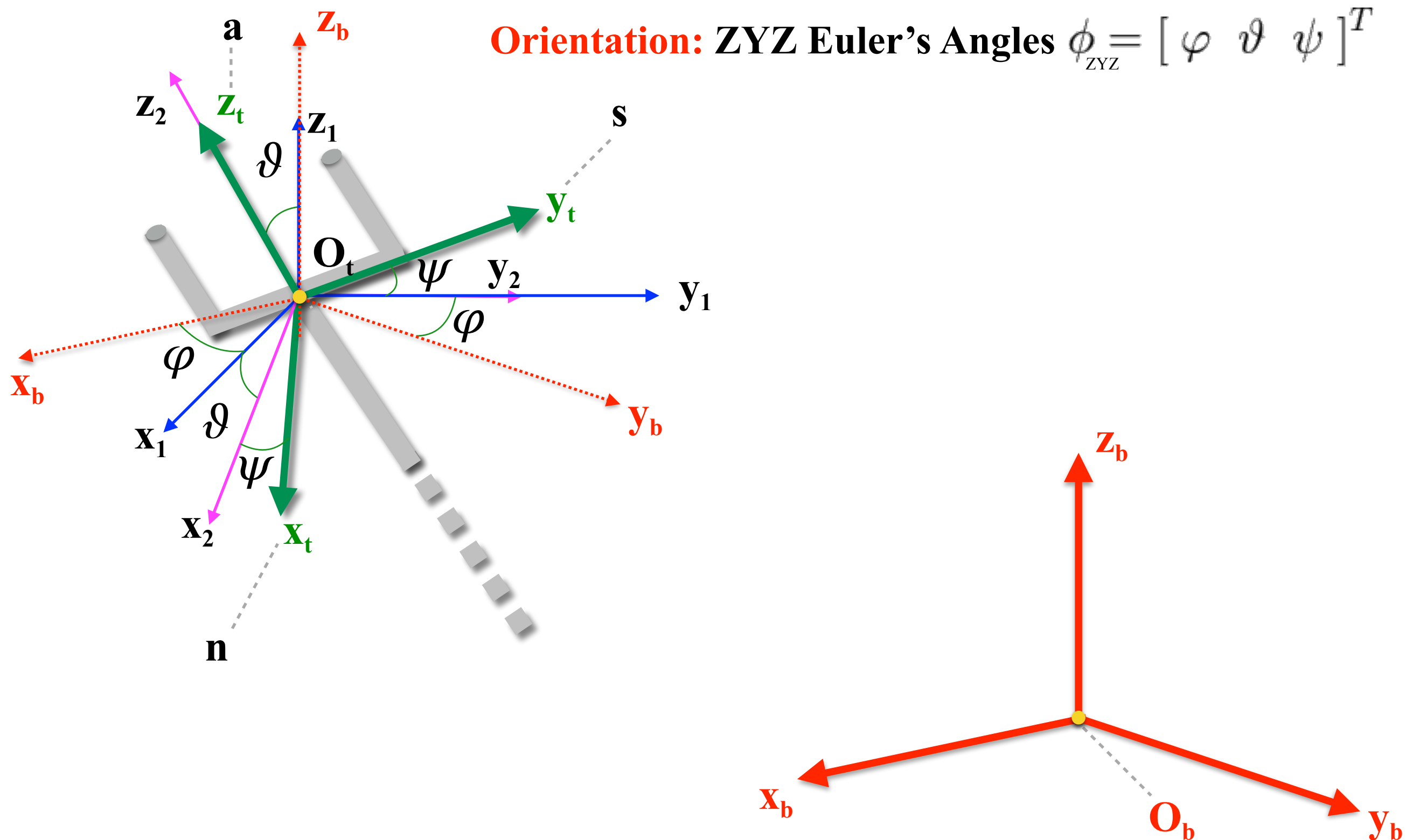
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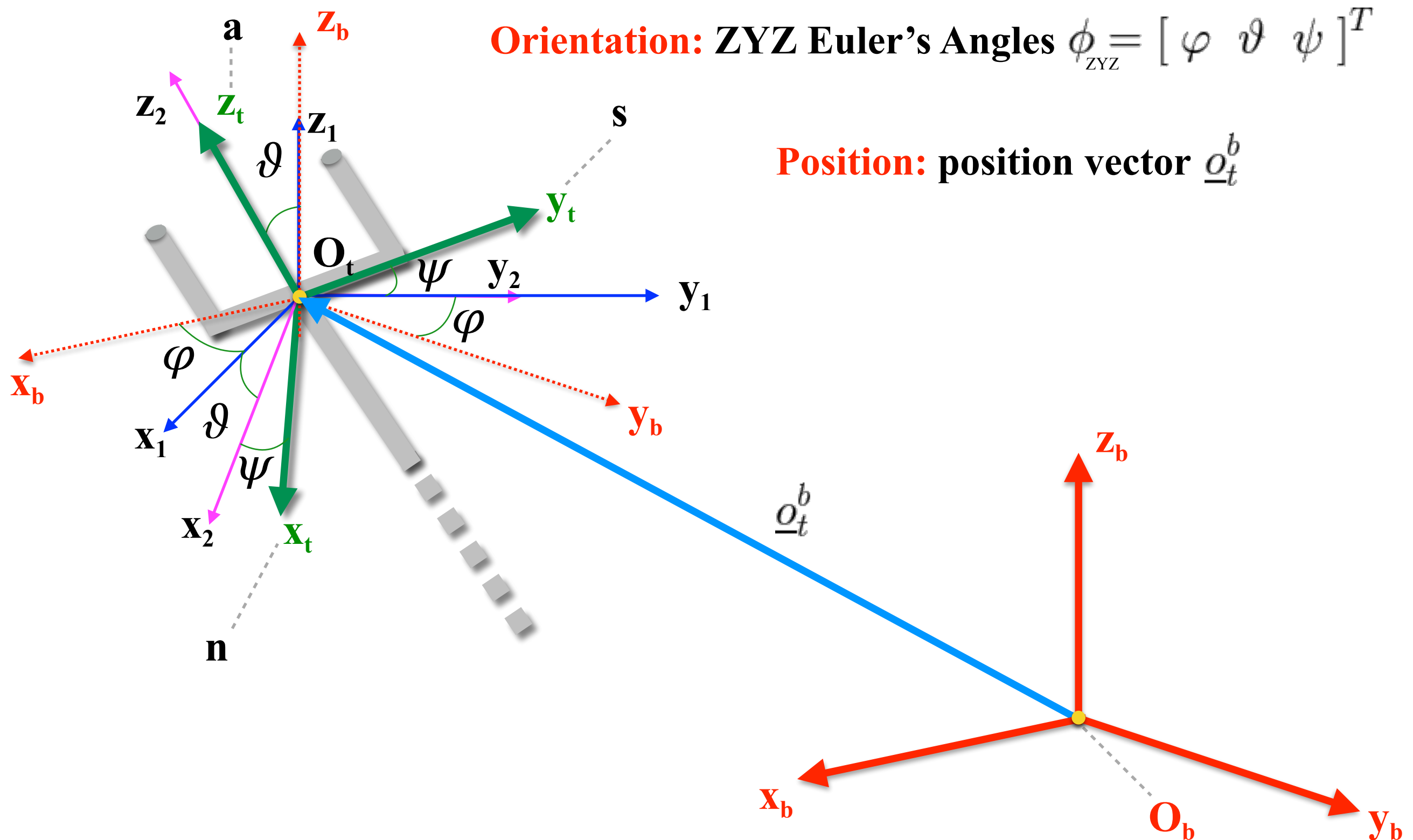
Euler's Angles and a New Pose Description



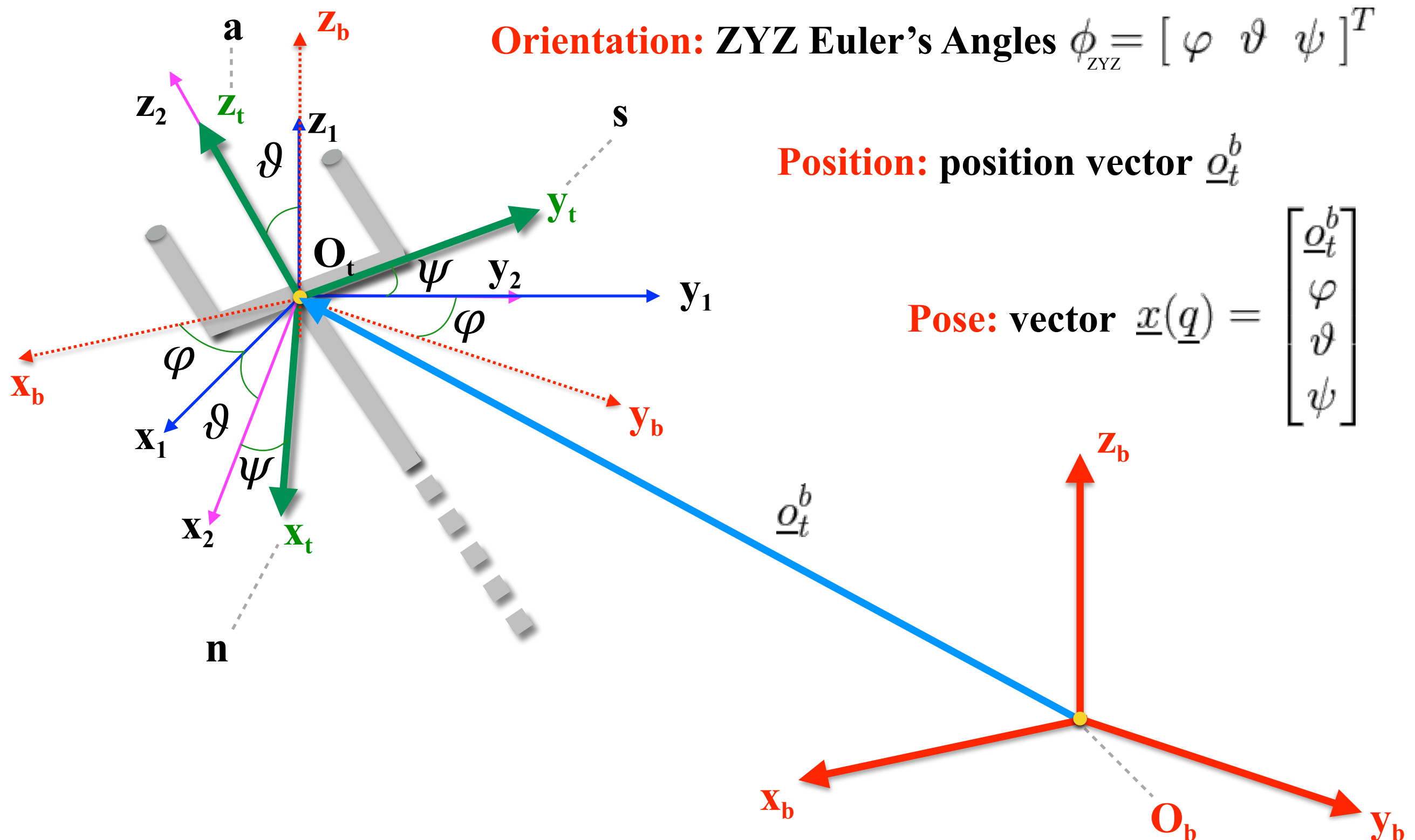
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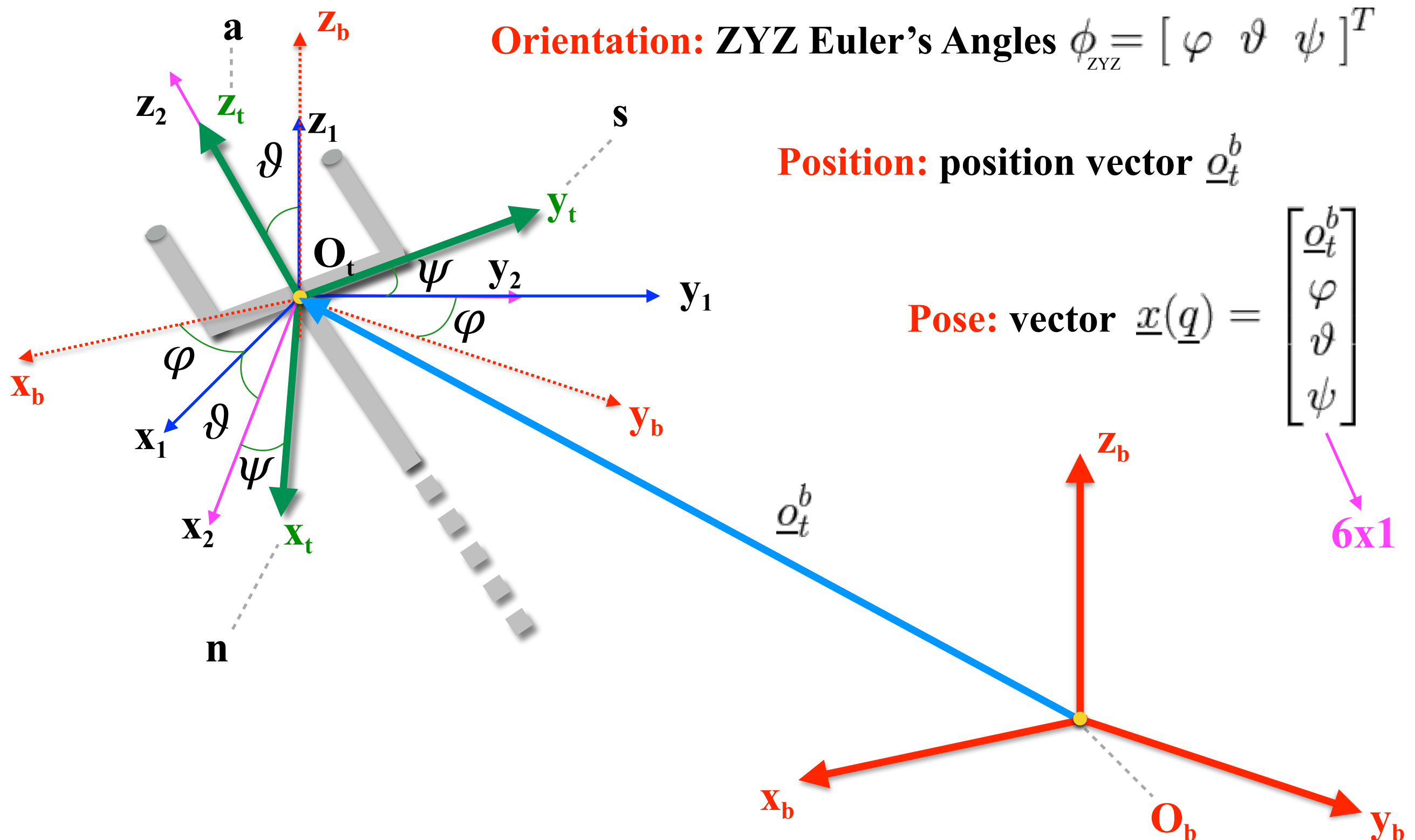
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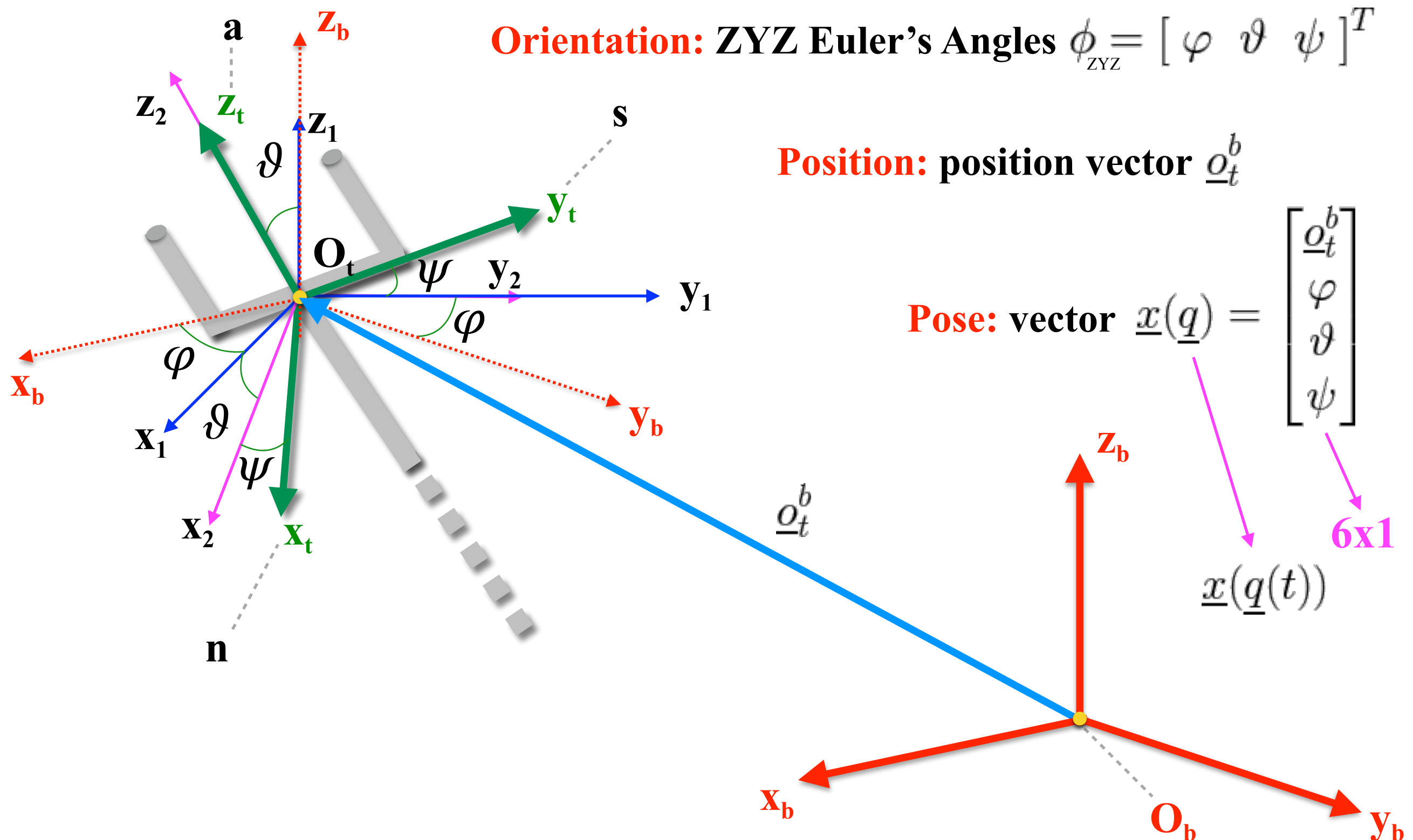
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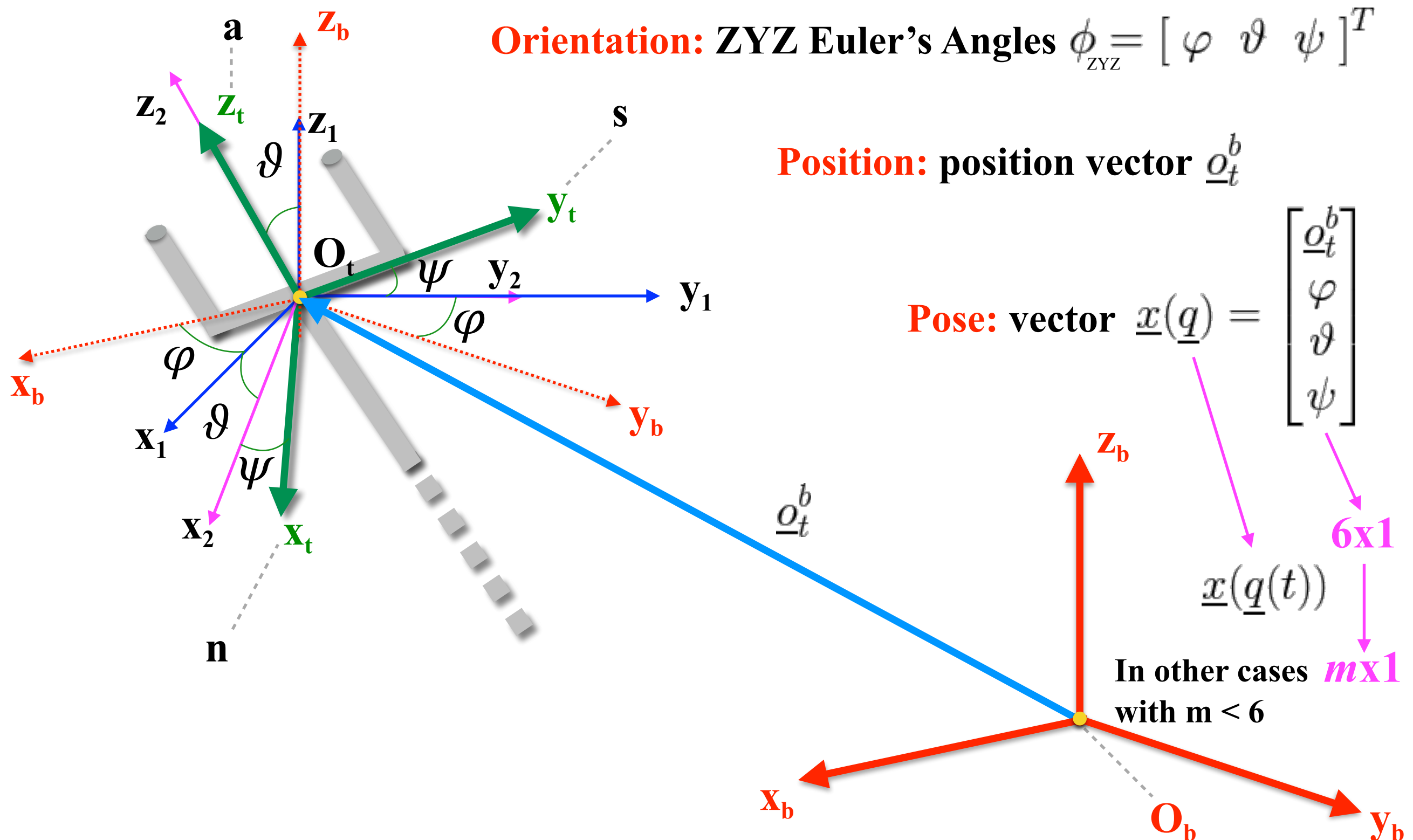
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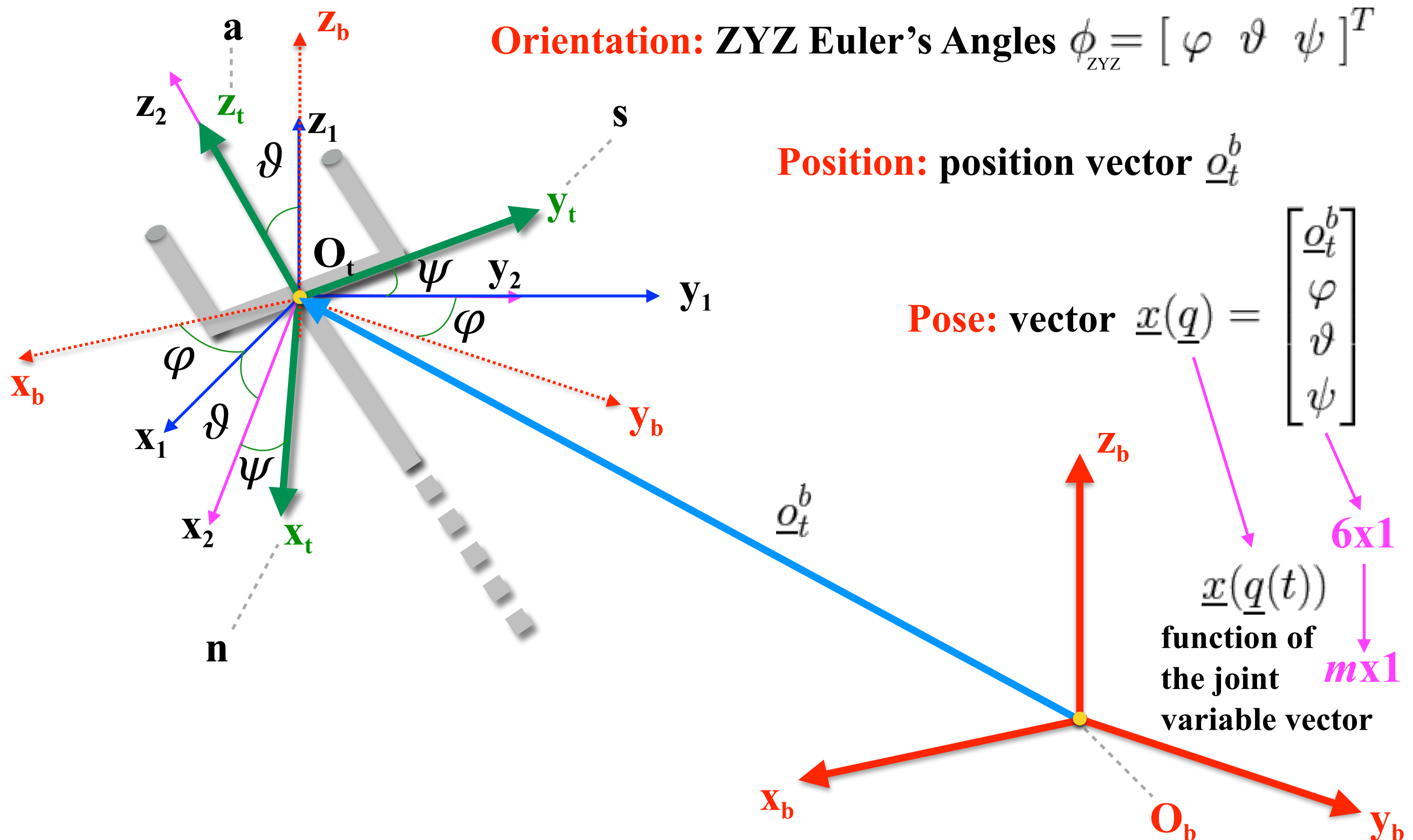
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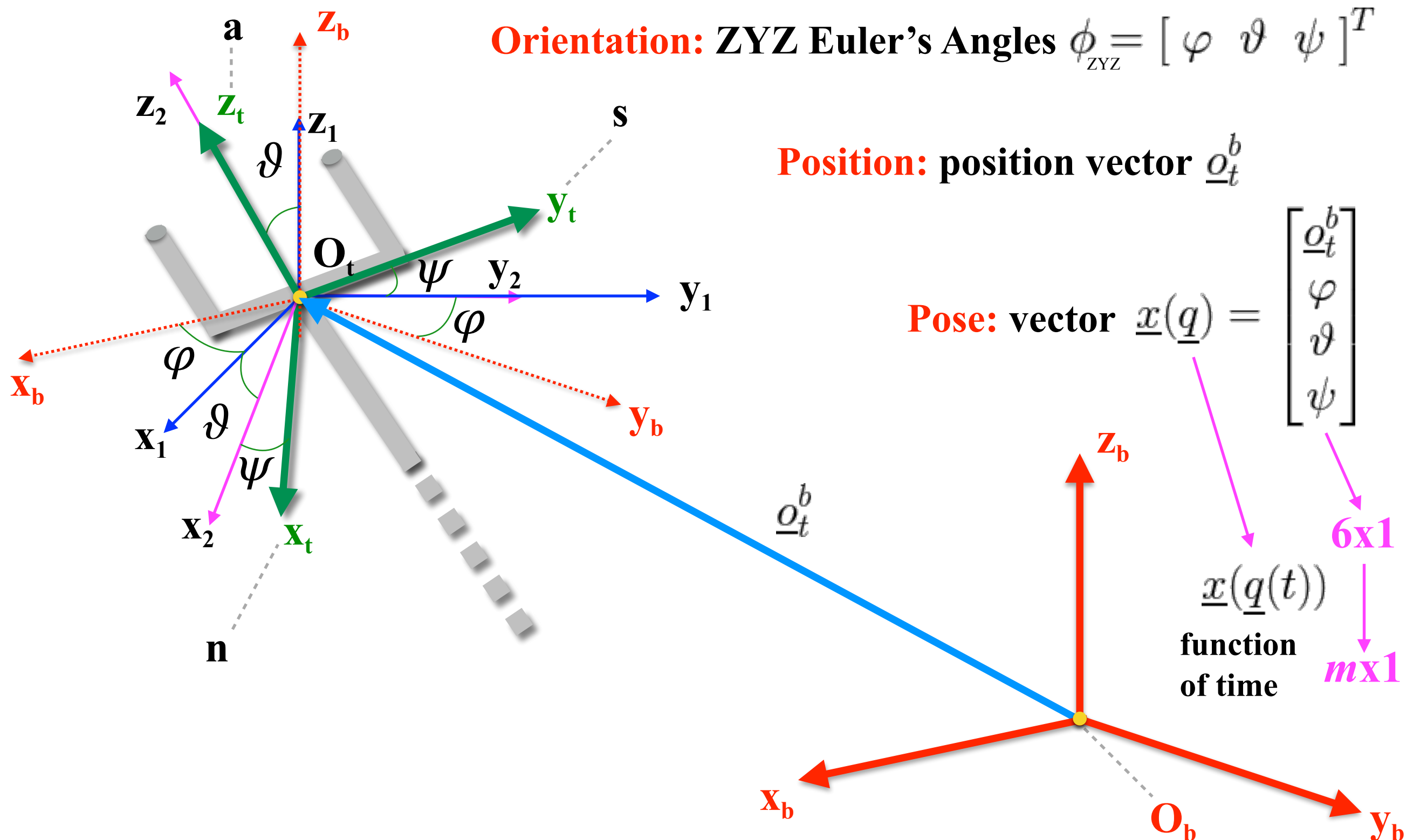
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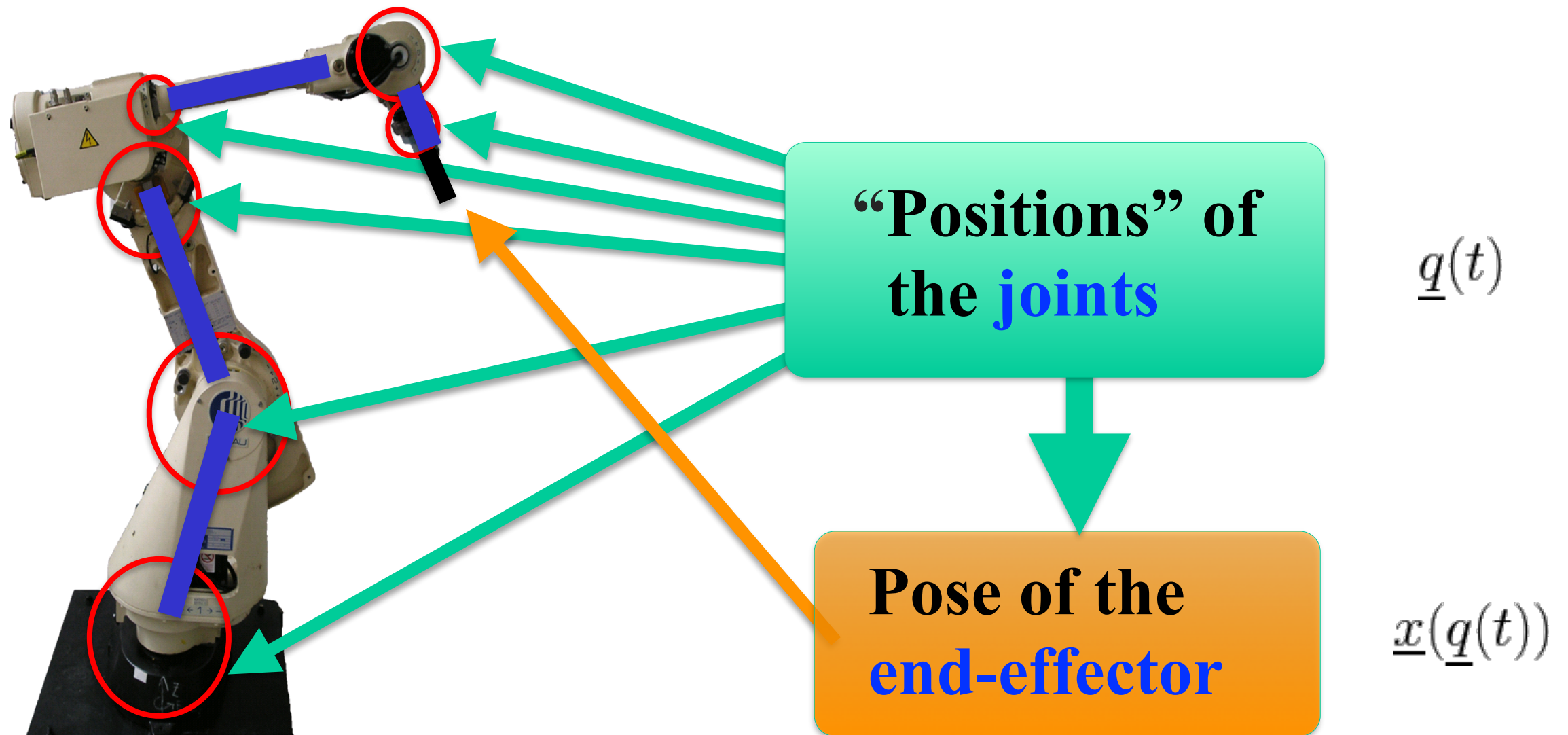
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Analytic Jacobian Matrix

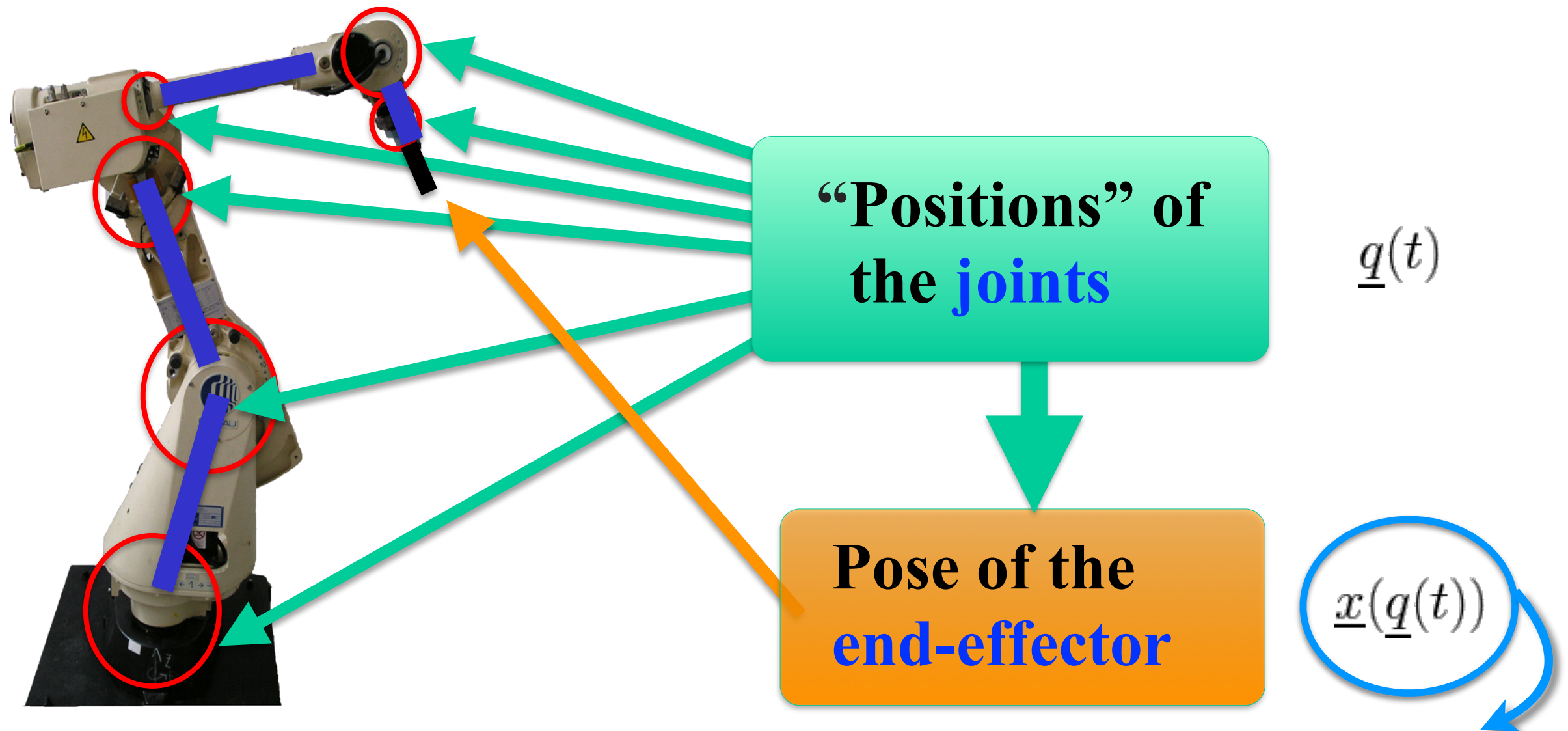
A Useful Definition: Analytic Jacobian Matrix

Pose vector $\underline{x}(\underline{q}) = \underline{x}(\underline{q}(t))$ $\longrightarrow$ **Direct Kinematics Function**



A Useful Definition: Analytic Jacobian Matrix

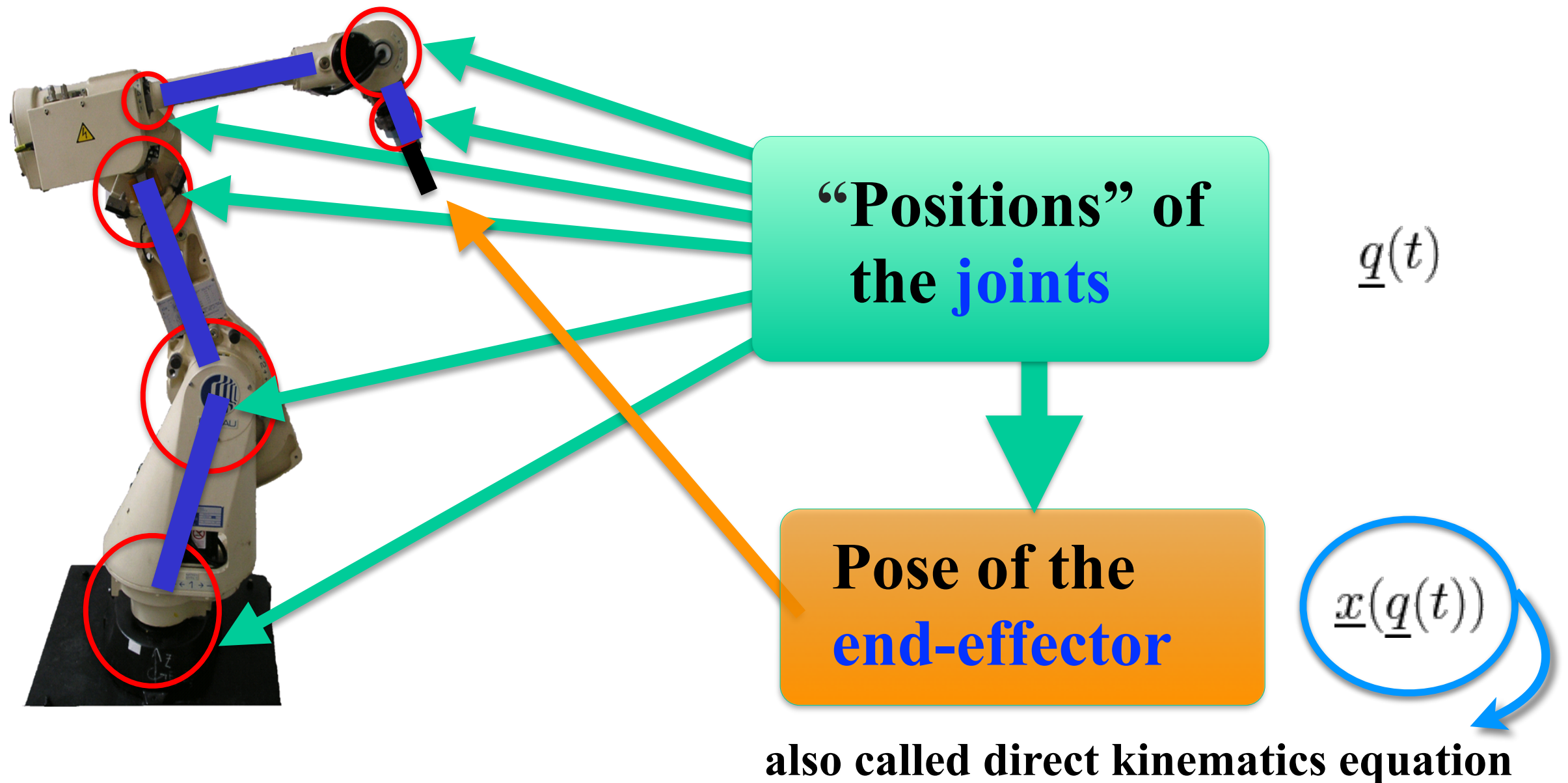
Pose vector $\underline{x}(\underline{q}) = \underline{x}(\underline{q}(t))$ $\longrightarrow$ **Direct Kinematics Function**



nonlinear function of the vector of the joint variables

A Useful Definition: Analytic Jacobian Matrix

Pose vector $\underline{x}(\underline{q}) = \underline{x}(\underline{q}(t))$ $\longrightarrow$ **Direct Kinematics Function**



A Useful Definition: Analytic Jacobian Matrix

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A Useful Definition: Analytic Jacobian Matrix

Pose vector $\underline{x}(\underline{q}) = \underline{x}(\underline{q}(t))$  **Direct Kinematics Function**



by differentiating w.r.t. time

$$\underline{x}(\underline{q}(t)) = \begin{bmatrix} \underline{o}_t^b(\underline{q}(t)) \\ \phi(\underline{q}(t)) \end{bmatrix} \quad \xrightarrow{\text{green arrow}} \quad \dot{\underline{x}}(\underline{q}(t)) = \begin{bmatrix} \frac{\partial \underline{o}_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix} \dot{\underline{q}}(t)$$

A Useful Definition: Analytic Jacobian Matrix

Pose vector $\underline{x}(\underline{q}) = \underline{x}(\underline{q}(t))$ $\longrightarrow$ **Direct Kinematics Function**

by differentiating w.r.t. time

$$\underline{x}(\underline{q}(t)) = \begin{bmatrix} \underline{o}_t^b(\underline{q}(t)) \\ \phi(\underline{q}(t)) \end{bmatrix} \xrightarrow{\quad} \dot{\underline{x}}(\underline{q}(t)) = \begin{bmatrix} \frac{\partial \underline{o}_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix} \underline{\dot{q}}(t)$$

The expression $\dot{\underline{o}}_t^b(\underline{q})$ is indicated by a blue arrow pointing to the top row of the Jacobian matrix.

A Useful Definition: Analytic Jacobian Matrix

Pose vector $\underline{x}(\underline{q}) = \underline{x}(\underline{q}(t))$ $\longrightarrow$ **Direct Kinematics Function**



by differentiating w.r.t. time

$$\underline{x}(\underline{q}(t)) = \begin{bmatrix} \underline{o}_t^b(\underline{q}(t)) \\ \phi(\underline{q}(t)) \end{bmatrix} \longrightarrow \dot{\underline{x}}(\underline{q}(t)) = \begin{bmatrix} \frac{\partial \underline{o}_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix} \dot{\underline{q}}(t)$$

A blue oval highlights the Jacobian matrix $\begin{bmatrix} \frac{\partial \underline{o}_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix}$, with an arrow pointing to $\dot{\phi}(\underline{q})$.

A Useful Definition: Analytic Jacobian Matrix

Pose vector $\underline{x}(\underline{q}) = \underline{x}(\underline{q}(t))$ $\longrightarrow$ **Direct Kinematics Function**

by differentiating w.r.t. time

**Analytic
Jacobian
Matrix**

$$J_A(\underline{q}(t))$$

$$\dot{\underline{x}}(\underline{q}(t)) = \begin{bmatrix} \frac{\partial o_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix} \dot{\underline{q}}(t)$$

A Useful Definition: Analytic Jacobian Matrix

Pose vector $\underline{x}(\underline{q}) = \underline{x}(\underline{q}(t))$ $\longrightarrow$ Direct Kinematics Function

by differentiating w.r.t. time

Analytic
Jacobian
Matrix

$$J_A(\underline{q}(t))$$

$$\dot{\underline{x}}(\underline{q}(t)) = \begin{bmatrix} \frac{\partial o_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix} \dot{\underline{q}}(t)$$



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Operational Space

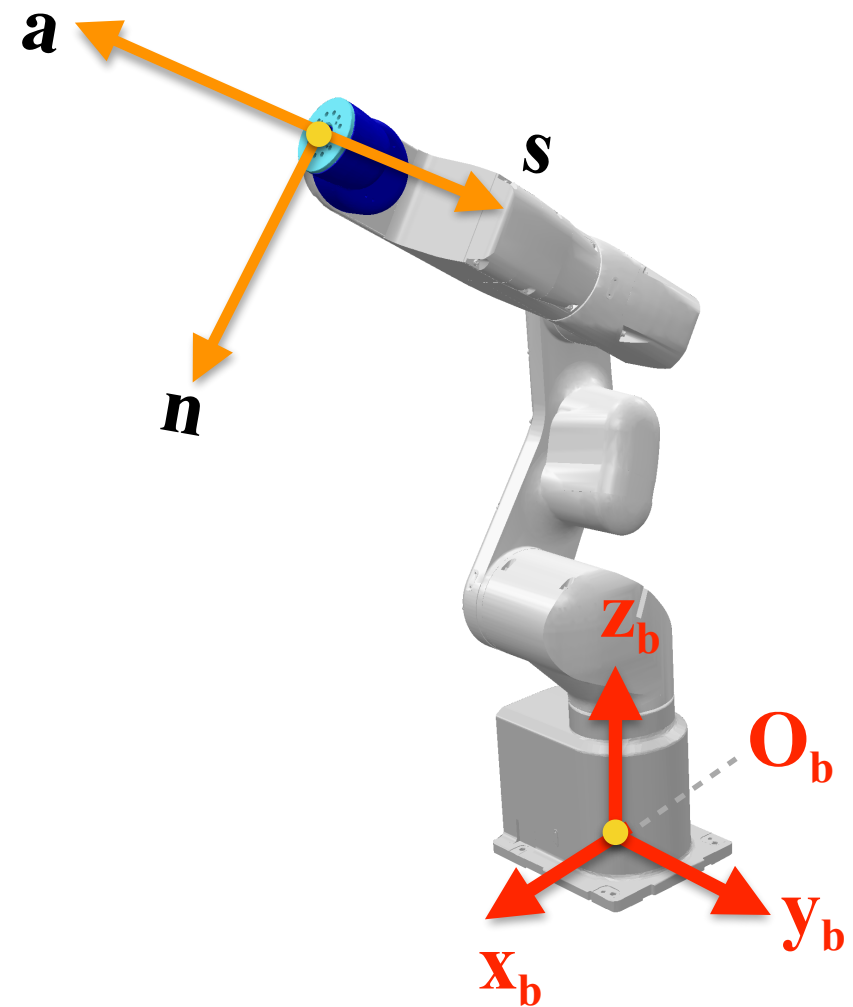
A Useful Definition: Operational Space

Task Space

Cartesian Space

Operational Space

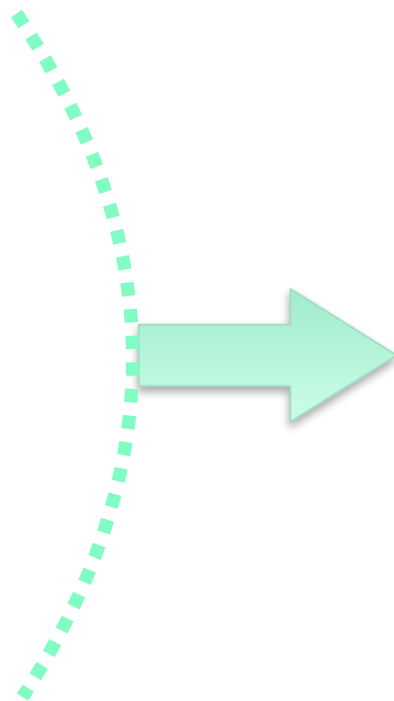
The space in which the pose vector $\underline{x}(\underline{q}(t))$ is defined is the space in which typically the operations required to the robot manipulator are specified



Operational Space Dimension

Task Space

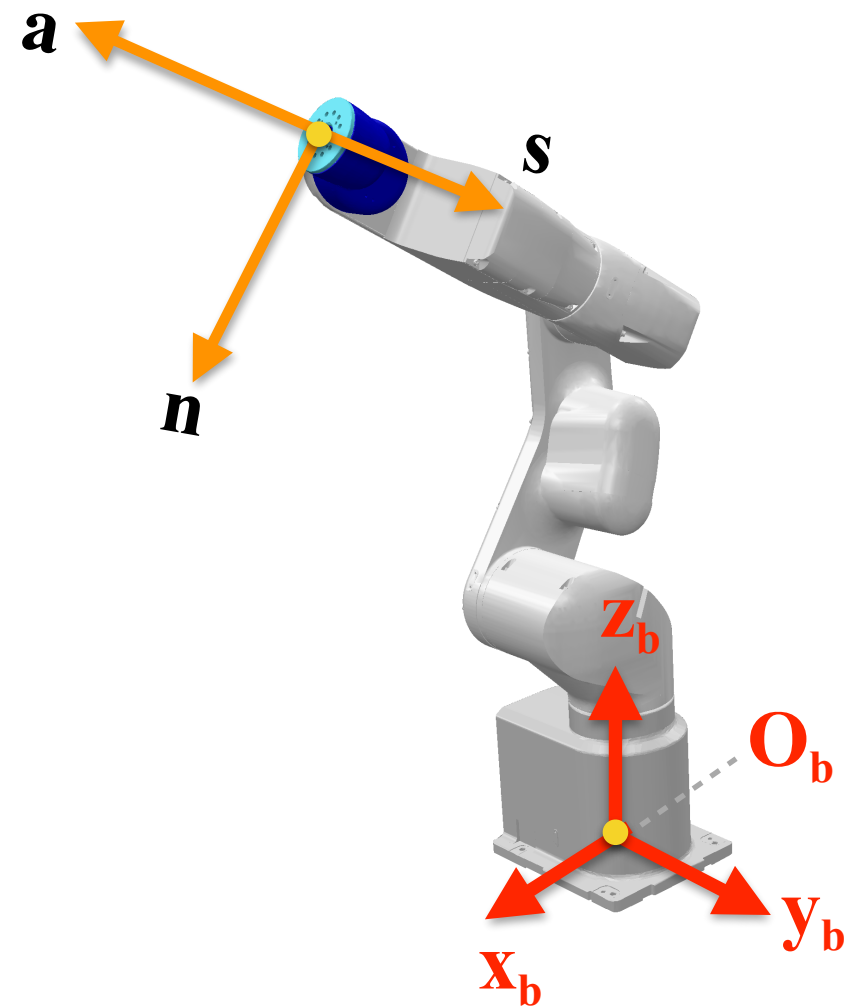
Cartesian Space



Operational Space

Pose vector $\underline{x}(\underline{q}) = \begin{bmatrix} \underline{O}_t^b \\ \varphi \\ \vartheta \\ \psi \end{bmatrix} \rightarrow m \times 1$

Operational Space Dimension : m





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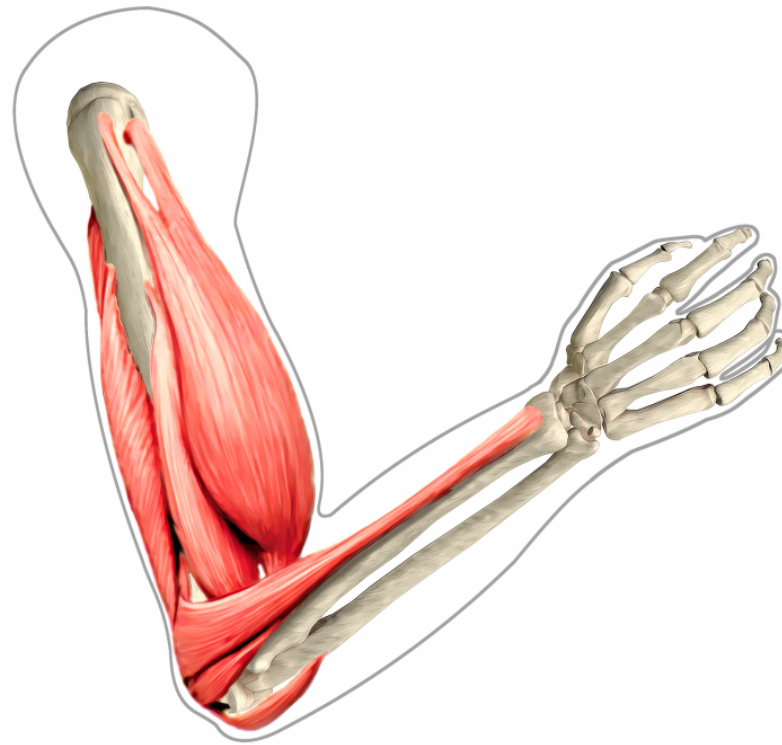


Kinematic Redundancy

Kinematic Redundancy

- A manipulator is defined **kinematically redundant** when it has a number of DOFs which is greater than the number of variables that are necessary to describe a given task
- A manipulator is **intrinsically redundant** when the dimension of the operational space is smaller than the dimension of the joint space ($m < n$).

EXAMPLES:



Panda by Franka Emika

$$n = 7, m = 6$$

Kinematic Redundancy

- **Redundancy** is a concept relative to the task assigned to the manipulator; a manipulator can be redundant with respect to a task and non-redundant with respect to another.
- Even in the case of $m = n$, a manipulator can be **functionally redundant** when only a number of r components of operational space are of concern for the specific task, with $r < m$.

EXAMPLE: 6-axes industrial robot used for laser-cutting

not intrinsically redundant ($n = m = 6$)

functional redundant since the end-effector rotation about the approach direction is irrelevant to perform the task ($r = 5$)





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Kinematic Singularity

A Useful Concept: Kinematic Singularity

A kinematic singularity is a configuration within the robot workspace in correspondence of which the robot Jacobian matrix loses rank



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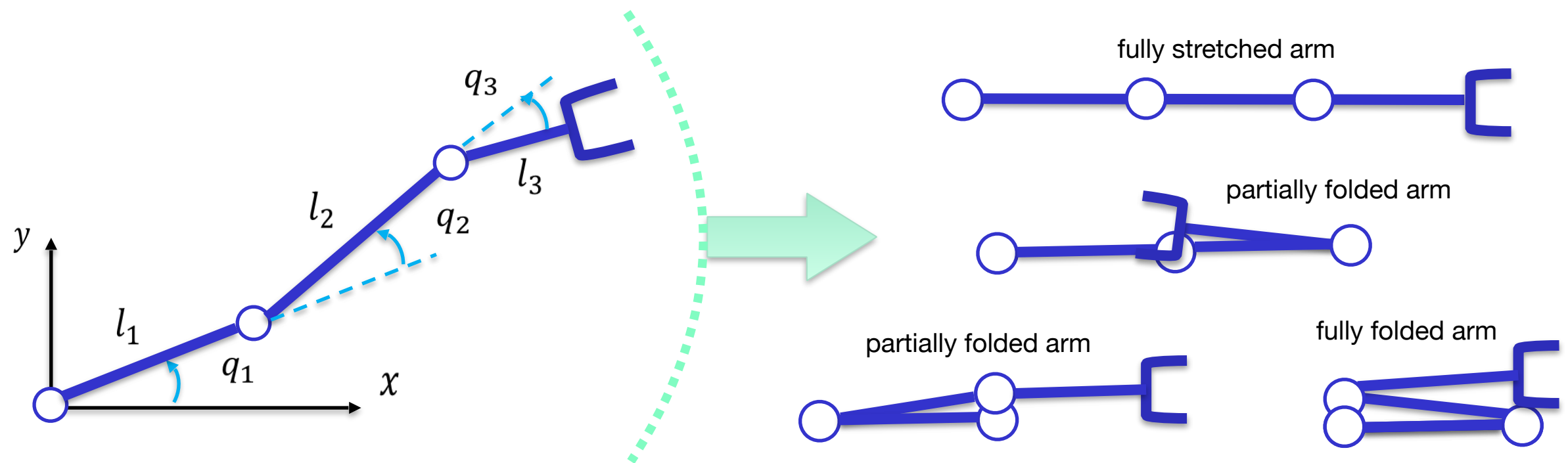


It typically occurs when two of the manipulator joints line up (but not only!)

A Useful Concept: Kinematic Singularity

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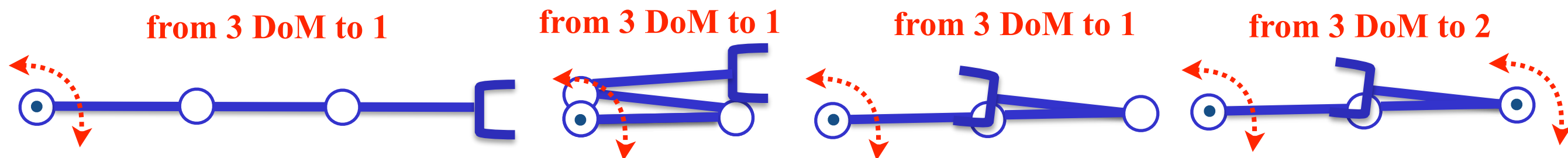


A Useful Concept: Kinematic Singularity

A kinematic singularity is a configuration within the robot workspace in correspondence of which the robot Jacobian matrix loses rank

It typically occurs when two of the manipulator joints line up (but not only!)

The mobility of a manipulator is reduced: “Loss of DoM”



A Useful Concept: Kinematic Singularity

EXAMPLE: 6-DoM robot manipulator Mecademic's Meca500



Payload: 500 g

Total weight: 4.5 Kg

Reach (at wrist center): 260 mm

the axes of joints 2 and 3 are parallel

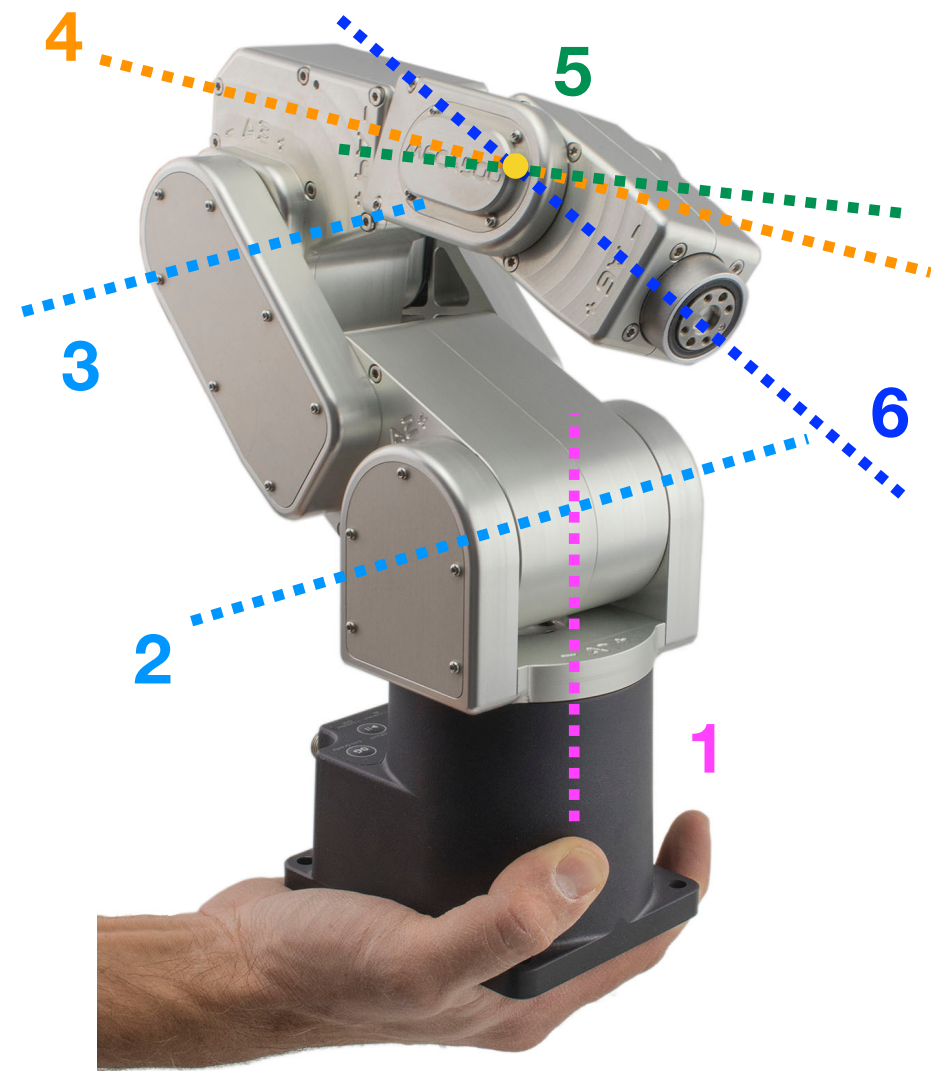
the axes of joints 1 and 4 are normal to the axes of joints 2 and 3

the axis of joint 5 is normal to the axes of joints 4 and 6,
and these last three axes intersect at one point.

<https://www.mecademic.com/products/Meca500-small-robot-arm>

A Useful Concept: Kinematic Singularity

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**Architecture often referred to as:
“vertically-articulated robot arm”**

<https://www.mecademic.com/products/Meca500-small-robot-arm>

A Useful Concept: Kinematic Singularity

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Robots referred to as:
“wrist-partitioned” or as “having inline wrist”

<https://www.mecademic.com/products/Meca500-small-robot-arm>

A Useful Concept: Kinematic Singularity

EXAMPLE: 6-DoM robot manipulator Mecademic's Meca500



Source: <https://www.mecademic.com/resources/Singularities/Robot-singularities>

Outline of the Course

Basic Terminology and Concepts

Reference Frames and Rotation Matrices

Robot Direct Kinematics: the Geometric Approach

Robot Direct Kinematics: the Denavit-Hartenberg Convention Based Approach

Robot Inverse Kinematics

Robot Differential Kinematics

Robot Dynamical Model

Robot Control



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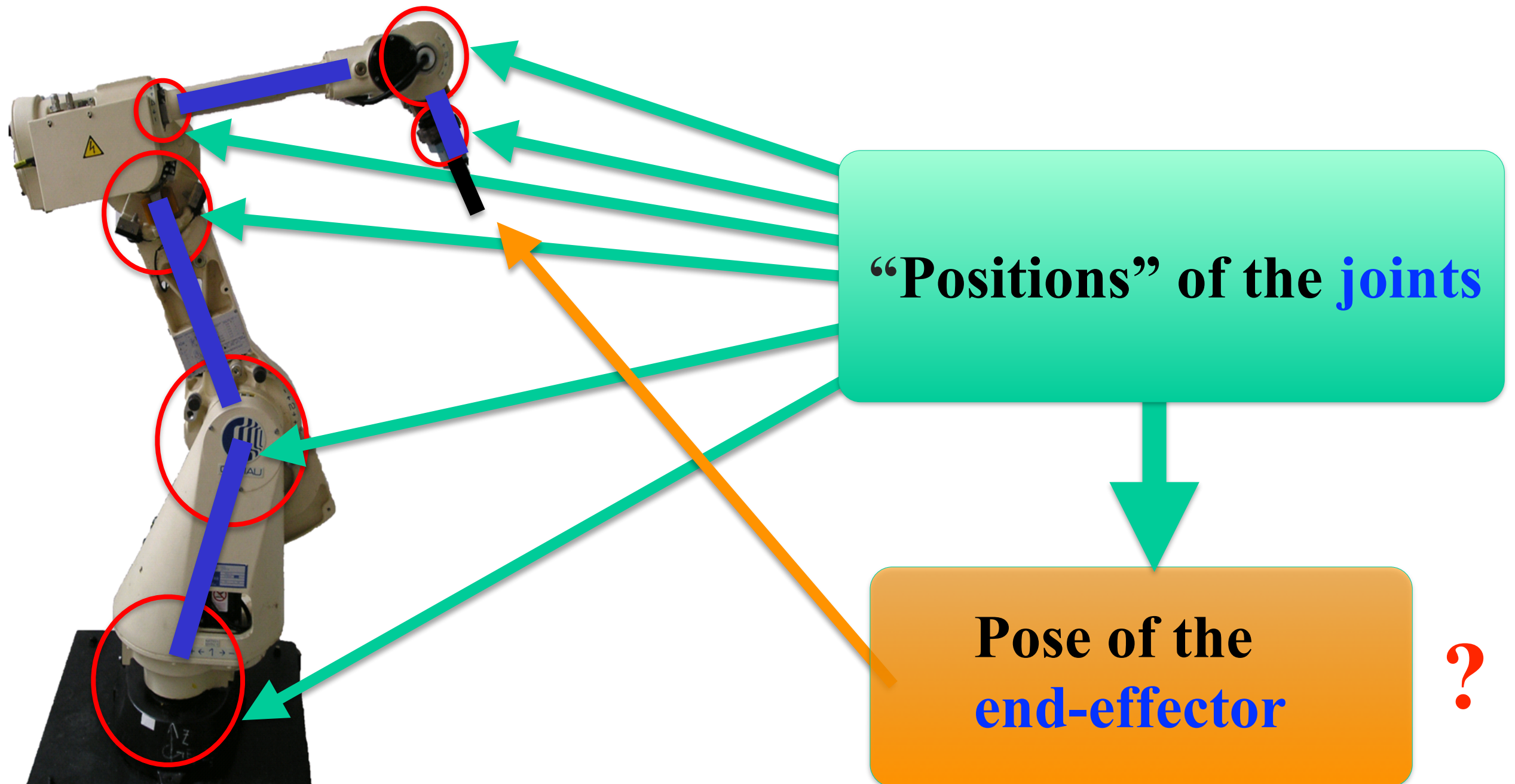
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The Inverse Kinematics Problem in Robotics

Direct Kinematics

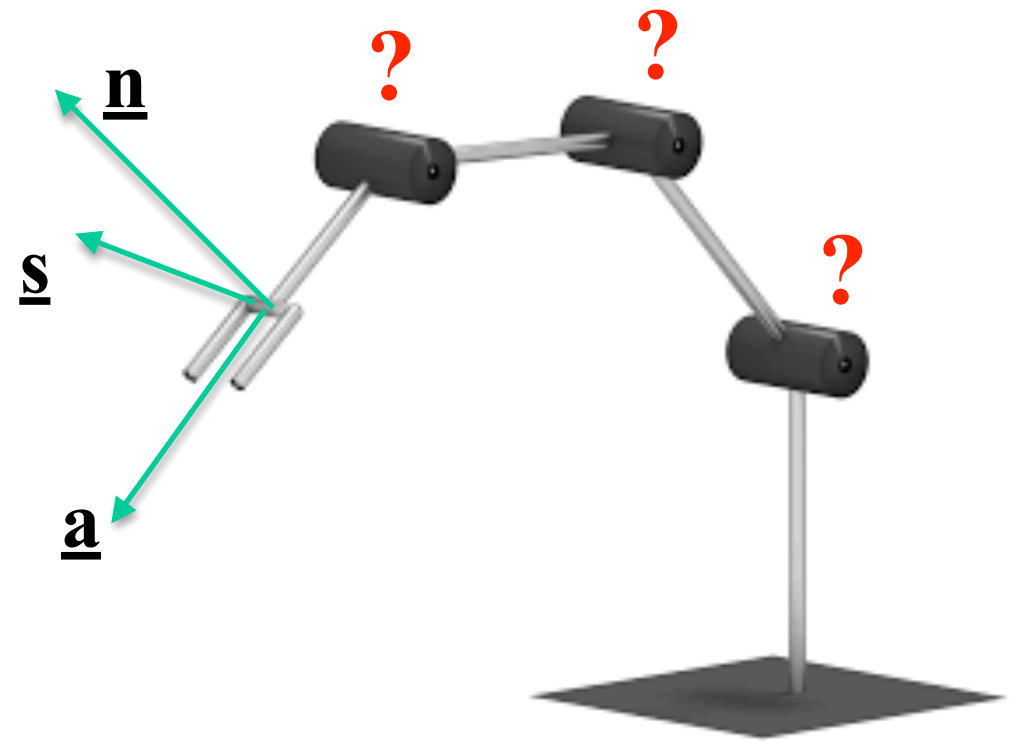


Inverse Kinematics

“Pose” of the
end-effector



“Positions” of the joints

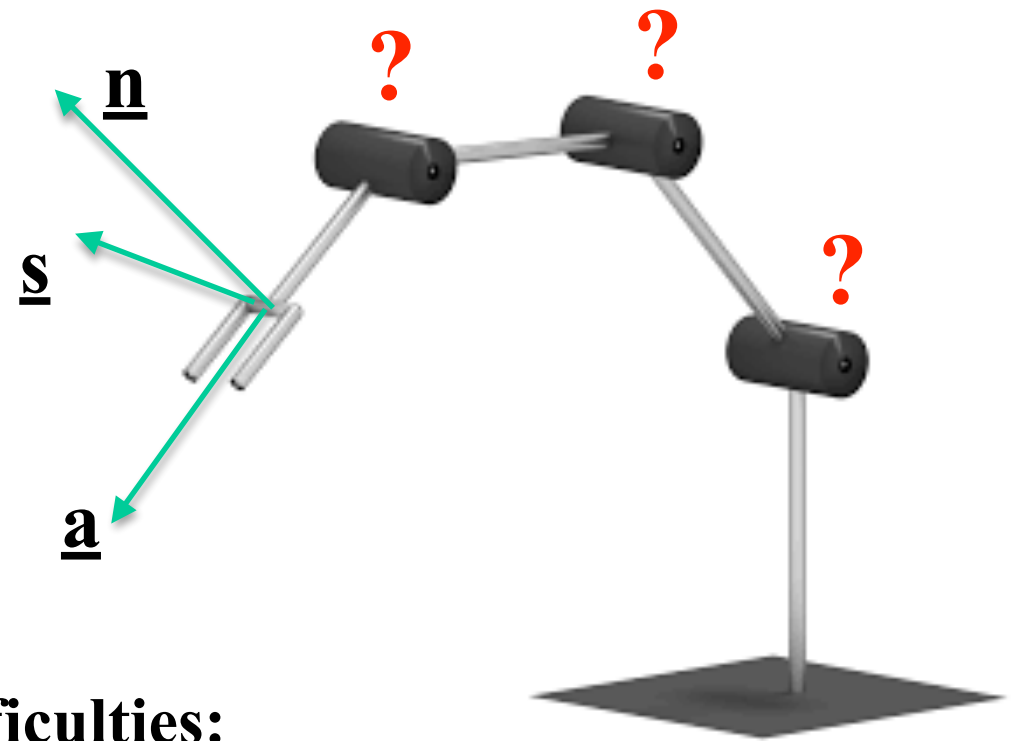


Inverse Kinematics

“Pose” of the
end-effector



“Positions” of the joints



Difficulties:

- Equations to solve are in general NL
- A closed form analytic solution is not always found
- Multiple solutions are possible
- Infinite solutions are possible (redundant manipulators)
- No admissible solutions given the robot kinematic structure

Primary and Secondary Workspace

Primary workspace WS_1 : set of points $\underline{p}$ which are reachable with at least one orientation

Reachable Workspace

- out of WS_1 there is no solution
- $\exists \underline{p} \in WS_1$, and a given orientation such that a solution exists

Secondary workspace WS_2 (dexterity space): set of points $\underline{p}$ which are reachable with any possible orientation

Dexterous Workspace

- $\forall \underline{p} \in WS_2$, there exists at least a solution for any orientation



$$WS_2 \subseteq WS_1$$



Primary and Secondary Workspace

Primary workspace WS_1 : set of points $\underline{p}$ which are reachable with at least one orientation

Reachable Workspace

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- $\exists \underline{p} \in WS_1$, and a given orientation such that a solution exists

For an n-DOF manipulator, the **reachable workspace** is the geometric locus of the points that can be achieved by considering the direct kinematics equation $\underline{x}(\underline{q}(t))$ for the sole position part $\underline{q}_t^b$ with:

$$q_{im} \leq q_i \leq q_{iM} \quad i = 1, \dots, n,$$



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continuous function

$$q_{im} \leq q_i \leq q_{iM} \quad i = 1, \dots, n,$$

**This volume is finite, closed, connected.
It is defined by its bordering surface**



Primary and Secondary Workspace

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Reachable Workspace

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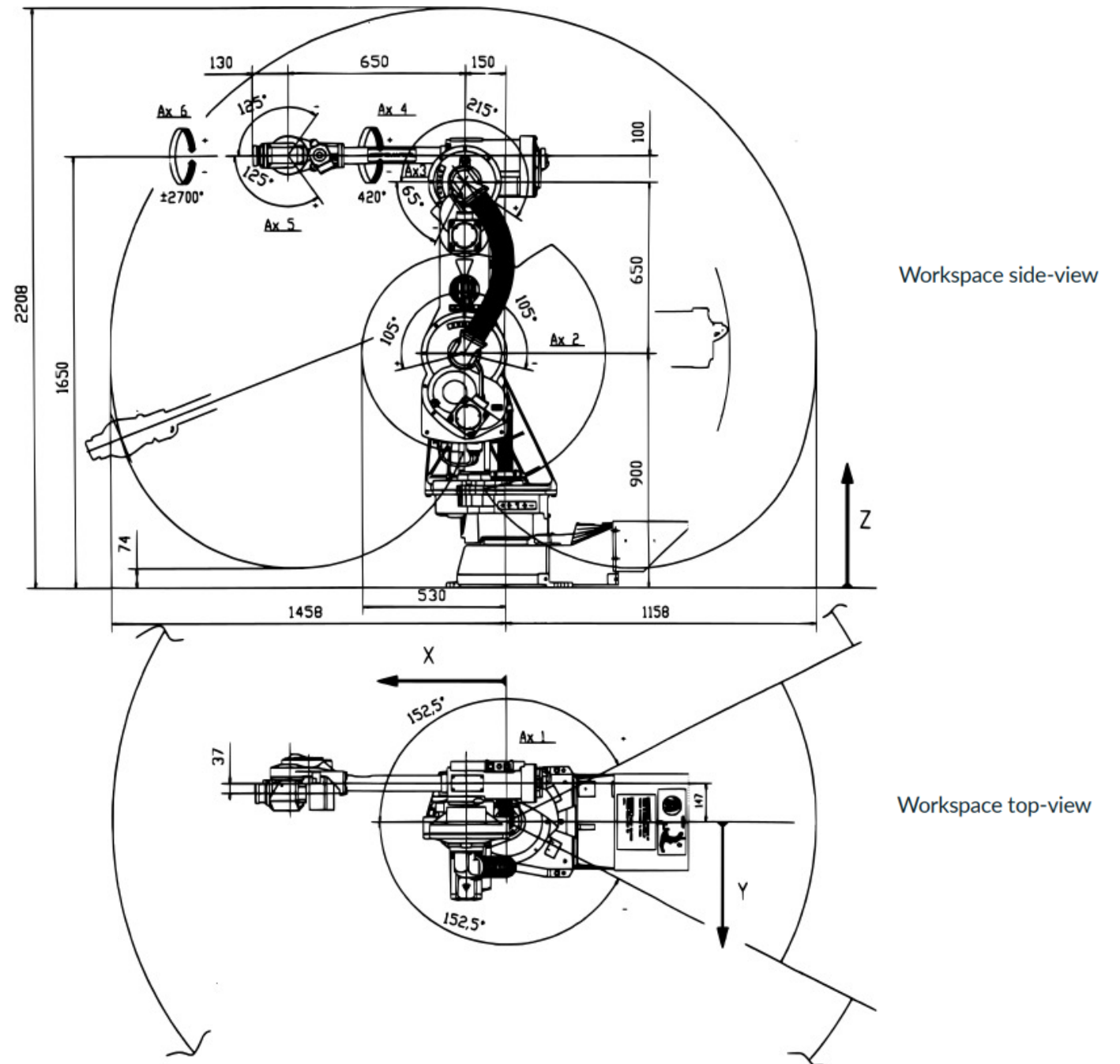
$$q_{im} \leq q_i \leq q_{iM} \quad i = 1, \dots, n,$$

It is reported in the data sheet given by the robot manufacturer in terms of a top view and a side view



The Robot Workspace in Data Sheets

PRIMARY WORKSPACE OF COMAU SMART3-S2

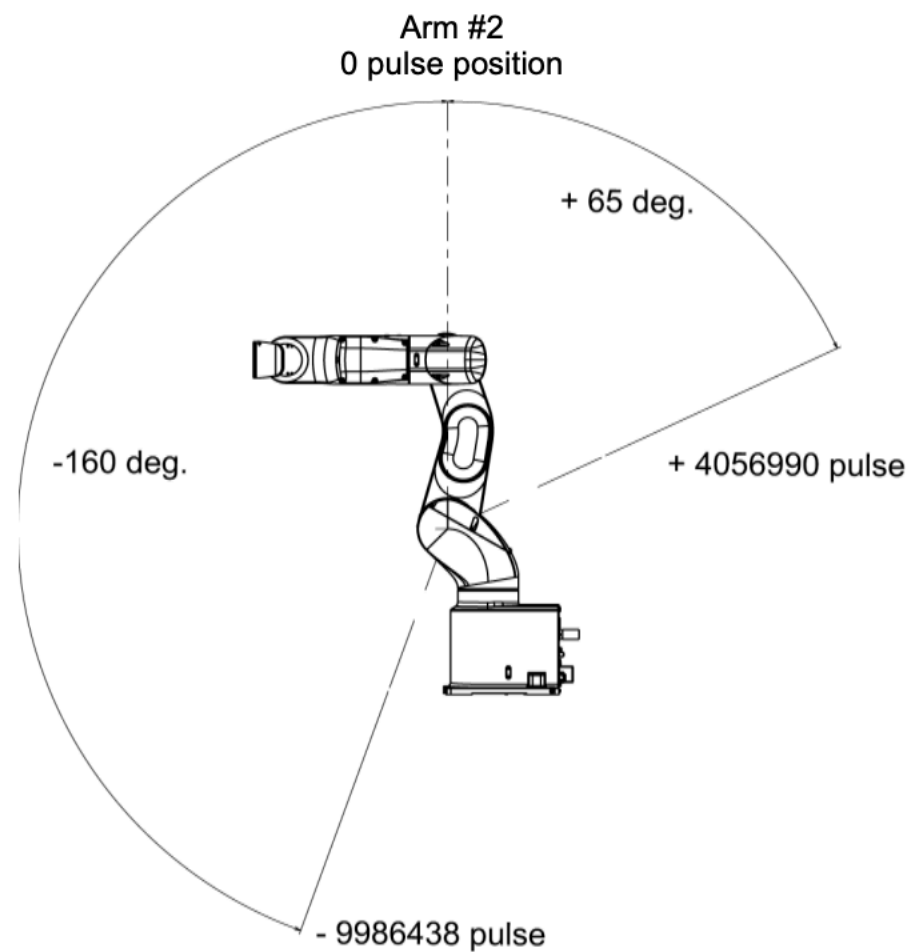


The Robot Workspace in Data Sheets

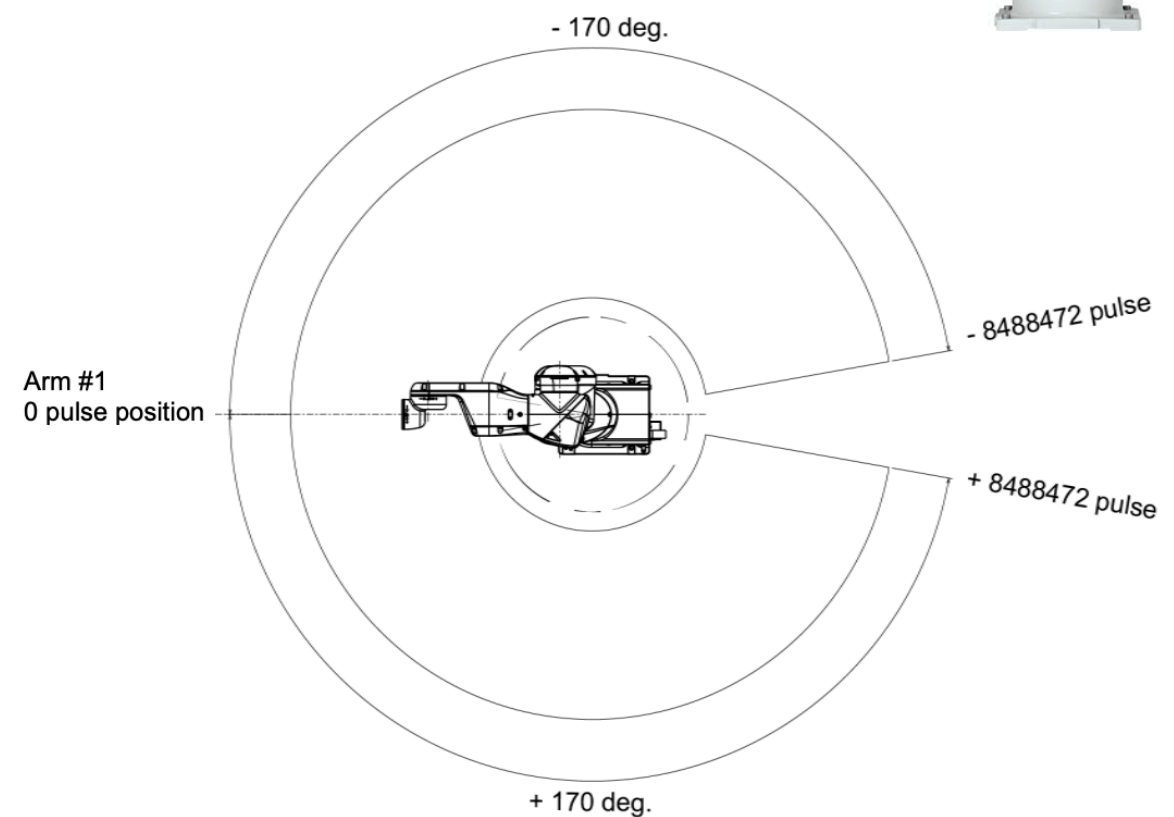
PRIMARY WORKSPACE OF EPSON VT6



Workspace side-view

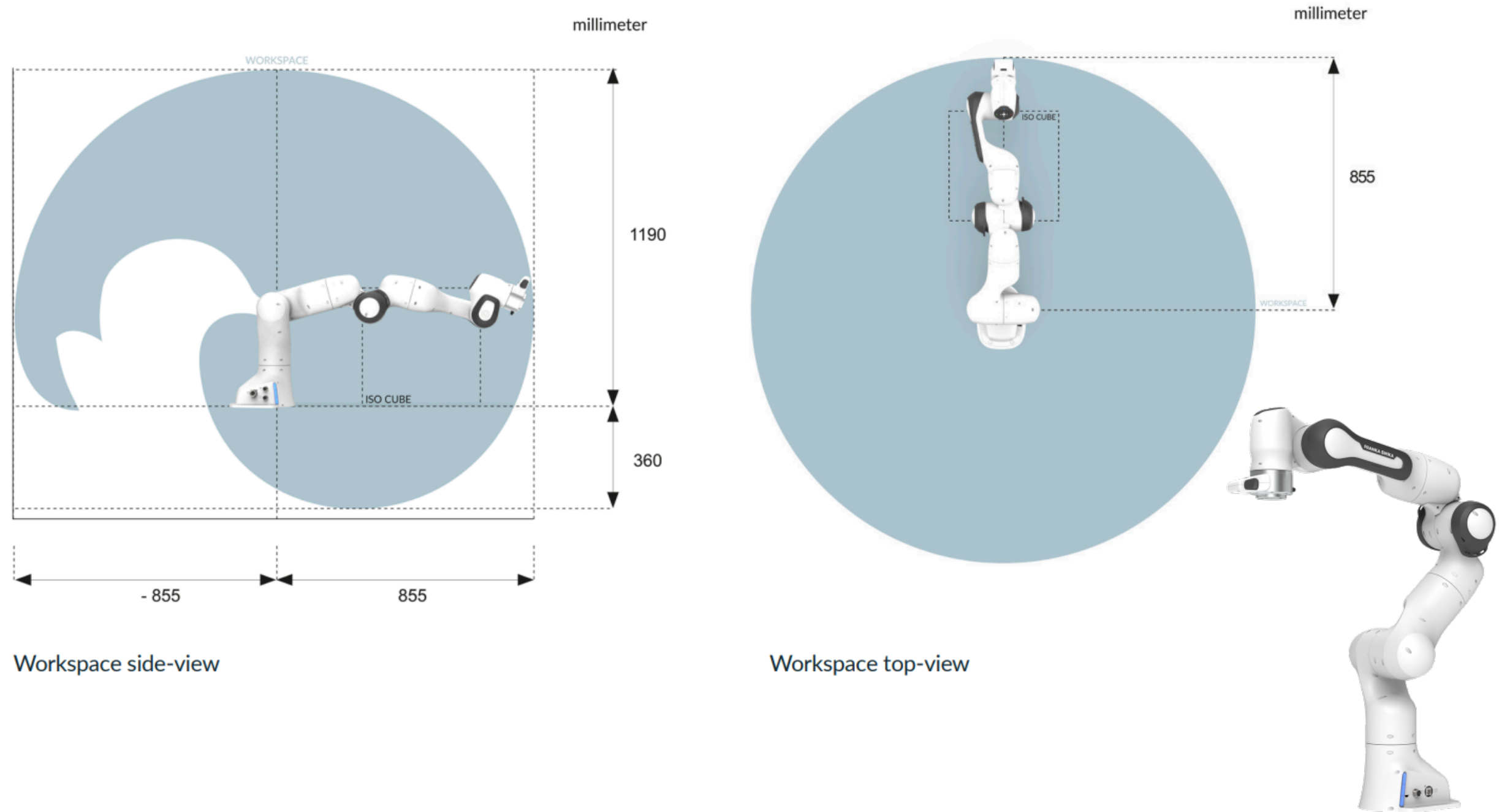


Workspace top-view



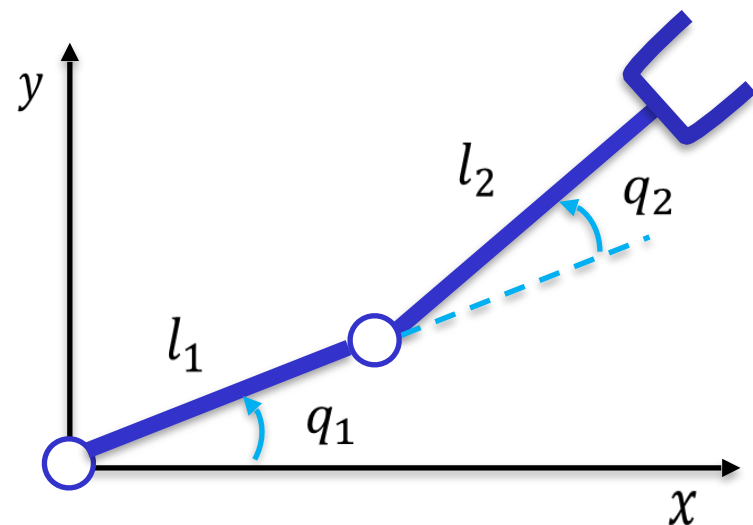
The Robot Workspace in Data Sheets

PRIMARY WORKSPACE OF FRANKA EMIKA PANDA

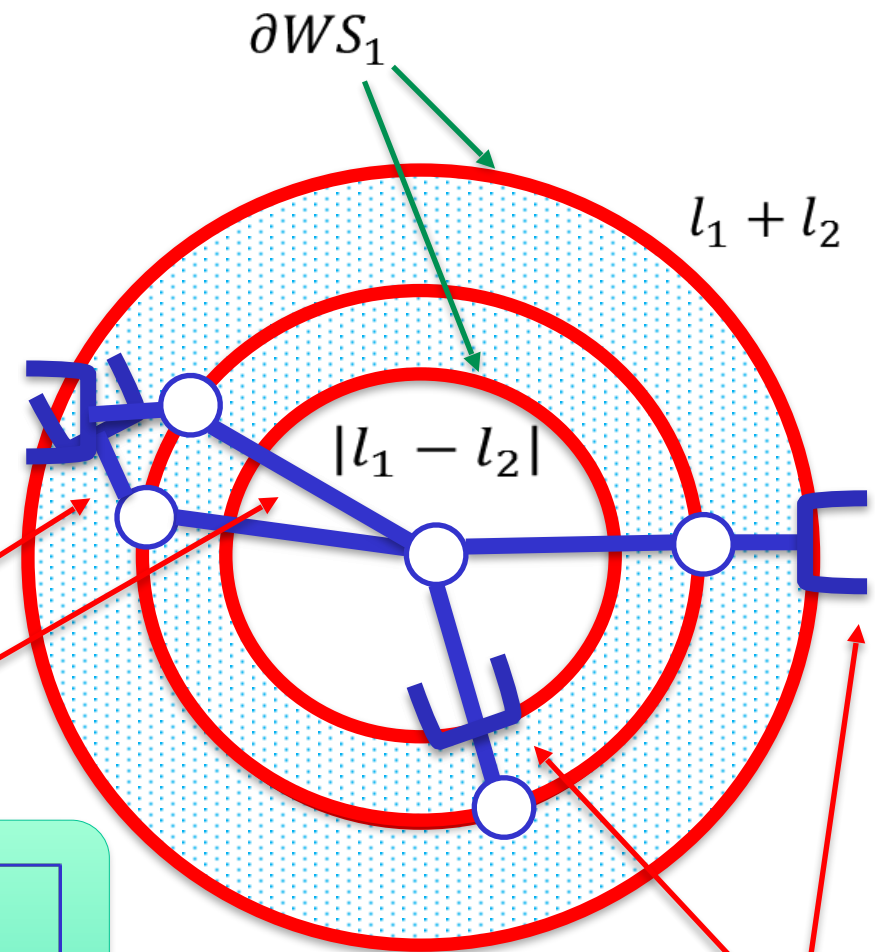


Primary and Secondary Workspace

EXAMPLE: 2 DoM PLANAR ROBOT



2 orientations



1 orientation

If $l_1 \neq l_2$

- $WS_1 = \{ \underline{p} \in \mathbb{R}^2 : |l_1 - l_2| \leq \| \underline{p} \| \leq l_1 + l_2 \}$
- $WS_2 = \emptyset$

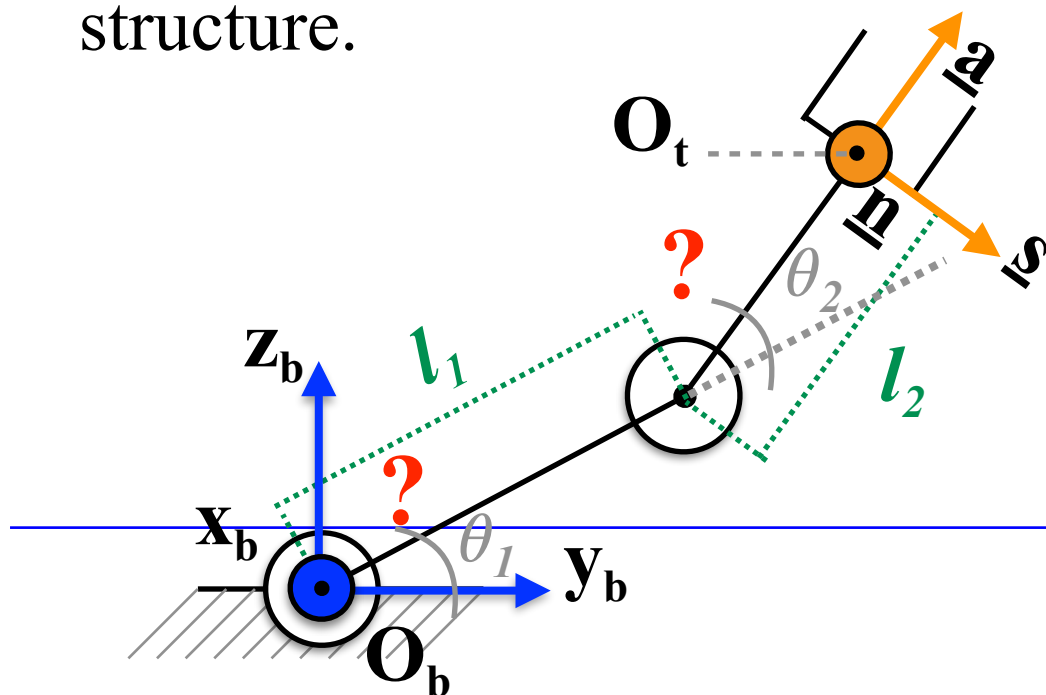
If $l_1 = l_2 = \ell$

- $WS_1 = \{ \underline{p} \in \mathbb{R}^2 : \| \underline{p} \| \leq 2\ell \}$
- $WS_2 = \{ \underline{p} = 0 \}$ (infinite orientation around the origin)

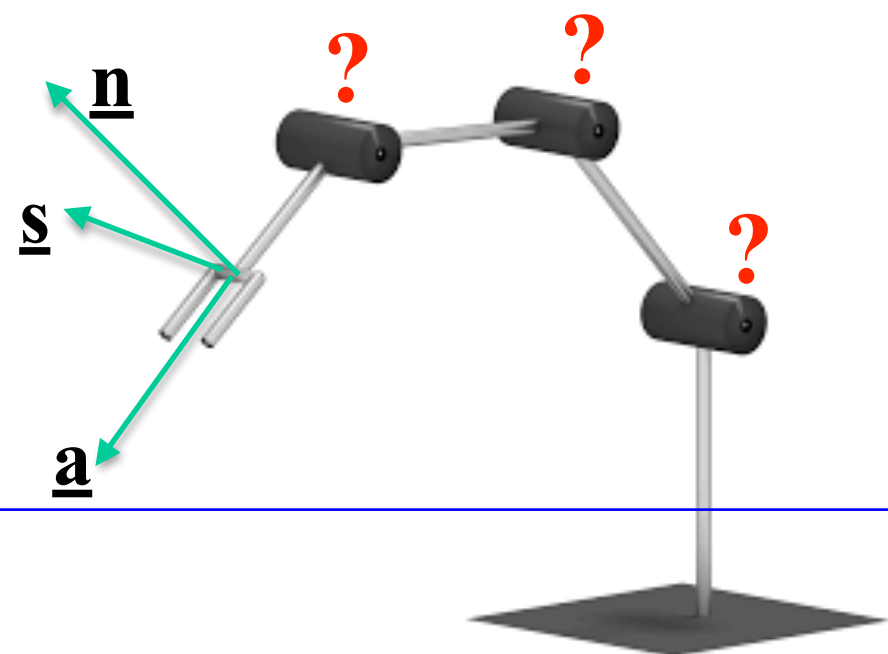
Inverse Kinematics: Possible Solutions

In general:

- The inverse kinematics problem is **much more complex** than the direct kinematics problem.
- The equations to solve are in general **nonlinear equations**: it is not always possible to find a closed-form solution.
- **Multiple solutions** may exist.
- **Infinite solutions** may exist, e.g., in the case of a kinematically redundant manipulator.
- There might be **no admissible solution**, in view of the manipulator kinematic structure.



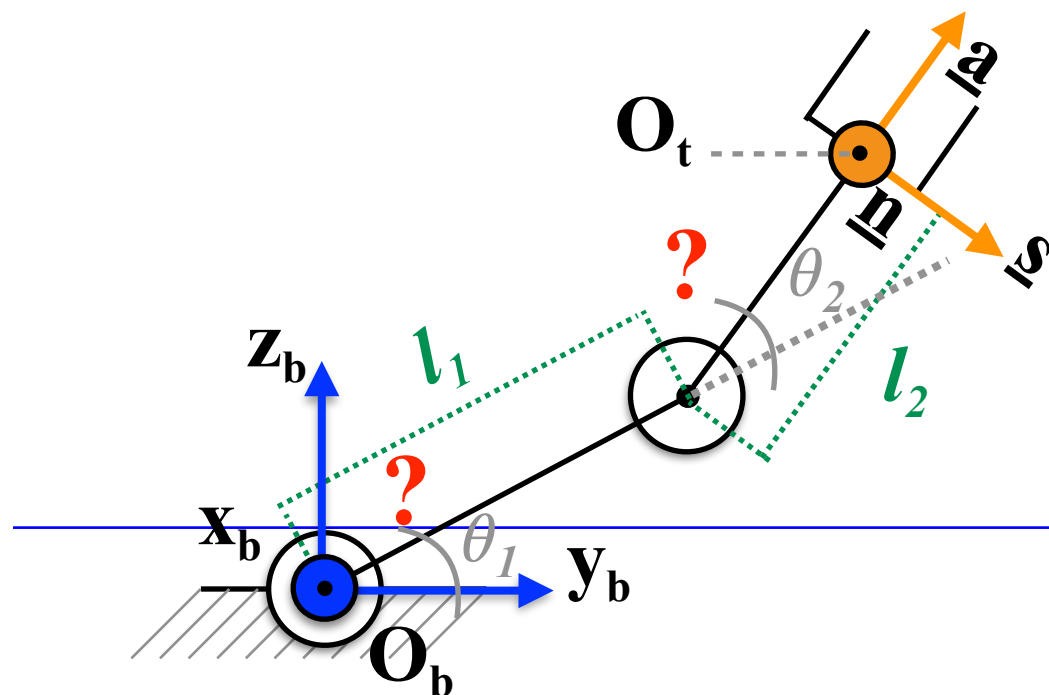
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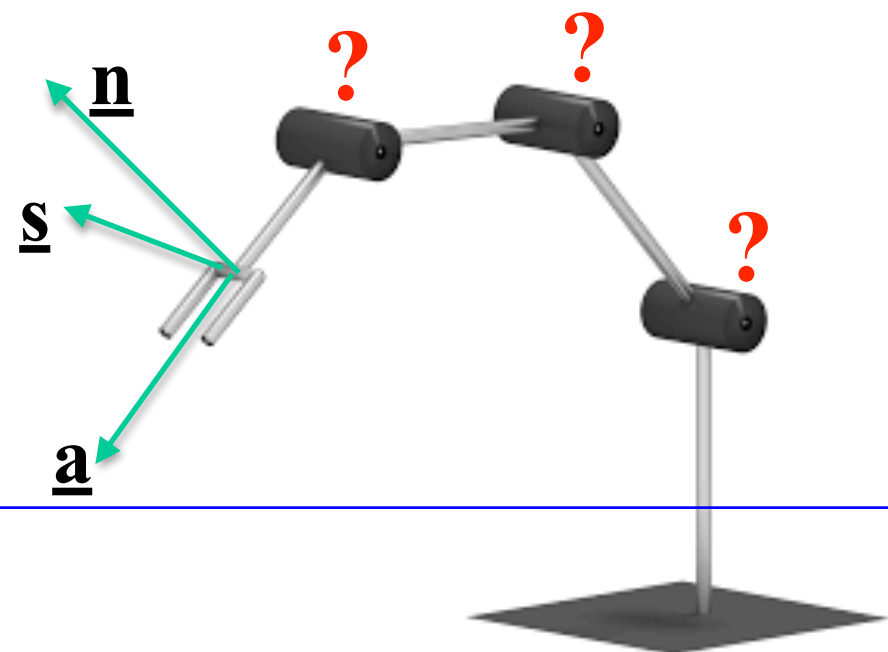
Inverse Kinematics: Possible Solutions

In general:

- The **existence of solution** is “guaranteed” if the prescribed pose belongs to the Secondary Workspace WS_2 , i.e. to the **dexterity space of the robot**.
- The **existence of multiple solution** depends on the number of DoMs but also by the number of DH parameters different from 0 (the larger the number of DH parameters different from 0, the larger the number of admissible solutions).
- The presence of **mechanical limit switches** may **reduce the number of multiple solution** in the actual robot manipulator.

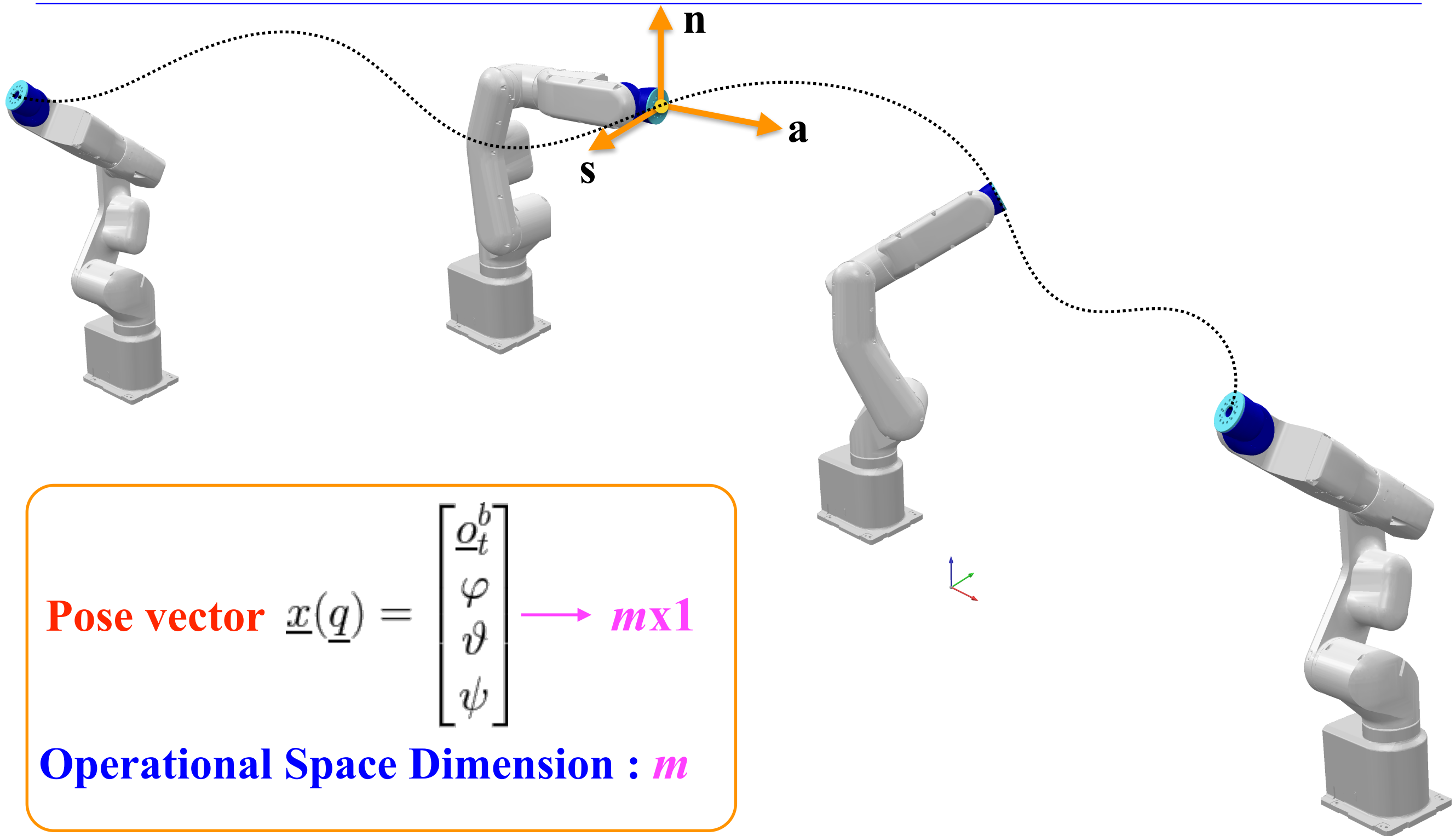


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In a previous lecture...

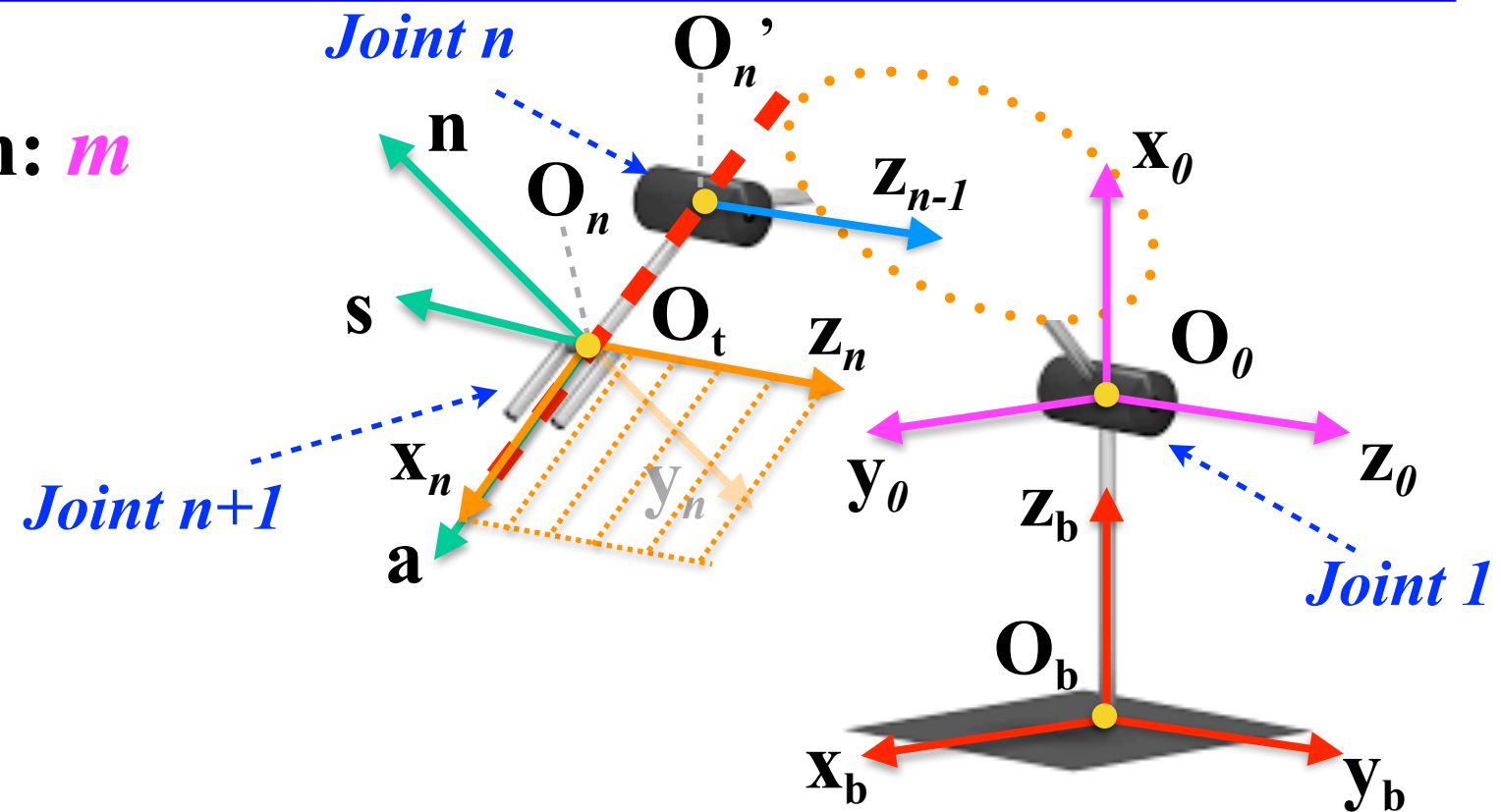
Operational Space Definition and Dimension



Inverse Kinematics: Possible Solutions

Operational Space Dimension: m

Number of joints: n



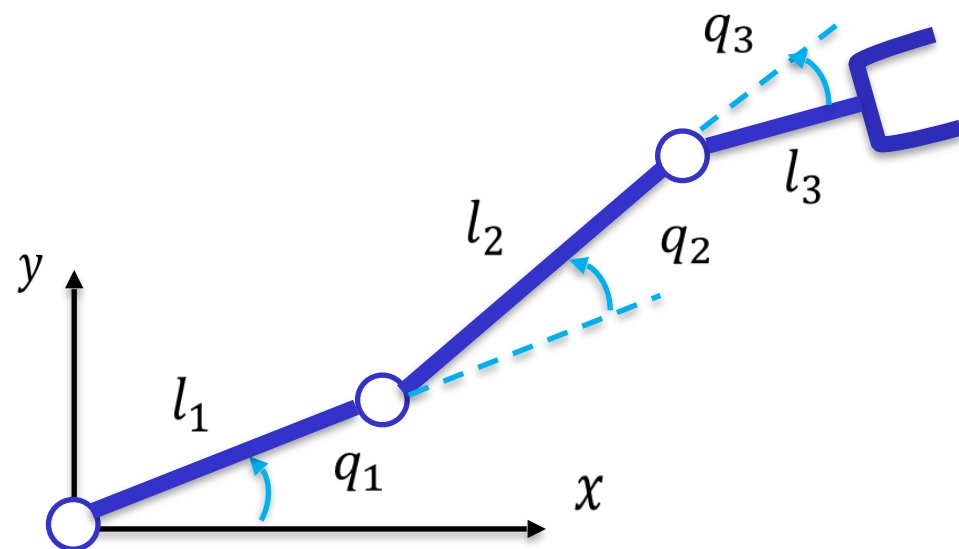
If $m = n$

- $\nexists$ solutions
- Finite number of multiple solutions
- Infinite solutions or a finite number of singular solutions

If $m < n$ (redundant robot for the task)

- $\nexists$ solutions
- ∞^{n-m} solutions
- Finite or infinite number of singular solutions

Inverse Kinematics: Possible Solutions

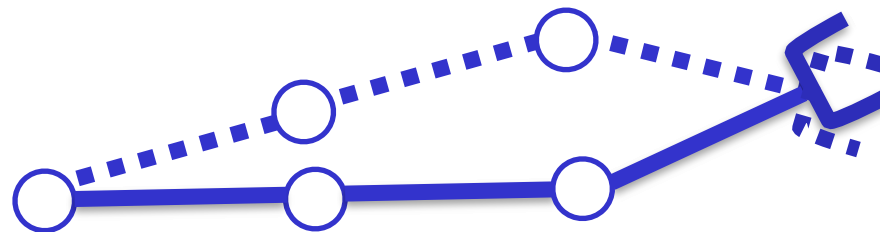


EXAMPLE: 3 DoM PLANAR ROBOT

If $l_1 = l_2 = l_3 = \ell$

- $WS_1 = \{ \underline{p} \in \mathbb{R}^2 : \|\underline{p}\| \leq 3\ell \}$
- $WS_2 = \{ \underline{p} \in \mathbb{R}^2 : \|\underline{p}\| \leq \ell \}$

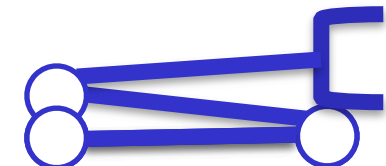
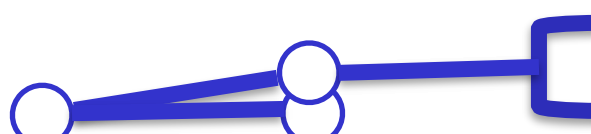
- In WS_1 one has ∞^1 non-singular solutions except for $\|\underline{p}\| = 3\ell$ and $\|\underline{p}\| = \ell$



- If $\|\underline{p}\| = 3\ell$ one has 1 singular solutions



- If $\|\underline{p}\| = \ell$ one has ∞^1 solutions among which 3 singular solutions

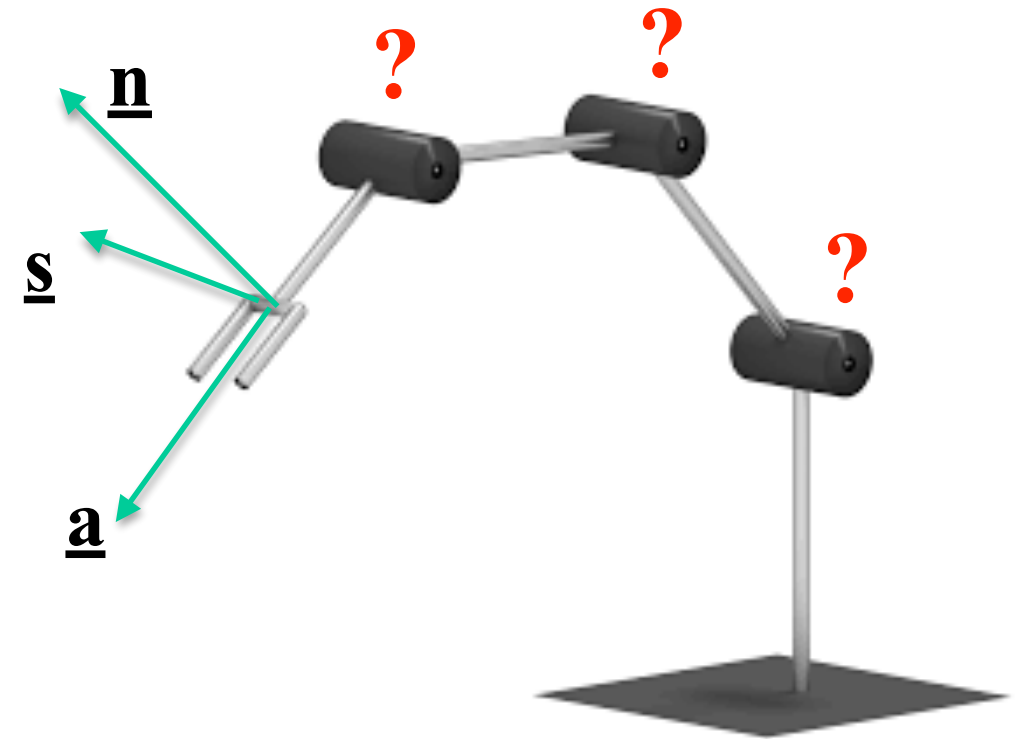


- If $\|\underline{p}\| < \ell$ one has ∞^1 solutions (never singular solutions)

Inverse Kinematics: Solution Methods

Analytical solutions

- Geometric inspection
- Algebraic methods
- Minimal set of equation to solve
- Fast solution if it is an easy case



Numerical solutions

- Necessary if $n > m$
- Slow solutions
- It is necessary to compute the analytical Jacobian

$$J_A(\underline{q}(t))$$

- Newton method, gradient method, etc.



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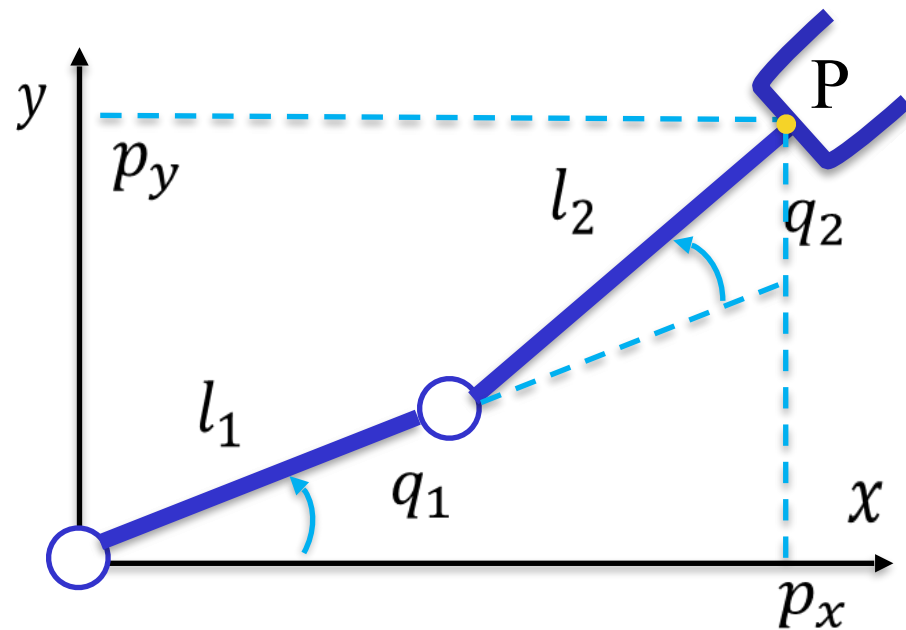
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Robot Inverse Kinematics: Analytic Solution Example

Inverse Kinematics: Solution Example

EXAMPLE: 2 DoM PLANAR ROBOT



Direct Kinematics

$$p_x = l_1 c_1 + l_2 c_{12}$$

$$p_y = l_1 s_1 + l_2 s_{12}$$

Solution

$$p_x^2 + p_y^2 - (l_1^2 + l_2^2) = 2l_1 l_2 (c_1 c_{12} + s_1 s_{12}) = 2l_1 l_2 c_2$$

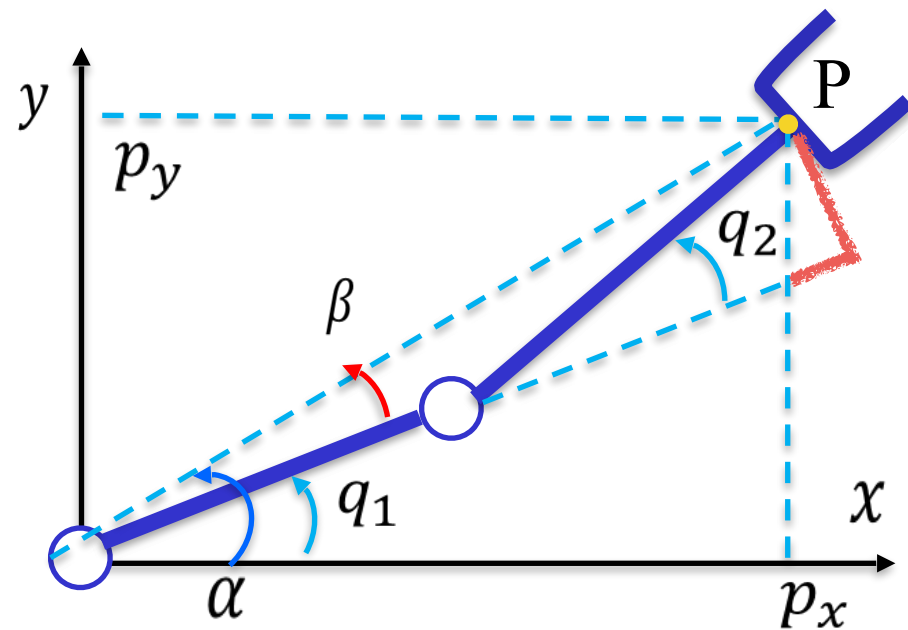
$$c_2 = \frac{p_x^2 + p_y^2 - (l_1^2 + l_2^2)}{2l_1 l_2}, c_2 \in [-1, 1]$$

$$s_2 = \pm \sqrt{1 - c_2^2}$$

$$q_2 = \text{atan2}\{c_2, s_2\}$$

Inverse Kinematics: Solution Example

EXAMPLE: 2 DoM PLANAR ROBOT



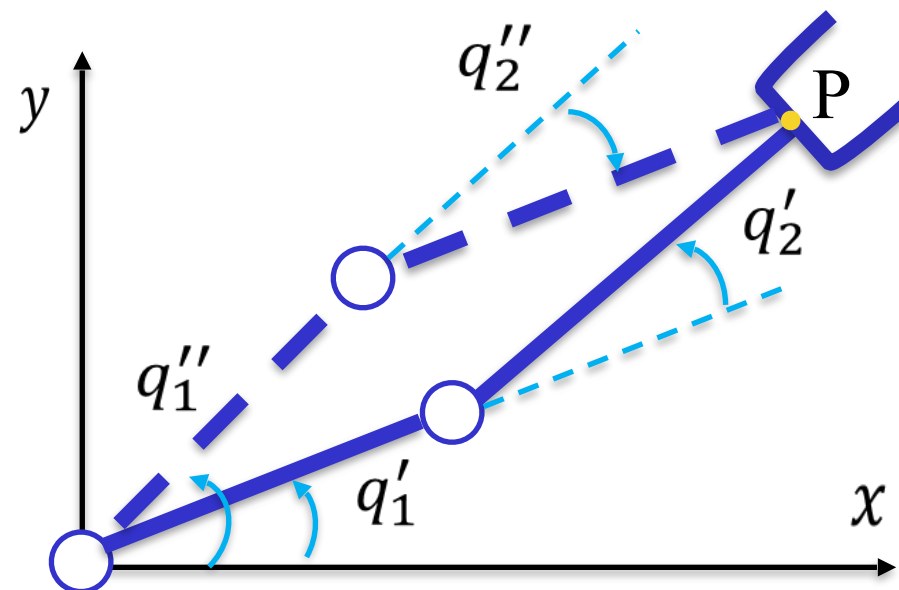
By geometric inspection find q_1

$$q_1 = \alpha - \beta$$

from which it holds

$$q_1 = \text{atan2}\{p_x, p_y\} - \text{atan2}\{l_2 s_2, l_1 + l_2 c_2\}$$

UP/LEFT

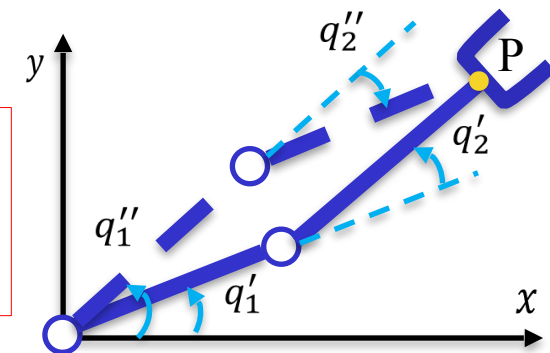


DW/RIGHT

Inverse Kinematics: Solution Example

EXAMPLE: 2 DoM PLANAR ROBOT

UP/LEFT



DW/RIGHT

Alternative solution to find q_1

Note that, it is possible to write a system linear in s_1 and c_1

$$p_x = l_1 c_1 + l_2 c_{12} = l_1 c_1 + l_2 (c_1 c_2 - s_1 s_2)$$

$$p_y = l_1 s_1 + l_2 s_{12} = l_1 s_1 + l_2 (s_1 c_2 + c_1 s_2)$$

$$\begin{bmatrix} l_1 + l_2 c_2 & -l_2 s_2 \\ l_2 s_2 & l_1 + l_2 c_2 \end{bmatrix} \begin{bmatrix} c_1 \\ s_1 \end{bmatrix} = \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

$$q_1 = \text{atan2}\{c_1, s_1\}$$

Note that the solution does not exist for $l_1 = l_2$ and $c_2 = -1$, since the determinant of the matrix is

$$\det = (l_1^2 + l_2^2) + 2l_1 l_2 c_2$$



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Inverse Kinematics: Numerical Methods

Recall the...

A Useful Definition: Analytic Jacobian Matrix

Pose vector $\underline{x}(\underline{q}) = \underline{x}(\underline{q}(t))$  **Direct Kinematics Function**



by differentiating w.r.t. time

**Analytic
Jacobian
Matrix**

$J_A(\underline{q}(t))$

$$\dot{\underline{x}}(\underline{q}(t)) = \begin{bmatrix} \frac{\partial o_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix} \dot{\underline{q}}(t)$$

Inverse Kinematics: Numerical Methods

Since the analytical solution of $\underline{x}(\underline{q}^*) = \underline{x}^*$ is difficult to find or it does not exist, **numerical solutions** are needed.

Newton method ($m = n$)

- Let $J_A = \frac{\partial \underline{x}(\underline{q})}{\partial \underline{q}}$ be the analytic Jacobian.
- Define $\underline{x}(\underline{q}) = \underline{x}(\underline{q}^k) + J_A(\underline{q}^k) (\underline{q} - \underline{q}^k) + o(\|\underline{q} - \underline{q}^k\|^2)$.
- The iterative approach can be written (by inversion) as
$$\underline{q}^{k+1} = \underline{q}^k + J_A^{-1}(\underline{q}^k) (\underline{x}^* - \underline{x}(\underline{q}^k)).$$
- It converges if the initial state $\underline{q}^0$ is very close to the real value $\underline{q}^*$, such that $\underline{x}(\underline{q}^*) = \underline{x}^*$.
- It does not converge if there are singularities of J_A .
- It is computationally complex since it requires to compute J_A^{-1} .
- It cannot be applied to redundant robot ($m < n$).
- It is a fast method.

Inverse Kinematics: Numerical Methods

Since the analytical solution of $\underline{x}(\underline{q}^*) = \underline{x}^*$ is difficult to find or it does not exist, **numerical solutions** are needed.

Target, i.e. pose to approximate
in finding the joint variable vector: $\underline{x}^*$

Newton method ($m = n$)

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- Define $\underline{x}(\underline{q}) = \underline{x}(\underline{q}^k) + J_A(\underline{q}^k)(\underline{q} - \underline{q}^k) + o(\|\underline{q} - \underline{q}^k\|^2)$.
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Inverse Kinematics: Numerical Methods

Since the analytical solution of $\underline{x}(\underline{q}^*) = \underline{x}^*$ is difficult to find or it does not exist, **numerical solutions** are needed.

Gradient method

- It consists of minimizing the following cost function

$$H(\underline{q}) = \frac{1}{2} \|\underline{x}^* - \underline{x}(\underline{q})\|^2 = \frac{1}{2} (\underline{x}^* - \underline{x}(\underline{q}))^T (\underline{x}^* - \underline{x}(\underline{q}))$$

$$\underline{q}^{k+1} = \underline{q}^k - \alpha \nabla_{\underline{q}} H(\underline{q}^k)$$

- Since $\nabla_{\underline{q}} H(\underline{q}) = -J_A^T(\underline{q}) (\underline{x}^* - \underline{x}(\underline{q}))$ one has $\underline{q}^{k+1} = \underline{q}^k + \alpha J_A^T(\underline{q}^k) (\underline{x}(\underline{q}^k) - \underline{x}^*)$.
- The value α must ensure that the cost function decreases at each step (for large α the minimum value could be skipped).
- For small α , the convergence is very slow.
- It is simpler from a computational point of view: it requires to compute J_A^T .
- It can be applied to redundant robot and it cannot diverge.

Inverse Kinematics: Numerical Methods

Since the analytical solution of $\underline{x}(\underline{q}^*) = \underline{x}^*$ is difficult to find or it does not exist, **numerical solutions** are needed.

Feature of an efficient numerical method

- Initial iterations with gradient method (slower) and final iterations with Newton method.
- Good choice of initial conditions q^0 .
- Optimal choice of the step α .
- Good choice of a stop criteria for the Newton method:

- Cartesian criterium:

$$\|\underline{x}^* - \underline{x}(q^*)\| \leq \varepsilon$$

- Algorithm increment criterium

$$\|\underline{q}^{k+1} - \underline{q}^k\| \leq \varepsilon_q$$

- Jacobian conditioning criterium

$$\sigma_{\min} \{J_A(\underline{q}^k)\} \geq \sigma_0$$

The number of non-zero singular values of matrix is equal to the rank of the matrix

or viceversa



UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

Course of Robot Control

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Inverse Kinematics: Software Tools

Inverse Kinematics: Matlab Tools

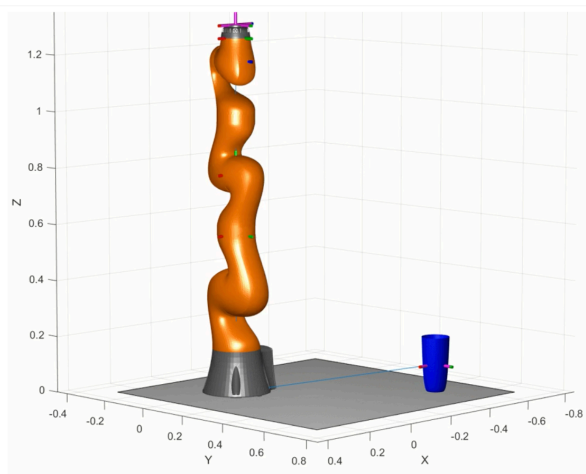


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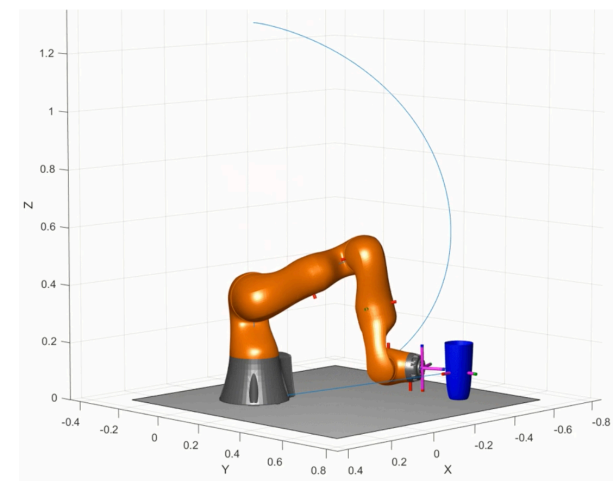
← All UNIPV students can access and use

→ **Robotics System Toolbox** provides tools and algorithms to design, simulate, test, and deploy mobile robot and manipulator applications. For manipulators, the toolbox includes algorithms for direct (forward) and inverse kinematics. It is possible to:

- Import robot definitions from DH parameters
- Building a multibody model based on the information defined in CAD
- Analyzing inverse kinematics
- Designing inverse kinematics solvers
- Solving for multiple-constraint kinematics configuration using generalized inverse kinematics solvers



Generalized inverse kinematics to plan a joint-space trajectory with multiple constraints.



Generalized inverse kinematics to plan a joint-space trajectory with multiple constraints.

For further info: <https://it.mathworks.com/discovery/inverse-kinematics.html>

Inverse Kinematics: CoppeliaSim

COPPELIA  **ROBOTICS**

CoppeliaSim

edu

⚠ Educational only

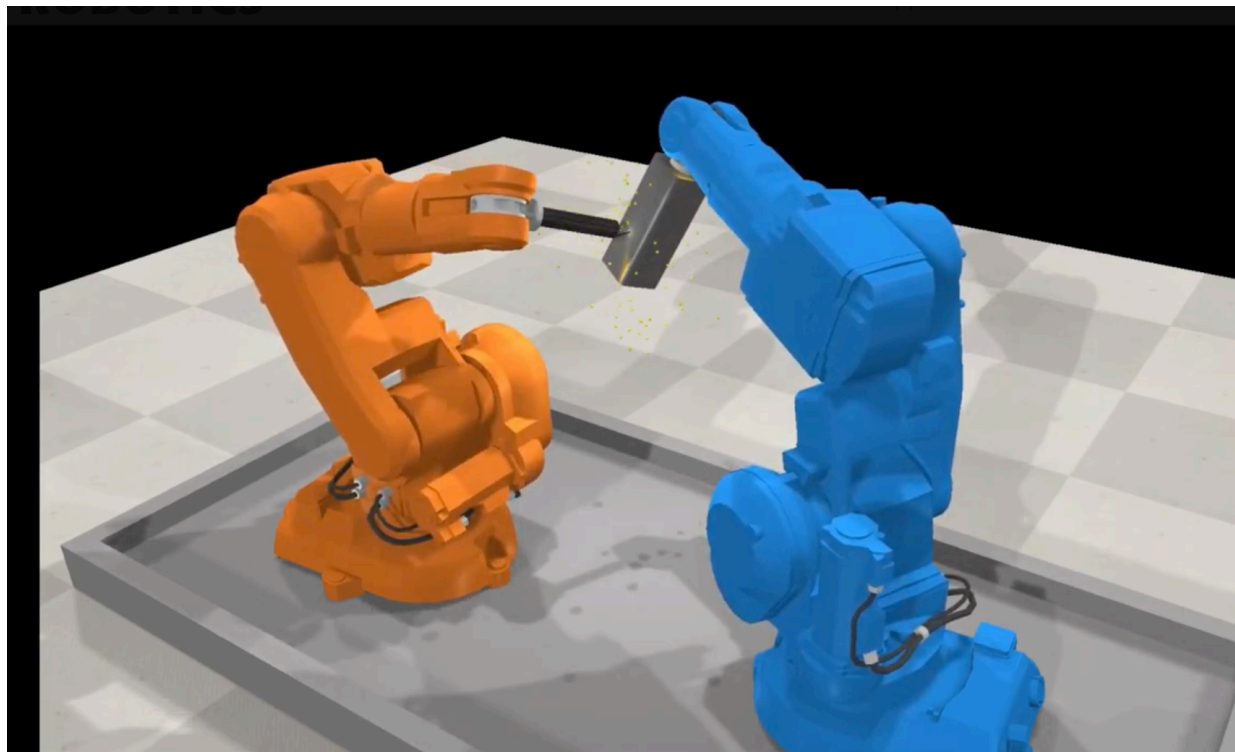
✓ Full simulation functionality

✓ Full editing capabilities

✗ Commercial usage

⬇ Download

← **All students can access and use**



Source: <https://www.coppeliarobotics.com/>



Forward / Inverse Kinematics

Forward/Inverse kinematics calculations for any type of mechanism (branched, closed, redundant, containing nested loops, etc.). An embeddable version of the IK/FK algorithms is available.

Inverse Kinematics: ROS



← **Everyone can access and use**

ROS - Robot Operating System

The Robot Operating System (ROS) is a set of software libraries and tools that help you build robot applications. From drivers to state-of-the-art algorithms, and with powerful developer tools, ROS has what you need for your next robotics project. And it's all open source.



Get ROS Noetic
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LTS)

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Get Iron Irwini on
Ubuntu Linux 22.04,
Windows 10

(Recommended for
Latest ROS 2)

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In the ROS ecosystem, there are several approaches to solve the inverse kinematics problem. e.g.:

- **IKFast**, the Robot Kinematics Compiler, is a powerful inverse kinematics solver provided within **Rosen Diankov's OpenRAVE** motion planning software.
- Unlike most inverse kinematics solvers, IKFast can analytically solve the kinematics equations of any complex kinematics chain and generate language-specific files (like C++) for later use.
- The end result is extremely stable solutions that can run as fast as **5 microseconds** on recent processors.

For further info: <https://www.ros.org/> <https://en.wikipedia.org/wiki/OpenRAVE>

https://abc-robotics.github.io/ros_course_materials_en/06_robotics_principles/